

THIS THESIS IS APPROVED BY:

Principal Supervisor: Ph.D. Vo Duc Hoang

Suggestions/Comments:

This image shows a full page of white paper with horizontal dotted lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

SUMMARY

Topic title: Build a job suggestion system and integrate a chatbot based on RAG

Student name: Dang Quang Nhat Linh

Student ID: 102200177

Class: 20TCLC_DT4

Nowadays, in the era of the Fourth Industrial Revolution, with the rapid development of information technology, the demand for human resources in enterprises is increasing day by day. Meanwhile, job seekers also desire positions that match their skills and expertise. However, traditional recruitment methods still face several challenges. Recruiters often struggle to identify applicants who meet mandatory requirements, and job seekers may spend a lot of time searching and applying for suitable jobs.

The main purpose of this project is to create a system that uses a chatbot assistant to optimize the user experience—helping users find suitable jobs and assisting with various concerns on the platform. This system aims to reduce the time and effort needed to secure a desired position. It covers all steps, including job postings, job applications, resume submissions, and even website integration. Simply put, the system is designed to overcome the limitations of traditional job searching. It is suitable for both part-time and full-time opportunities.

While this is not a new topic, I hope my solution can be helpful making it easier for users to find suitable jobs and enabling enterprises to manage applicants more efficiently. Ultimately, this project contributes to the development of digital services in Vietnam and around the world.

GRADUATION PROJECT COMMENT

I. General information:

1. Student name: DANG QUANG NHAT LINH
2. Class: 20TCLC_DT4 Student ID: 102200177
3. Topic title: BUILD A JOB SUGGESTION SYSTEM AND INTEGRATE A CHAT BOT BASED ON RAG
4. Instructor: Ph.D. VO DUC HOANG Academic title/ degree: Ph.D.

II. Reviews of graduation project

1. About the urgency, novelty and usability of the topic: (2 points)

.....
.....

2. About the results of solving the tasks required by the project: (4 points)

.....
.....

3. About the form, structure and layout of the graduation project: (2 points)

.....
.....

4. The topic includes scientific value/article/problem solving of the enterprise or school: (1 point)

.....
.....

5. Existing shortcomings need to be supplemented or modified:

.....
.....

III. Spirit and attitude of the student (1 point):

.....
.....

IV. Evaluation:

1. Evaluation point: /10
2. Suggest: Defense permitted/ Edit to defend/ Defense not permitted

Da Nang, date.....month..... 2025

Instructor

GRADUATION PROJECT REQUIREMENTS

Student Name: DANG QUANG NHAT LINH

Student ID: 102200177

Class: 20TCLC_DT4

Major: Information

Technology

1. *Topic title:* BUILD A JOB SUGGESTION SYSTEM AND INTEGRATE A CHAT BOT BASED ON RAG.

2. *Project topic:* ☐ has signed intellectual property agreement for final result

3. *Initial figure and data:*

For the data to server as recommendation project, it was crawled from official websites of Indeed.

Content of the explanations and calculations:

Content of the explanations contains 5 parts

- **Introduction:** This chapter gives information about the context and purpose of the project as well as giving the scope of the problems which will be focused on the thesis.
- **Chapter 1:** Theories and Technologies
- **Chapter 2:** Analysis and Design
- **Chapter 3:** Implementation and Evaluation
- **Chapter 4:** Conclusion

4. Drawing, charts (specify types and size of drawing)

- Use case diagram
- Activity diagram
- Sequence diagram
- Architecture structure diagram

5. *Instructor:*

Ph. D Vo Duc Hoang, Information Technology Faculty, Danang University of Science and Technology, The University of Danang, Vietnam.

6. *Date of assignment:* Dec 16th, 2024.

7. *Date of completion:* *Jun 5th, 2025.*

Danang, Jun, 2025

Head of Division.....

Instructor

PREFACE

To achieve good results for this project, I have received enthusiastic help from many people. With deep affection and sincerity, I want to express my gratitude to all the people who have helped me in my study and research.

I would like to thank the teachers at Da Nang University of Science and Technology in general and the faculty of information technology in particular, who have imparted valuable knowledge through classroom hours during the past years, creating a premise for me to do well on this graduation project. And especially, I would like to express my deep gratitude to **Ph. D Vo Duc Hoang** for enthusiastically guiding, sharing valuable experiences and facilitating in many ways for me to successfully complete this project.

With the limitation of knowledge, the report may face unwanted errors, I really hope for the advice of teachers and everyone so that I can learn from it. Sincerely,

CONFIRMATION

I guarantee:

1. The contents of this senior project are performed by myself following the guidance of instructor **PhD. Vo Duc Hoang**.
2. All references used in this senior project thesis are quoted with the author's name, project name, time and location to publish clearly and faithfully.
3. All invalid copies educated statue violation or cheating will be borne the full responsibility by myself.

Student Performed

TABLE OF CONTENTS

THIS THESIS IS APPROVED BY:	i
REFEREE’S COMMENTS	ii
SUMMARY	iii
GRADUATION PROJECT REQUIREMENTS	v
PREFACE	vii
CONFIRMATION	viii
TABLE OF CONTENTS	ix
FIGURES	xii
TABLES	xv
LIST OF SYMBOLS, ACRONYM	ix
INTRODUCTION	1
1. Purpose for doing thesis	1
2. Scope and objective	1
3. Methods	1
4. Main functions in application	2
Chapter 1 : THEORIES AND TECHNOLOGIES	3
1.1 Introduction	3
1.1.1 AI-Powered Chatbot using embedding and LLM (RAG)	3
1.1.2 AI for related job recommendations using TF-IDF and hybrid scoring	3
1.2 Retrieval-Augmented Generation system	4
1.2.1 Architecture Overview	4
1.2.2 Implementation in the Chatbot	5
1.2.3 Pros and Cons	5
1.3 Related jobs system	5
1.3.1 TF-IDF-Based Content Similarity	6
1.3.2 Hybrid Scoring Method	7
1.3.3 Pros and Cons	7
1.4 Technical constraint	7
1.4.1 Next.js	7
1.4.2 NestJs	8
1.4.3 FastAPI	9
1.4.4 Docker	9
1.4.5 Langchain	10
1.4.6 Third-party services (Paypal, Google, Facebook)	11

1.4.7	Firestore cloud messaging	11
1.4.8	Redis	12
1.4.9	Postgres and vector database extensions	13
Chapter 2 : SYSTEM OVERVIEW		14
2.1	Descriptive overview of objects	14
2.2	Descriptive functionality	14
2.2.1	For Admin	14
2.2.2	For enterprise	15
2.2.3	For candidate	15
2.3	Use-case diagram	16
2.3.1	General	16
2.3.2	Admin	17
2.3.3	Candidates	17
2.3.4	Enterprise	19
2.3.5	Authentication	20
2.4	Activity diagram	22
2.4.1	Login	22
2.4.2	Registration	23
2.4.3	Create Job	24
2.4.4	Update Job	25
2.4.5	Get user	26
2.4.6	Get Tags	27
2.5	Sequence diagrams	27
2.5.1	Login	27
2.5.2	Registration	29
2.5.3	Job posting	30
2.5.4	Apply job	31
2.5.5	Manage applicants	32
2.5.6	Enterprise registration	33
2.6	Entity Diagram	34
2.6.1	Table Overview	34
2.6.2	Entity Relationship Diagram	35
2.6.3	Entity details	36
Chapter 3 IMPLEMENTATION AND EVALUATION		43
3.1	Implementation	43
3.1.1	UI for Authentication	43

3.1.2	UI for user role	45
3.1.3	UI for enterprise role	58
3.1.4	Notifications outside	71
3.2	Evaluation	73
3.2.1	The effectiveness of TF-IDF algorithms	73
3.2.2	The embedding models	75
3.2.3	The LLMs	76
3.2.4	Advantages	77
3.2.5	Disadvantages	78
Chapter 4 CONCLUSION AND FUTURE WORK		79
4.1	Result	79
4.2	Future work	79
REFERENCES		80

FIGURES

Figure 1: Agentic RAG architecture	4
Figure 2: Formula for calculating IF	6
Figure 3: Formula for calculating IDF	6
Figure 4: Life cycle overview of Next.js	8
Figure 5: FastAPI architecture	9
Figure 6: Docker architecture	10
Figure 7: General Use-case diagram	16
Figure 8: Use-case for Admin features	17
Figure 9: Use-case for User features	18
Figure 10: Use-case for Enterprise feature	19
Figure 11: Use-case for Authenticate feature	20
Figure 12: Use-case for registration feature	21
Figure 13: Activity diagram for login	22
Figure 14: Activity diagram for registration	23
Figure 15: Activity diagram of Job creating	24
Figure 16: Activity diagram of Job updating	25
Figure 17: Activity diagram of get users	26
Figure 18: Activity diagram of get tags	27
Figure 19: Sequence diagram of login feature	28
Figure 20: Sequence diagram of registration	29
Figure 21: Sequence diagram of job posting	30
Figure 22: Sequence diagram for applying to a job	31
Figure 23: Sequence diagram for managing applicants	32
Figure 24: Sequence diagram for enterprise registration	33
Figure 25: Entity diagram	35
Figure 26: UI for registration	43
Figure 27: UI for email verification	43
Figure 28: UI for login feature	44
Figure 29: UI for forgot password feature	44
Figure 30: UI for reset password feature	45
Figure 31: UI home page	45
Figure 32: UI for find job feature	46
Figure 33: UI for filter job feature	46
Figure 34: UI for list view job	47
Figure 35: UI for job detail	48
Figure 36: UI for apply job feature	49
Figure 37: UI enterprise list	49
Figure 38: UI enterprise detail	50
Figure 39: UI user profile	50

Figure 40: UI for view applied job	51
Figure 41: UI for view detail applied job	51
Figure 42: UI list favorite job view	52
Figure 43: UI for set up personal user	53
Figure 44: UI for setting profile	53
Figure 45: UI for posting CV	54
Figure 46: UI for managing CV	54
Figure 47: UI for managing social network	55
Figure 48: UI for enterprise registration	55
Figure 49: UI chat bot	56
Figure 50: Find job follow user experience	57
Figure 51: UI candidate list	58
Figure 52: UI for filtering user	58
Figure 53: UI for pricing plan	59
Figure 54: UI for complete payment	60
Figure 55: UI for enterprise overview	61
Figure 56: UI for posting job	61
Figure 57: UI for managing job	62
Figure 58: UI for boosting job	63
Figure 59: UI for managing applications	63
Figure 60: UI for filter job	64
Figure 61: UI for application detail	64
Figure 62: UI for notifications	65
Figure 63: UI for setting enterprise	65
Figure 64: UI for setting up a founding enterprise	66
Figure 65: UI for social networking	67
Figure 66: UI for admin dashboard	67
Figure 67: UI for listing users	68
Figure 68: UI for change status	68
Figure 69: UI for managing enterprise	68
Figure 70: UI for change status enterprise	69
Figure 71: UI for enterprise detail	69
Figure 72: UI for list job	70
Figure 73: UI for managing tag	70
Figure 74: UI for managing category	70
Figure 75: UI for code verification email	71
Figure 76: UI for reset password	71
Figure 77: UI for admin reject form enterprise registration	71
Figure 78: UI for job expried	72

Figure 79: UI when admin accepted enterprise form	72
Figure 80: UI when admin block enterprise	72
Figure 81: Word cloud	74
Figure 82: Industry type distribution	75

TABLES

Table 1: Pros and Cons of using RAG	5
Table 2: Pros and Cons of using TF-IDF	7
Table 3: Details of Table Overview	34
Table 4: Accounts entity detail	36
Table 5: Profile entity detail	36
Table 6: Address entity detail	37
Table 7: Enterprise entity detail	38
Table 8: Jobs entity detail	39
Table 9: Tags entity detail	40
Table 10: Boosted Job entity detail	40
Table 11: Categories entity detail	40
Table 12: CVs entity detail	41
Table 13: Notifications entity detail	41
Table 14: Transaction entity detail	41
Table 15: Website entity detail	42
Table 16: Performance Evaluation	76
Table 17: Other LLMs Evaluated	76

LIST OF SYMBOLS, ACRONYM

SYMBOL:

ACRONY

No	Acronym	Explanation
1	OOP	Object Oriented Programming
2	FP	Functional Programming
3	AI	Artificial Intelligence
4	SQL	Structured Query Language
5	RAG	Retrieval-Augmented Generation
6	LLM	Large Language Model
7	TF-IDF	Term Frequency - Inverse Document Frequency

INTRODUCTION

1. Purpose for doing thesis

Today, the number of job-seeking people is increasing. Along with that, the demand for human resources of enterprise in this field is increasing day by day. Accordingly, the workload is also diverse and rich. However, besides that, there still exist some problems of traditional recruitment, as Enterprises cannot determine the applicants that match with their requirements, and it may take a lot of time for applicants to find and apply suitable jobs. Moreover, with the strong development of AI, large corporations are racing to develop their own models. Taking advantage of that, I have integrated chat bot assistant to assist users, optimize users to suggest users' suitable jobs and guide users to use web applications.

2. Scope and objective

Based on that idea, I have chosen this topic with the goal of solving those problems. The main objective is to create software integrated chat bot system that allows candidates to quickly find suitable jobs according to the requirements of the jobs as well as their skill. All steps from job posting, applying, submitting resumes, even the website system for candidate roles, managing jobs managers, managing applicants, promoting jobs and their profiles for the enterprise role. And the manager all for the administrator.

3. Methods

Observation method: Observing the general situation and the problems that enterprise is facing.

Theories analyzing method: Research about related papers, analyze system and domain business.

Experimental method: Use current applications is related to the problem of providing the necessary functionality for my application in this project.

Modeling Method: After implementing and getting information from above methods, I built a model by drawing diagrams and figures to show the big picture to develop the final usable application.

4. Main functions in application

- Users can search for and apply for any job they want.
- Candidates and enterprises can quickly register and log in to the application.
- Includes a system management function for enterprises and administrators.
- Users can interact with a chatbot assistant.
- Users will receive notifications from the system.
- Users will also receive certain information via email.
- Enterprises can boost their job postings to top visibility and manage their listings.
- Both enterprises and administrators can view statistics through a dashboard.
- Enterprises can register for a subscription to purchase points.

Chapter 1 : THEORIES AND TECHNOLOGIES

1.1 Introduction

This system integrates multiple advanced technologies, particularly in the field of artificial intelligence and machine learning, to deliver intelligent, responsive, and user-focused functionality. In this section, I will present the key concepts and technologies used throughout the project.

The application incorporates two distinct AI components, each designed to serve a specific purpose:

1.1.1 AI-Powered Chatbot using embedding and LLM (RAG)

The first AI component is an intelligent chatbot system built using the LangChain framework. It utilizes text embedding techniques to convert user queries and documents into vector representations, allowing for semantic understanding. These embeddings are then used in combination with a large language model (LLM) in a Retrieval-Augmented Generation (RAG) architecture. This enables the chatbot to retrieve relevant context from a document database and generate accurate, human-like responses based on both the retrieved data and the power of the LLM.

1.1.2 AI for related job recommendations using TF-IDF and hybrid scoring

The second AI module focuses on job recommendation. It applies the TF-IDF (Term Frequency-Inverse Document Frequency) algorithm to analyse job descriptions and user profiles by extracting important keywords. Based on these keyword vectors, the system calculates the similarity between jobs and users. A hybrid scoring method is then used, combining content-based filtering (e.g., from TF-IDF) with collaborative features (e.g., user interactions or behaviours) to generate a ranked list of related job suggestions that are both personalized and relevant.

These two AI components work independently but complement each other, enhancing the overall functionality of the platform: one improves user interaction via conversational AI, and the other provides intelligent job discovery through advanced recommendation techniques.

1.2 Retrieval-Augmented Generation system

Retrieval-Augmented Generation (RAG) is a powerful architecture that combines two core components of natural language processing: retrieval and generation. Unlike traditional language models that rely solely on pre-trained knowledge, RAG enhances the model's capabilities by allowing it to dynamically retrieve external information from a document store at the time of answering a query.

1.2.1 Architecture Overview

A typical RAG system consists of two main stages:

- **Retriever:** This module searches for relevant documents or pieces of information from a knowledge base. It often uses text embedding models to convert both the query and documents into vectors and applies similarity metrics (e.g., cosine similarity) to find the most relevant documents.
- **Generator:** Once relevant documents are retrieved, the generator—typically a **large language model (LLM)**—uses both the query and the retrieved context to produce a coherent and accurate answer.

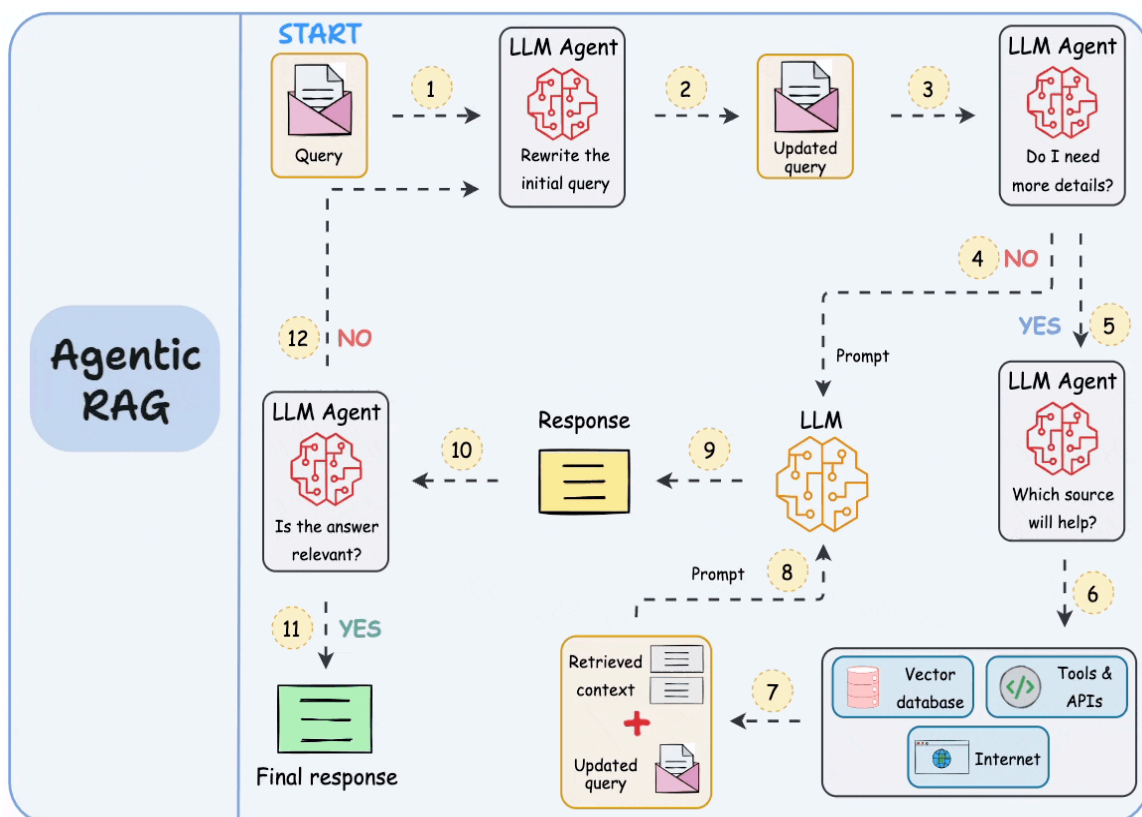


Figure 1: Agentic RAG architecture

1.2.2 Implementation in the Chatbot

The chatbot in this system is built using the LangChain framework, which supports seamless integration of RAG components. The implementation process includes:

- Creating a document vector store using a text embedding model (e.g., Sentence Transformers).
- Using similarity search to retrieve relevant documents based on user queries.
- Passing the retrieved documents and user query into an LLM (e.g., GPT or similar) to generate the final response.

This setup enables the chatbot to respond to user inquiries with high contextual relevance, particularly in domains such as job searching, application instructions, or enterprise-specific information.

1.2.3 Pros and Cons

Table 1: Pros and Cons of using RAG

Pros	Cons
Enhances language model responses with up-to-date or domain-specific information	More complex to implement compared to standalone LLMs
Reduces hallucination by grounding answers in retrieved documents	Depends on the quality and relevance of the document store
Enables dynamic updates of knowledge without retraining the language model	Retrieval step may return irrelevant or noisy documents if not properly tuned
Scalable to large corpora using vector databases and embedding search	Requires good vector indexing (e.g., FAISS) and text embedding for effective performance

1.3 Related jobs system

The related jobs system in this project is a specialized recommendation engine designed to suggest relevant job postings to users based on content similarity and user interaction data. It plays a crucial role in helping users discover opportunities that match their skills and interests efficiently.

Unlike traditional recommendation systems that focus on general products or ratings, this system emphasizes job matching, where textual content (job

descriptions, skills, titles, etc.) is central. The system is built using a combination of TF-IDF (Term Frequency-Inverse Document Frequency) and a hybrid scoring mechanism that balances content relevance with other factors such as popularity or user engagement.

1.3.1 TF-IDF-Based Content Similarity

TF-IDF is a statistical measure used to evaluate the importance of a word in a document relative to a corpus. In the context of job recommendation:

- Each job posting is treated as a document.
- Important keywords (skills, technologies, responsibilities) are extracted from the job description.
- A TF-IDF vector is generated for each job, forming a high-dimensional representation of its content.

The similarity between two jobs (or between a job and a user profile) is calculated using cosine similarity between their TF-IDF vectors. Jobs with similar textual content are considered related.

Term frequency works by looking at the frequency of a *particular term* you are concerned with relative to the document. There are multiple measures, or ways, of defining frequency.

$$tf(i, j) = \frac{n_{ij}}{\sum_k n_{kj}}$$

Figure 2: Formula for calculating IF

Inverse document frequency looks at how common (or uncommon) a word is amongst the corpus. IDF is calculated as follows where t is the term (word) we are looking to measure the commonness of and N is the number of documents (d) in the corpus (D). The denominator is simply the number of documents in which the term, t , appears in.

$$idf(i, D) = \log \log \left(\frac{N}{count(d \in D: t \in d)} \right)$$

Figure 3: Formula for calculating IDF

1.3.2 Hybrid Scoring Method

While TF-IDF captures textual similarity, it may overlook user behaviour patterns or job trends. Therefore, a hybrid scoring approach is applied to improve recommendation accuracy.

The **Hybrid Score** is computed as a combination of:

- Content Similarity Score (from TF-IDF)
- Interaction Score (based on click, view, or apply data)
- Popularity Score (based on how often a job is interacted with by all users)

1.3.3 Pros and Cons

Table 2: Pros and Cons of using TF-IDF

Pros	Cons
Simple and easy to implement	Does not capture semantic meaning or context
Computationally efficient and lightweight	Treats each word independently (bag-of-words model)
Effective for basic similarity tasks (e.g., document or job matching)	Sensitive to vocabulary mismatch (e.g., synonyms not recognized)
Requires no labeled data or training	Can perform poorly on short or ambiguous text

1.4 Technical constraint

1.4.1 Next.js

❖ Introduction

Next.js is a popular open-source React framework developed by Vercel. It enables developers to build server-side rendered (SSR) and statically generated web applications using React. Next.js provides features like file-based routing, automatic code splitting, API routes, and built-in support for performance optimization, making it suitable for modern web development.

❖ Architecture and Environments

Next.js runs on the Node.js runtime environment and supports both server-side and client-side rendering. It leverages React components for UI building and uses a hybrid rendering approach — combining Static Site Generation (SSG), Server-Side Rendering (SSR), and Client-Side Rendering (CSR) where necessary. Understanding how Next.js handles API routes,

middleware, and rendering behavior is essential for optimizing performance and building scalable applications.

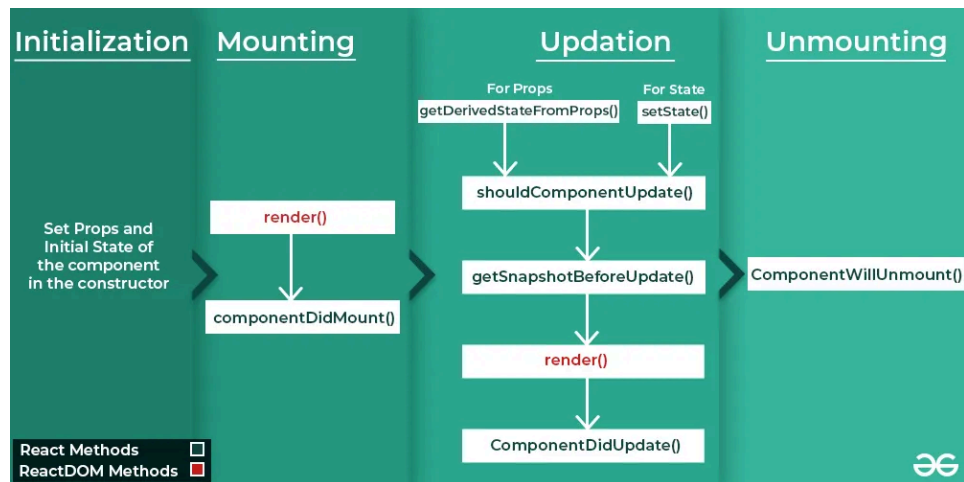


Figure 4: Life cycle overview of Next.js

❖ **Use case**

Build a server-client architecture for an application with UI responsibilities.

1.4.2 NestJs

❖ **Introduction**

Nest (NestJs) is a framework for building efficient, scalable NodeJS server-side applications. It uses progressive JavaScript, is built with and fully supports Typescript and combine elements of OOP(Object oriented Programming), FP (Functional Programming)

Especially, Essentially, NestJS is a layer on top of Node that has supercharged methods and implementations that can help us write server-side applications quick and easy. Nest is very convenient to content all your needs.

❖ **Why we need NestJs when we have Node?**

- Nest provide a layer of abstraction on top of node, which means that it can now leverage the features of Node as well as expose APIs for better performance and efficiency
- Nest also provide compatible modules interacting with ORM's which can take advantage of working with database

❖ **Use case**

Build a server-side application with modular and scalable architecture.

1.4.3 FastAPI

❖ Introduction

FastAPI is a modern, high-performance web framework built with Python for developing web applications and APIs. Based on Python standards like type hinting, FastAPI helps developers quickly build maintainable and efficient RESTful APIs. The goal of FastAPI is to make API development fast, simple, and reliable, while offering an excellent developer experience.

❖ Architecture and environment

FastAPI is built on powerful libraries such as Starlette (for web handling) and Pydantic (for validation and serialization). When launching a FastAPI application, it starts a development server with asynchronous processing, automatically generates API documentation using Swagger UI or ReDoc, and validates input data based on Python type hints.

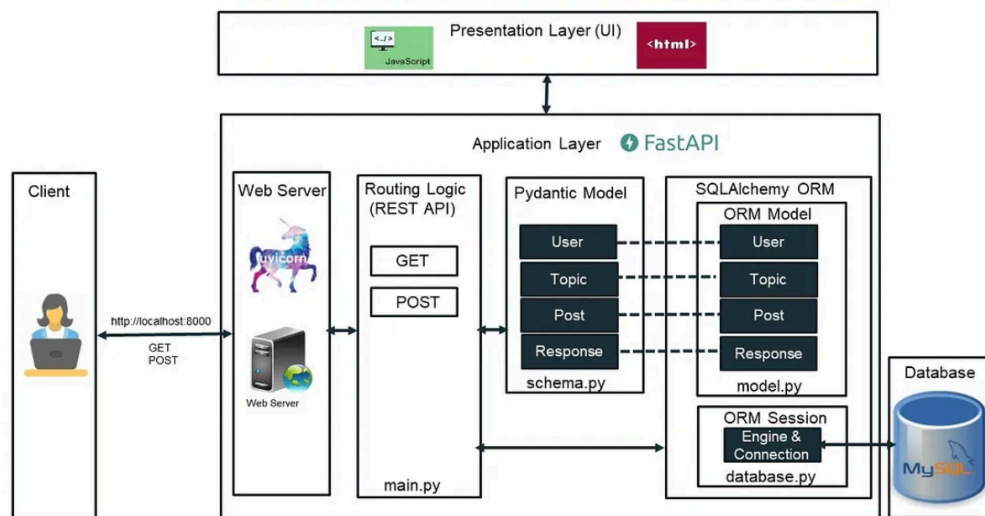


Figure 5: FastAPI architecture

❖ Use case

Build a high-performance backend API for AI-powered services.

1.4.4 Docker

❖ Introduction

Docker is an open-source platform designed to automate the deployment, scaling, and management of applications using containerization. It allows developers to package an application and all of its dependencies into a standardized unit called a container, which ensures that the application runs

consistently across different environments.

❖ Architecture and Environment

Docker uses a client-server architecture. The Docker client communicates with the Docker daemon, which is responsible for building, running, and managing containers. Applications are defined in a Dockerfile, which includes instructions for building the container image, such as installing dependencies and setting up the environment.

Once the image is built, it can be deployed consistently across development, staging, and production environments using Docker Engine.

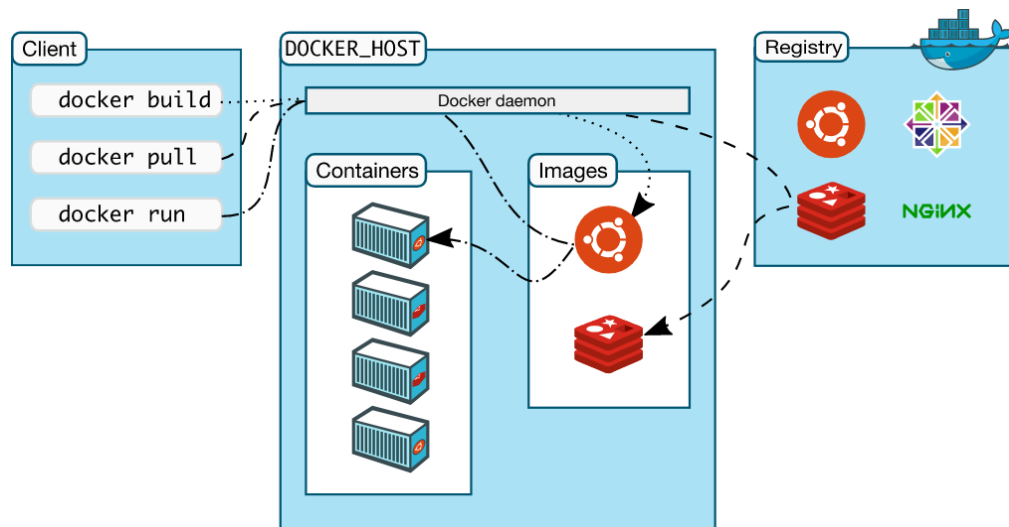


Figure 6: Docker architecture

❖ Use case

Containerize and deploy applications consistently across different environments.

1.4.5 Langchain

❖ Introduction

LangChain is open-source framework designed to simplify the development of applications powered by llms such as OpenAI's GPT, Anthropic's Claude, or Hugging Face models. LangChain provides tools to help developers build intelligent, multi-step, context-aware applications by connecting language models with external data sources and computation environments.

❖ Architecture and Environment

LangChain introduces a modular architecture that allows developers to combine different components into complex workflows. These include:

- Prompt Templates – Predefined structures for consistent LLM interaction.

Application of a job suggestion system and integration of a chatbot based on RAG

- Chains – Sequences of steps (e.g., prompts → model calls → actions).
- Agents – LLMs that can decide what actions to take (such as querying a database or calling an API) based on user input.
- Memory – Enables applications to remember past interactions and use them in future prompts.

LangChain can run in various environments, including local machines, cloud services, or containerized platforms like Docker.

❖ Use case

Develop agentic systems that combine retrieval-augmented generation (RAG) with reasoning capabilities.

1.4.6 Third-party services (Paypal, Google, Facebook)

❖ Introduction

Modern web applications often require integration with third-party platforms to extend functionality, improve user experience, and handle services such as authentication, payments, and analytics. Among the most commonly integrated platforms are PayPal, Google, and Facebook, each providing robust APIs to support these needs.

These third-party services help developers offload complex functionalities like payment processing, OAuth-based login, and user data synchronization, allowing them to focus more on the core features of their application.

❖ Architecture and Environment

Third-party integrations typically follow client-server architecture, where the frontend (e.g., Next.js) interacts with the service provider's APIs, and the backend (e.g., Fast API) handles secure data processing, storage, and validation.

- OAuth 2.0 – Used by Google and Facebook for secure user authentication and authorization.
- REST APIs / Webhooks – Used by PayPal to handle payment events and updates in real time.

These integrations often require secure API keys, token management, and callback URL configuration to ensure safe and reliable communication.

❖ Use case

Enable secure user authentication through third-party identity providers.

1.4.7 Firebase cloud messaging

❖ Introduction

Firebase Cloud Messaging (FCM) is a cross-platform messaging solution provided by Firebase that enables developers to send push notifications and in-app messages to users on Android, iOS, and web platforms. FCM supports both targeted messages (e.g., user-specific updates) and broadcast messages (e.g., promotions or alerts), making it a powerful tool for enhancing user engagement and real-time communication

❖ **Architecture and Environment**

FCM operates on a client-server model where messages are sent from a backend server (or Firebase Console) to client applications via device tokens. These tokens uniquely identify user devices and are used to route notifications accurately.

The typical flow includes:

- Client app registers with FCM and receives a unique device token.
- Backend stores this token and uses it to send push notifications via FCM HTTP or legacy APIs.
- FCM routes the message to the correct device using its infrastructure.

❖ **Use case**

Send real-time push notifications to users across platforms.

1.4.8 Redis

❖ **Introduction**

Redis (REmote DIctionary Server) is an open-source, in-memory data structure store, used as a database, cache, and message broker. It supports a wide range of data structures such as strings, hashes, lists, sets, sorted sets, bitmaps, hyperloglogs, and geospatial indexes. Known for its high performance, low latency, and simplicity, Redis is widely used in scenarios that require fast access to data, real-time analytics, pub/sub messaging, and caching.

Redis is written in ANSI C and typically operates as a single-threaded process. It is optimized for speed, often serving millions of read and write operations per second in production environments. Redis also supports persistence through snapshots and append-only files (AOF), allowing data to survive server restarts.

❖ **Architect and environment**

Redis follows a client-server model and uses a single-threaded, event-driven architecture. The core architectural components include:

Application of a job suggestion system and integration of a chatbot based on RAG

- Redis Server: The main daemon process that handles client requests.
- Clients: Applications that connect to the Redis server to perform operations.
- Command Dispatcher: Redis processes commands sequentially using an event loop, ensuring simplicity and high throughput.
- Data Structures: Stored in memory using efficient algorithms and custom encoding to optimize for memory usage and speed.

❖ Use case

Ideal for caching frequently accessed data, managing cronjob scheduling, and implementing message queues to enable efficient and scalable application workflows.

1.4.9 Postgres and vector database extensions

❖ Introduction

PostgreSQL is a powerful, open-source object-relational database system known for its robustness, extensibility, and compliance with SQL standards. With the rise of AI and large language model (LLM) applications, PostgreSQL has evolved to support vector similarity search through various extensions, making it suitable for storing and querying high-dimensional vector embeddings.

Docker image: ankane/pgvector

❖ Architecture and Environment

PostgreSQL supports vector search through specialized extensions that integrate vector storage and similarity search into the traditional relational database structure. Common extensions include:

- pgvector – A popular PostgreSQL extension that enables storage and similarity search of vectors (e.g., embeddings from LLMs).
- PostGIS – Adds support for geographic objects, useful in geospatial queries.
- TimescaleDB – An extension for time-series data (not vector-specific, but often used alongside).

The pgvector extension allows storing vectors as native data types and performing operations such as cosine similarity, inner product, and Euclidean distance efficiently using index structures like IVFFlat.

❖ Use case

Store and manage relational data with strong consistency and complex

Application of a job suggestion system and integration of a chatbot based on RAG
querying.

Chapter 2 : SYSTEM OVERVIEW

1.1 Descriptive overview of objects

The system has three main user Groups

- **Administrator:** A person has the highest privilege in system. Admins operate functions of user management, enterprise management, item management, admin still take responsibly for security tracking and manage tools of the system and statistics.
- **Enterprise:** (recruiter) When Enterprise register new user and accepted by system, they could change the information of company, post new jobs, manage the jobs, boost jobs, and manage the applicants applied for job.
- **User:** People who are looking for a job in the system. Once a user has completed registration and authentication, they can update personal information, upload their CV, interact with the chatbot assistant, and save favorite enterprises or job postings.

1.2 Descriptive functionality

1.2.1 For Admin

- ❖ Login with admin: User must be logged in with an Admin account to use all the functions and benefits that this role possesses.
- ❖ Statistics
 - Admin can statistically total jobs, users, enterprises, and revenue.
- ❖ User Management
 - Admin has the access to view, edit, delete, update, and search any user account.
 - Admin confirms the information comes from a company and can accept new one registration coming from that company. Then an email will be sent from the system to provide the password for the registered company account.
- ❖ Enterprise management
 - Admin can view, approve, reject form enterprise registration.
 - Admin can view, edit, block enterprise
- ❖ Job management
 - Admin can view, edit, delete, search any existing job in system.
- ❖ Items Management
 - Admin can view, edit, delete, and update any existing categories in

system.

- Admin can view, edit, delete, and update any existing tags in system.

1.2.2 For enterprise

❖ Register

- Enterprises must complete the registration steps and must be confirmed by the administrator.

❖ Login

- Enterprise must sign in with enterprise accounts to use the system with all the features of the enterprise.

❖ Profile management

- Enterprise can view, edit their profile

❖ Job management

- Enterprise can view list of jobs, search, edit, or delete any the job posted on the company site.

❖ Applicant management

- Enterprise can view applicant's listings, search, edit, delete an applicant.

1.2.3 For candidate

❖ Register

- Job seekers must complete registration operations to have a user account in the system.

❖ Login

- Job seekers must login into the system with a valid account to use the features for job seekers.

❖ Manage profile

- Build up their profile, and social network.
- Register enterprise form if they want to be enterprise.

❖ Chat bot assistant

- The chat suggests jobs for candidates based on user behaviour.
- Interact chat bot with 4 tools: search jobs, search enterprise, website information.

❖ Search for a job

- Candidates can search for job information by name, location, salary, job requirements.

❖ Posting and manage resumes

Application of a job suggestion system and integration of a chatbot based on RAG

- o Candidates can upload resumes to the system, This CV will be store in the AWS S3.
- ❖ Jobs favorite management
 - o Job seekers can see the recent job view, favourite job, applied job
 - o They can also see the status of that job, and will be notified if it is approved or rejected by the Enterprise or when the job has ended the recruitment period

1.3 Use-case diagram

1.3.1 General

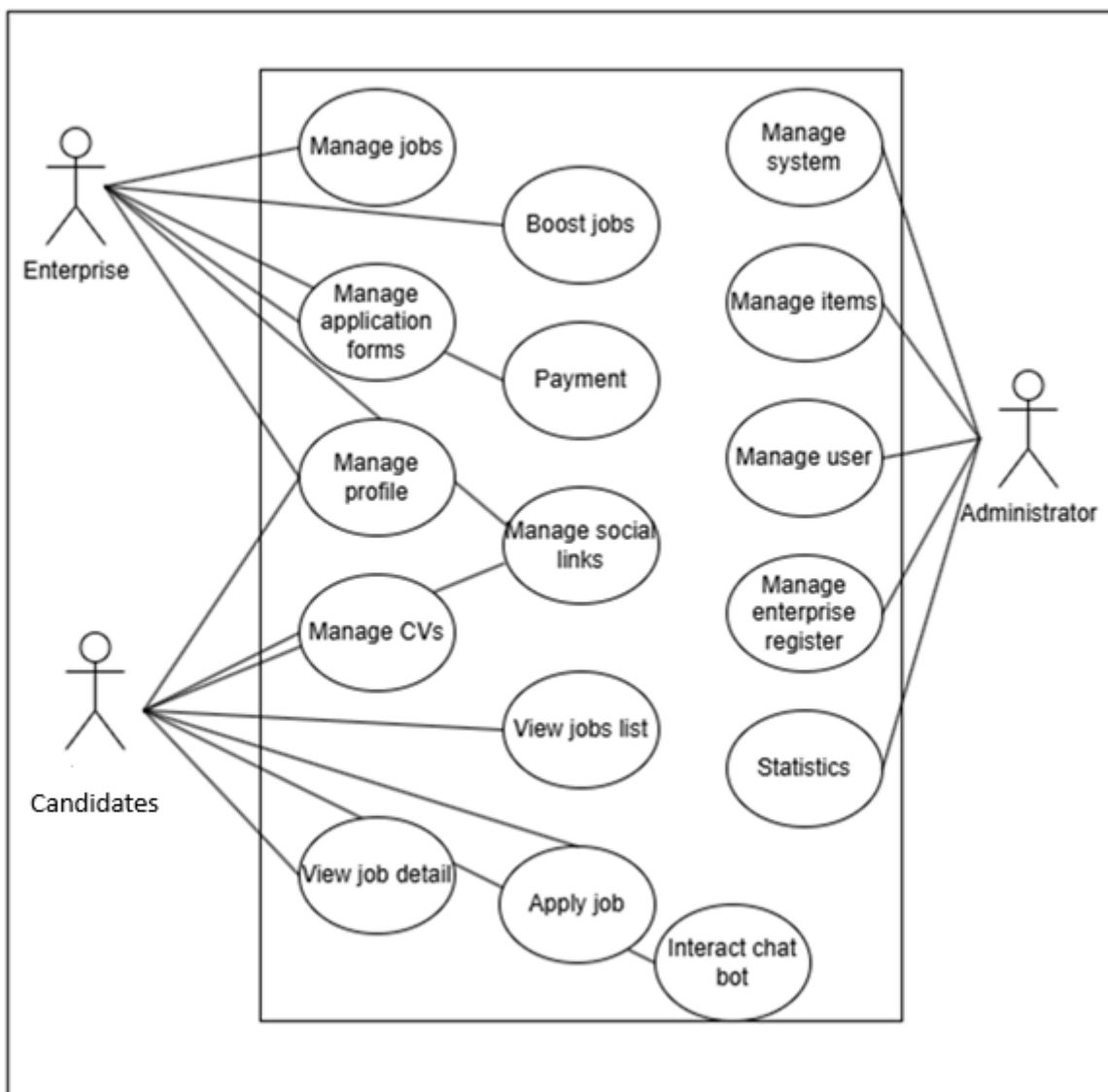


Figure 7: General Use-case diagram

1.3.2 Admin

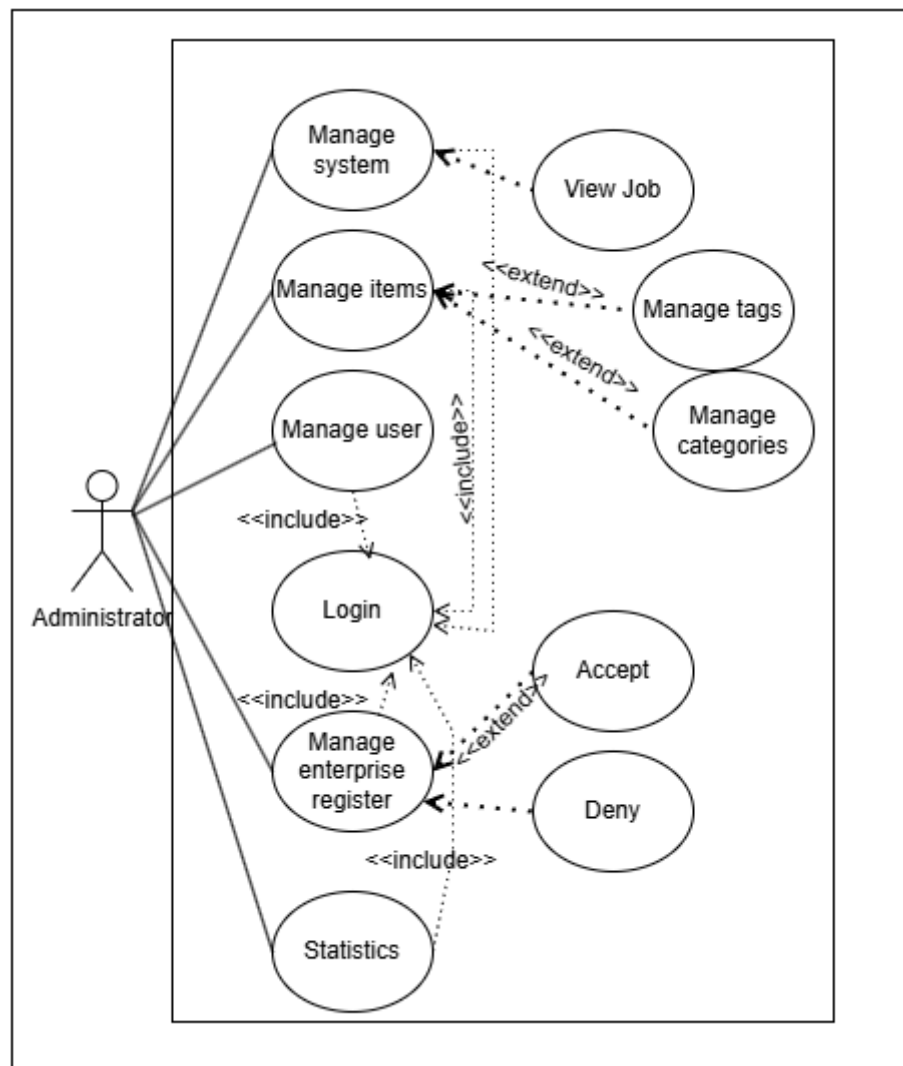


Figure 8: Use-case for Admin features

1.3.3 Candidates

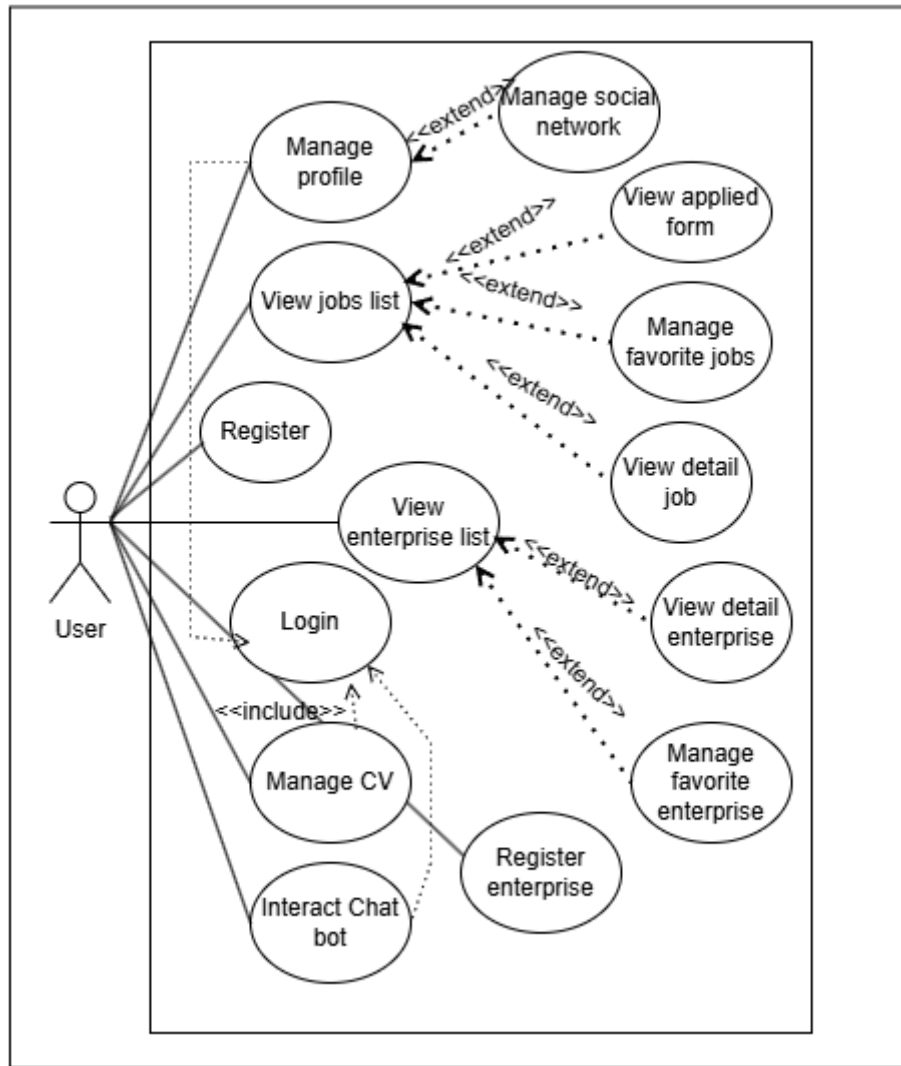


Figure 9: Use-case for User features

1.3.4 Enterprise

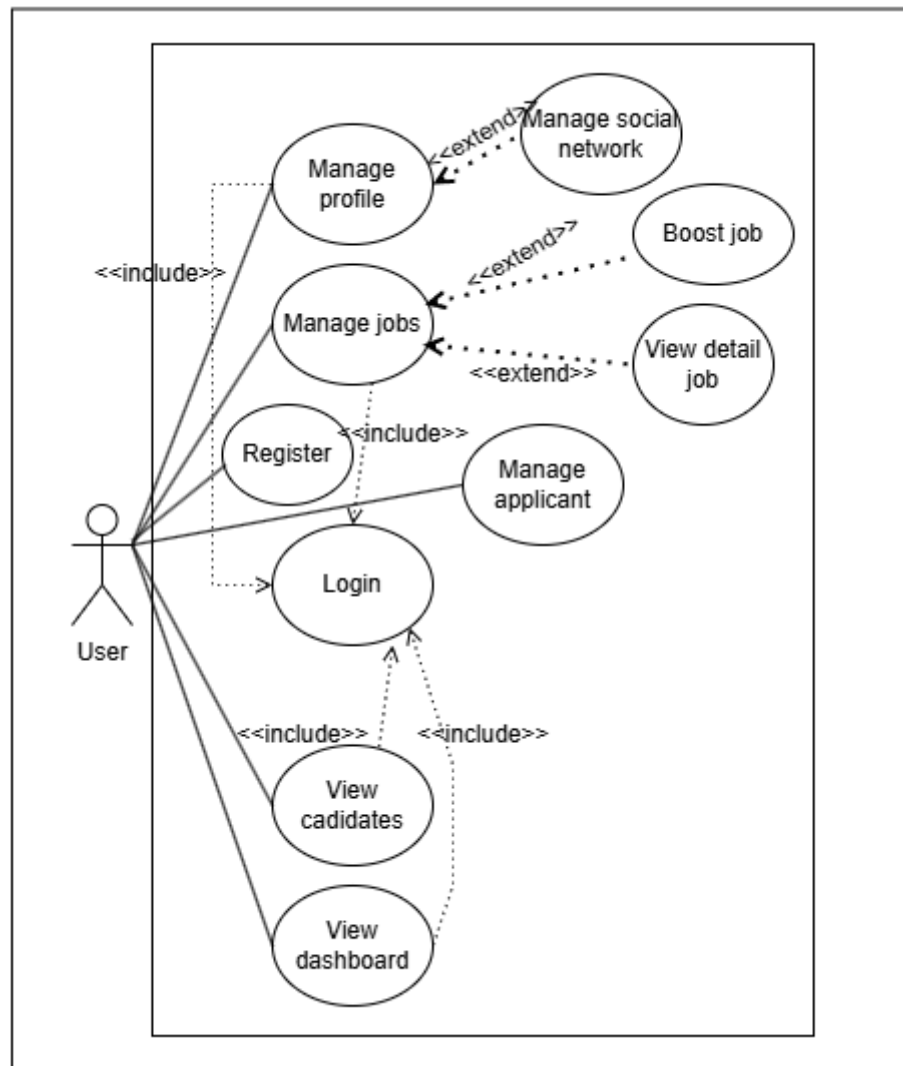


Figure 10: Use-case for Enterprise feature

1.3.5 Authentication

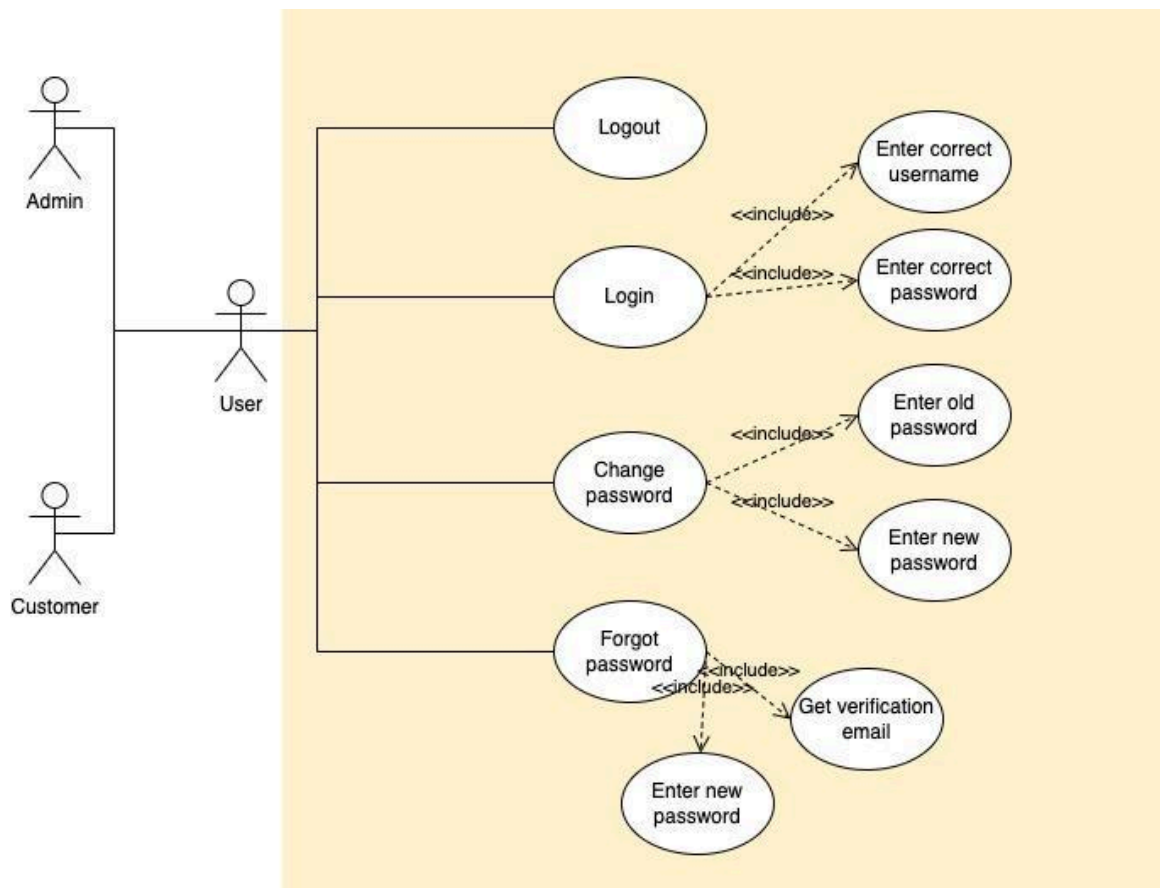


Figure 11: Use-case for Authenticate feature

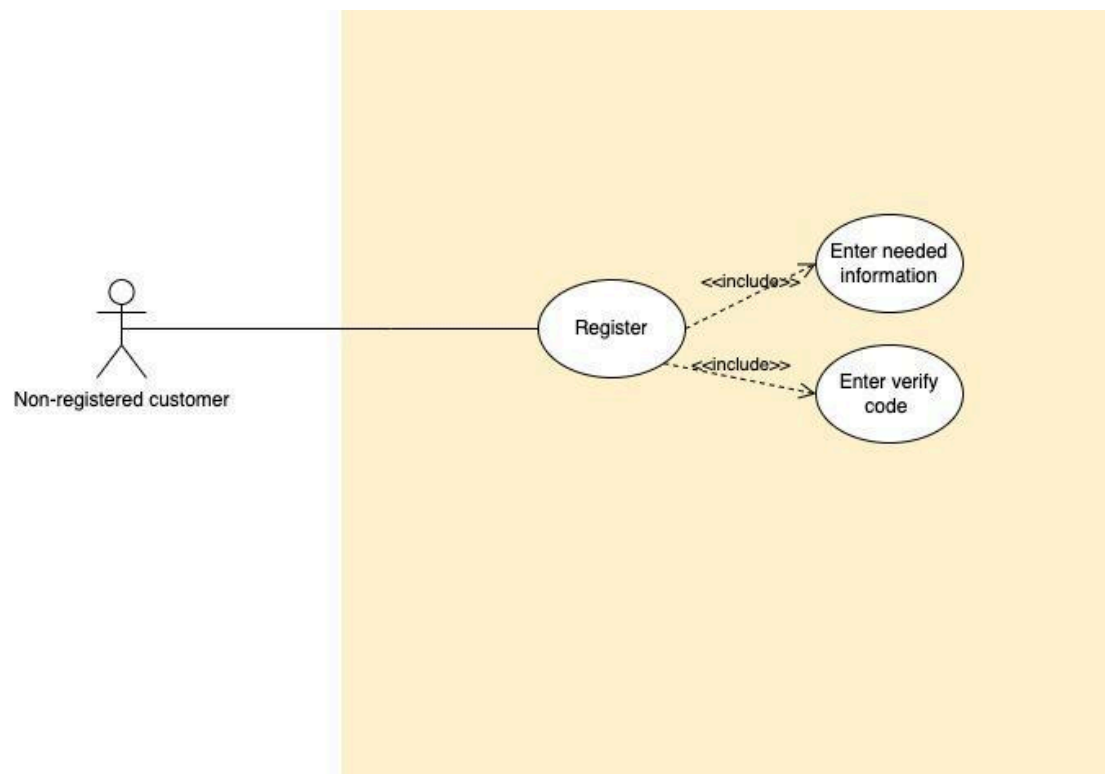


Figure 12: Use-case for registration feature

0.4 Activity diagram

0.4.1 Login

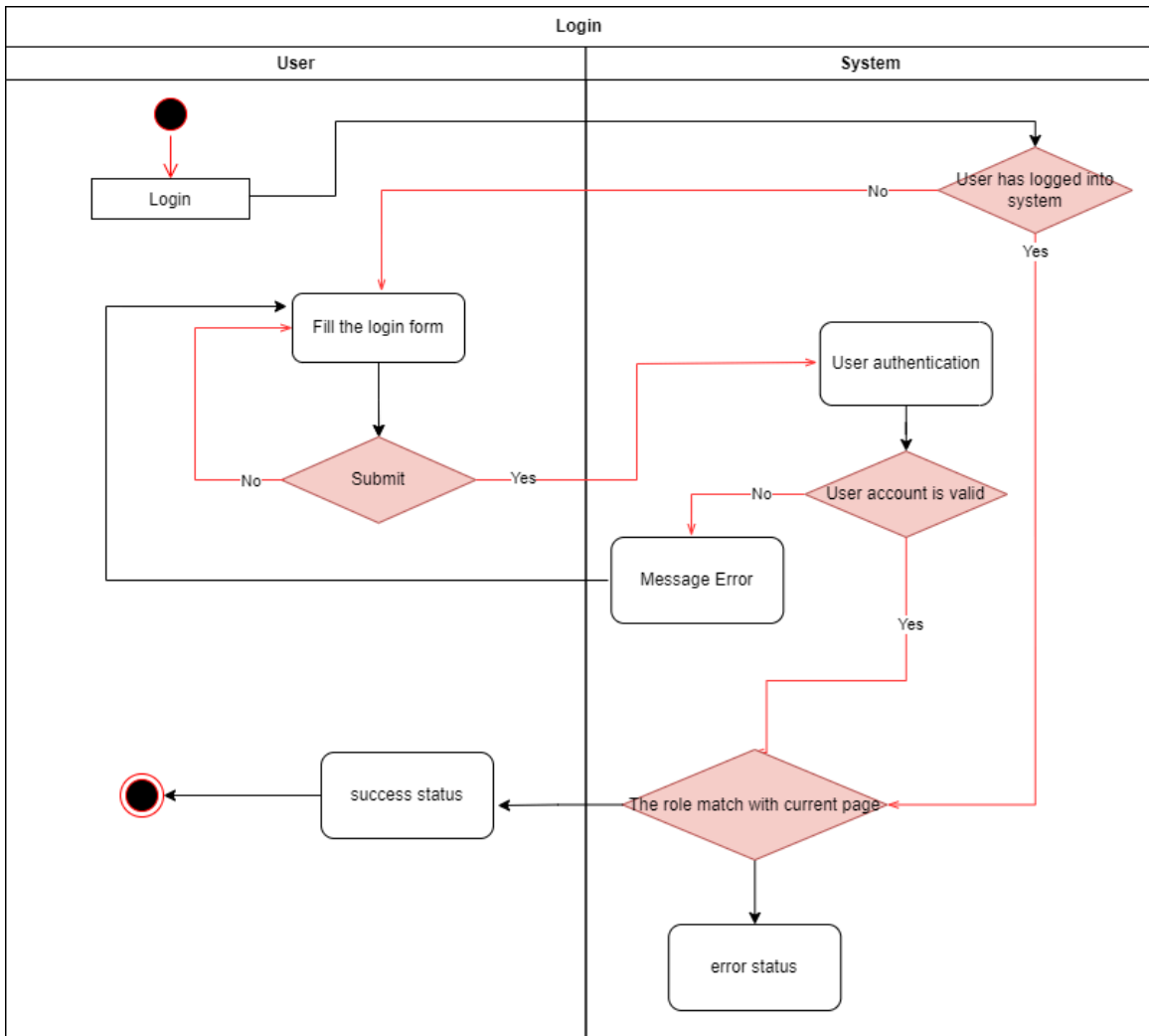


Figure 13: Activity diagram for login

0.4.2 Registration

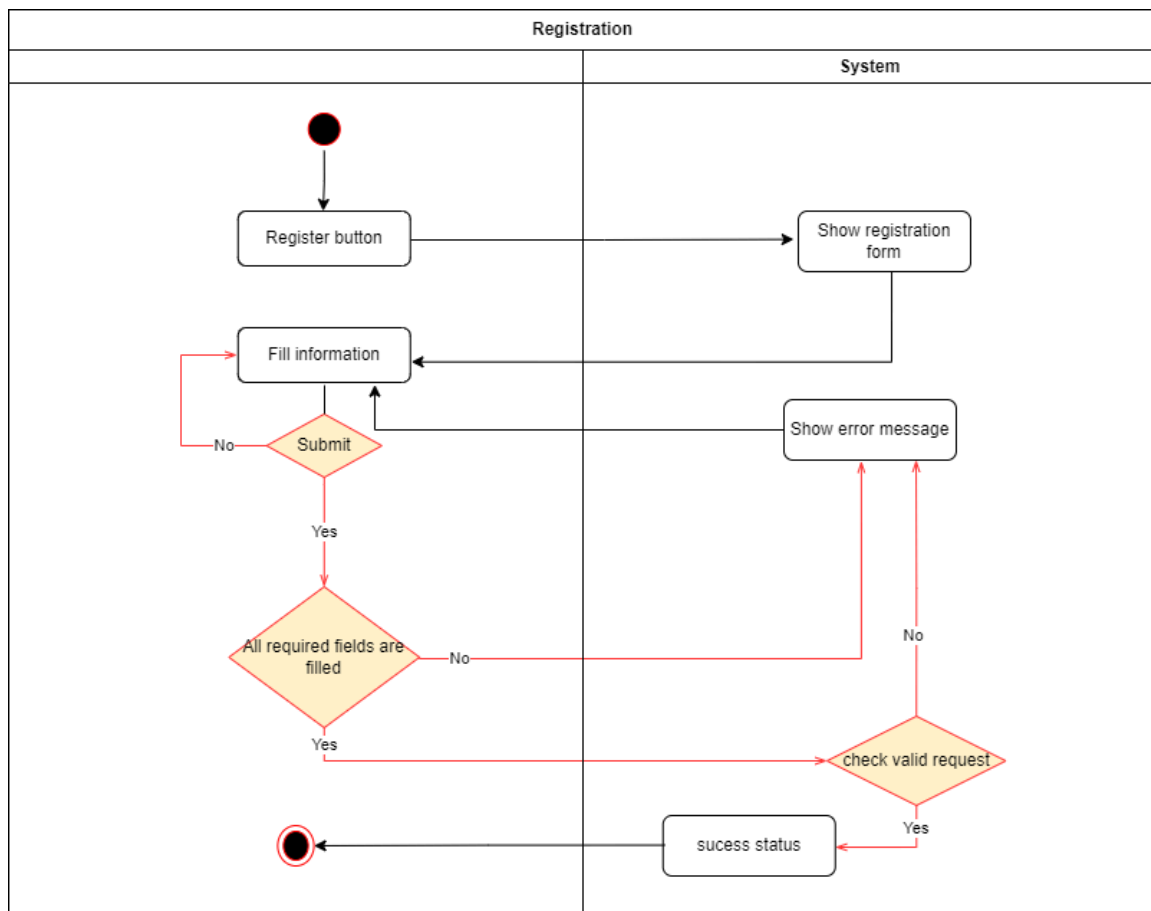


Figure 14: Activity diagram for registration

0.4.3 Create Job

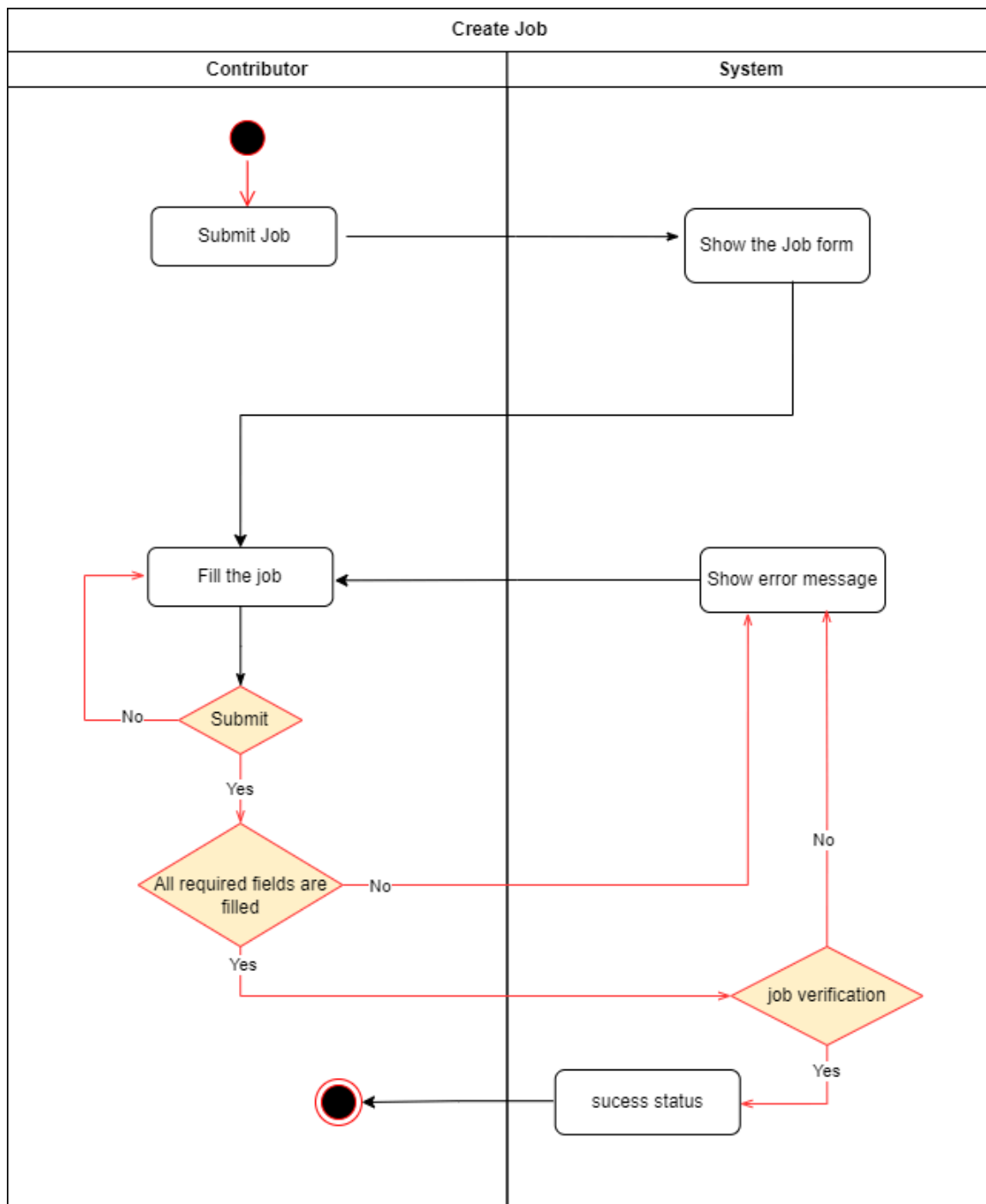


Figure 15: Activity diagram of Job creating

0.4.4 Update Job

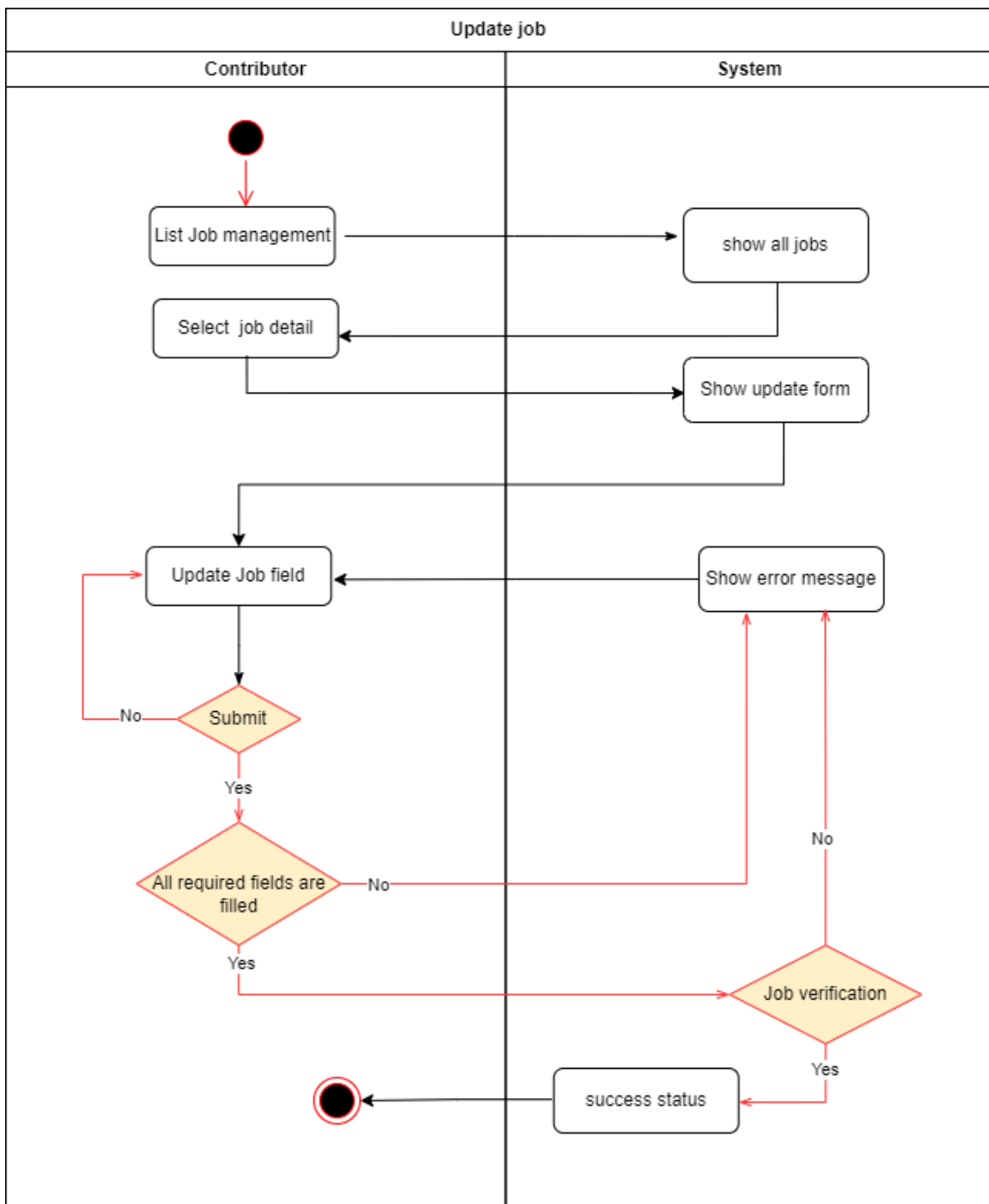


Figure 16: Activity diagram of Job updating

0.4.5 Get user

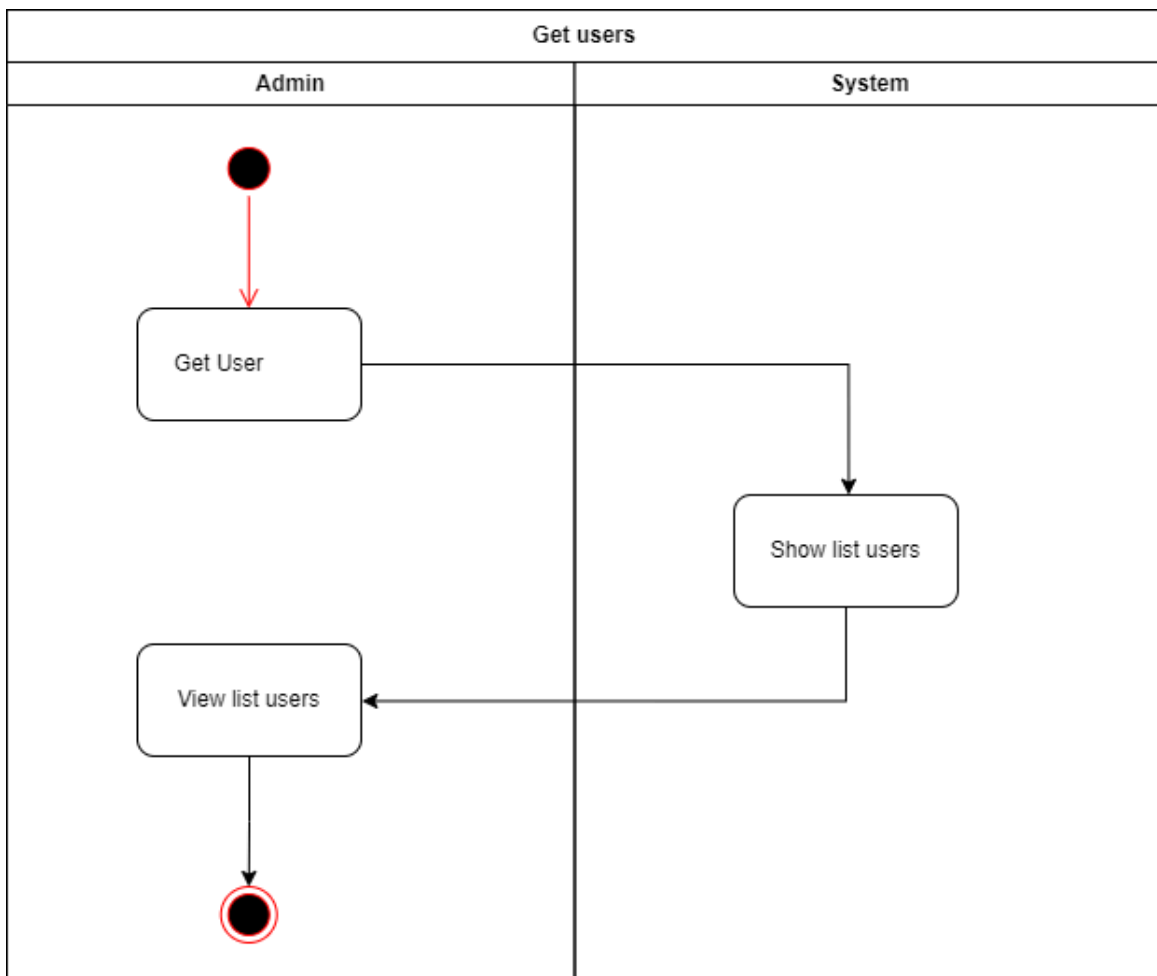


Figure 17: Activity diagram of get users

0.4.6 Get Tags

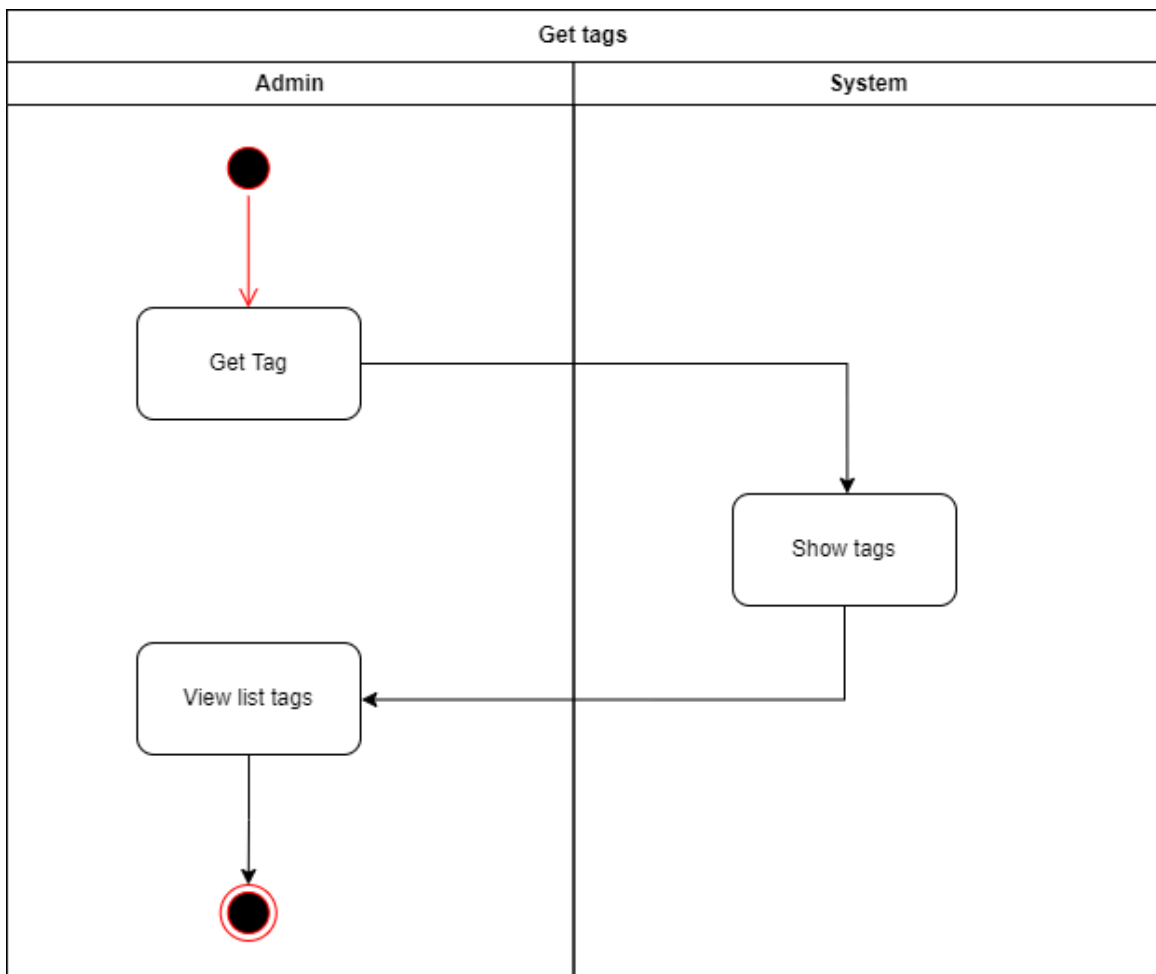


Figure 18: Activity diagram of get tags

0.5 Sequence diagrams

0.5.1 Login

Application of a job suggestion system and integration of a chatbot based on RAG

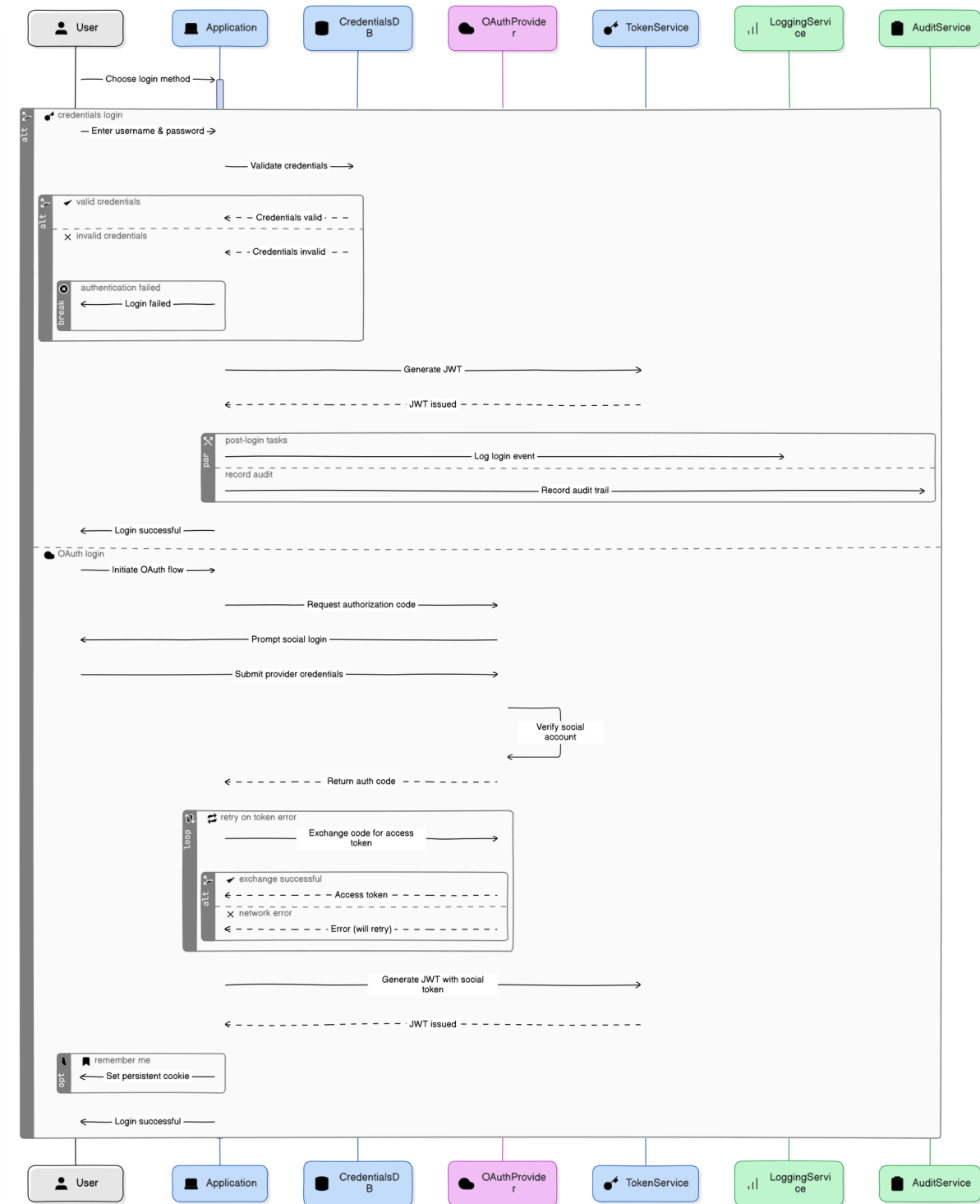


Figure 19: Sequence diagram of the Login Feature

0.5.2 Registration

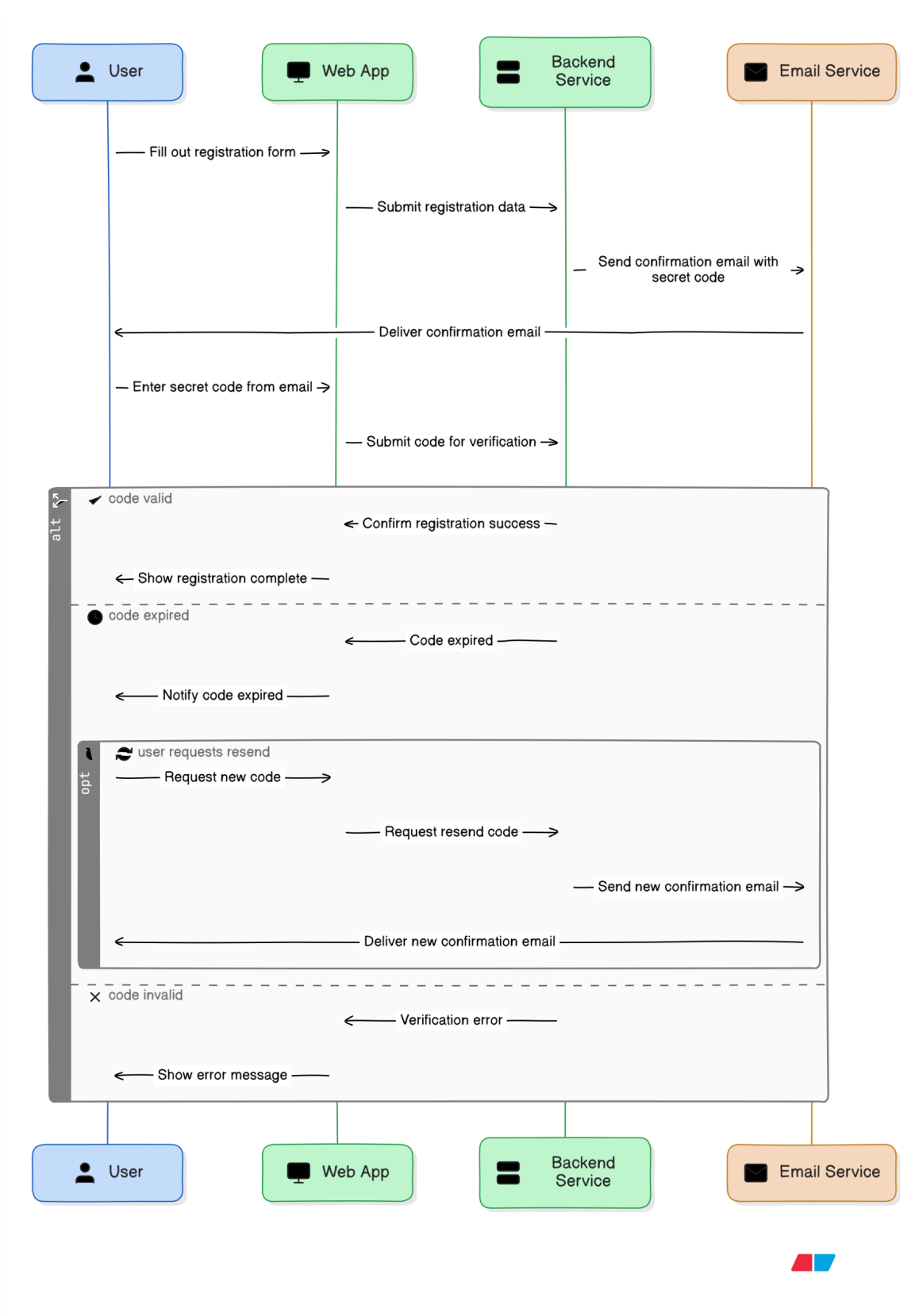


Figure 20: Sequence diagram of the registration Feature

0.5.3 Job posting

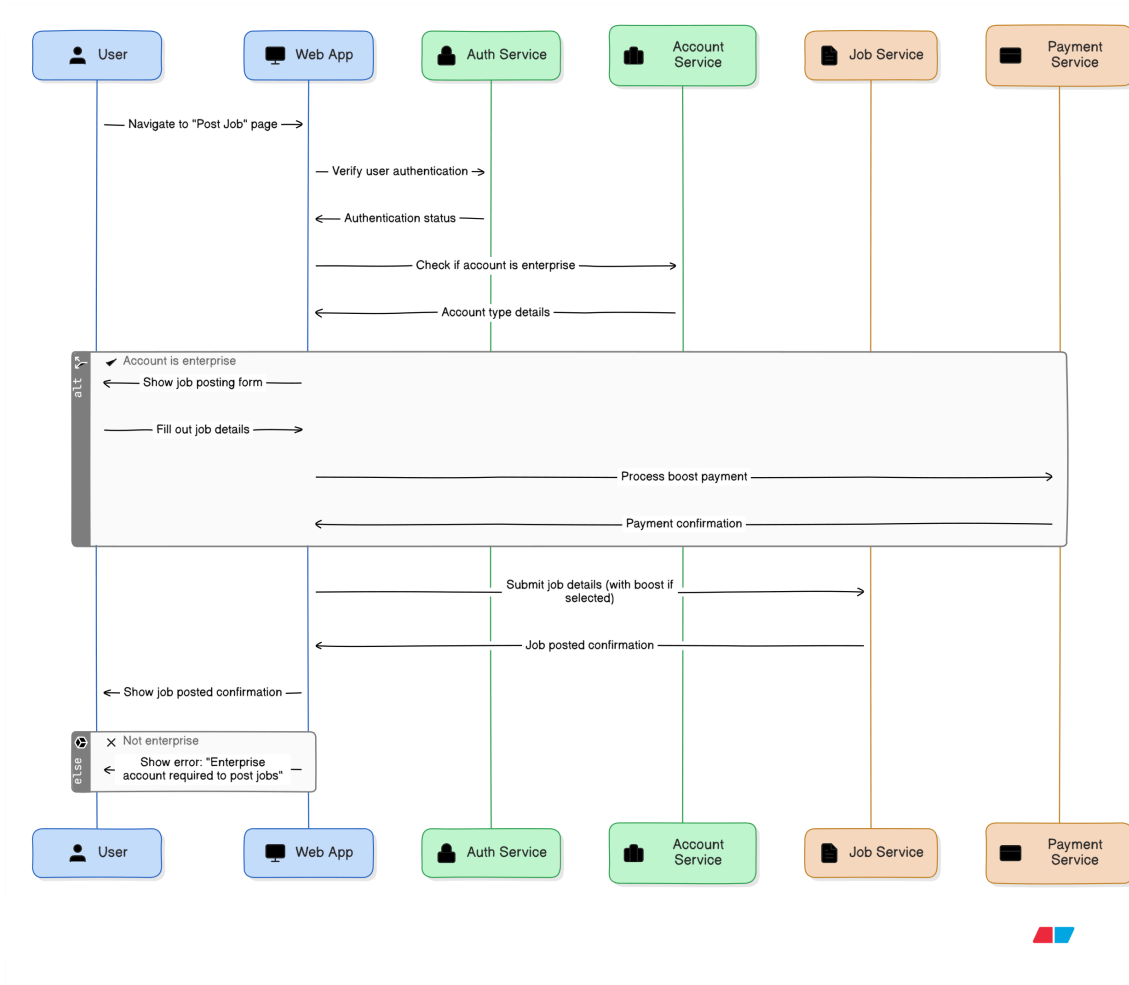


Figure 21: Sequence diagram of the Job Posting Feature

0.5.4 Apply job

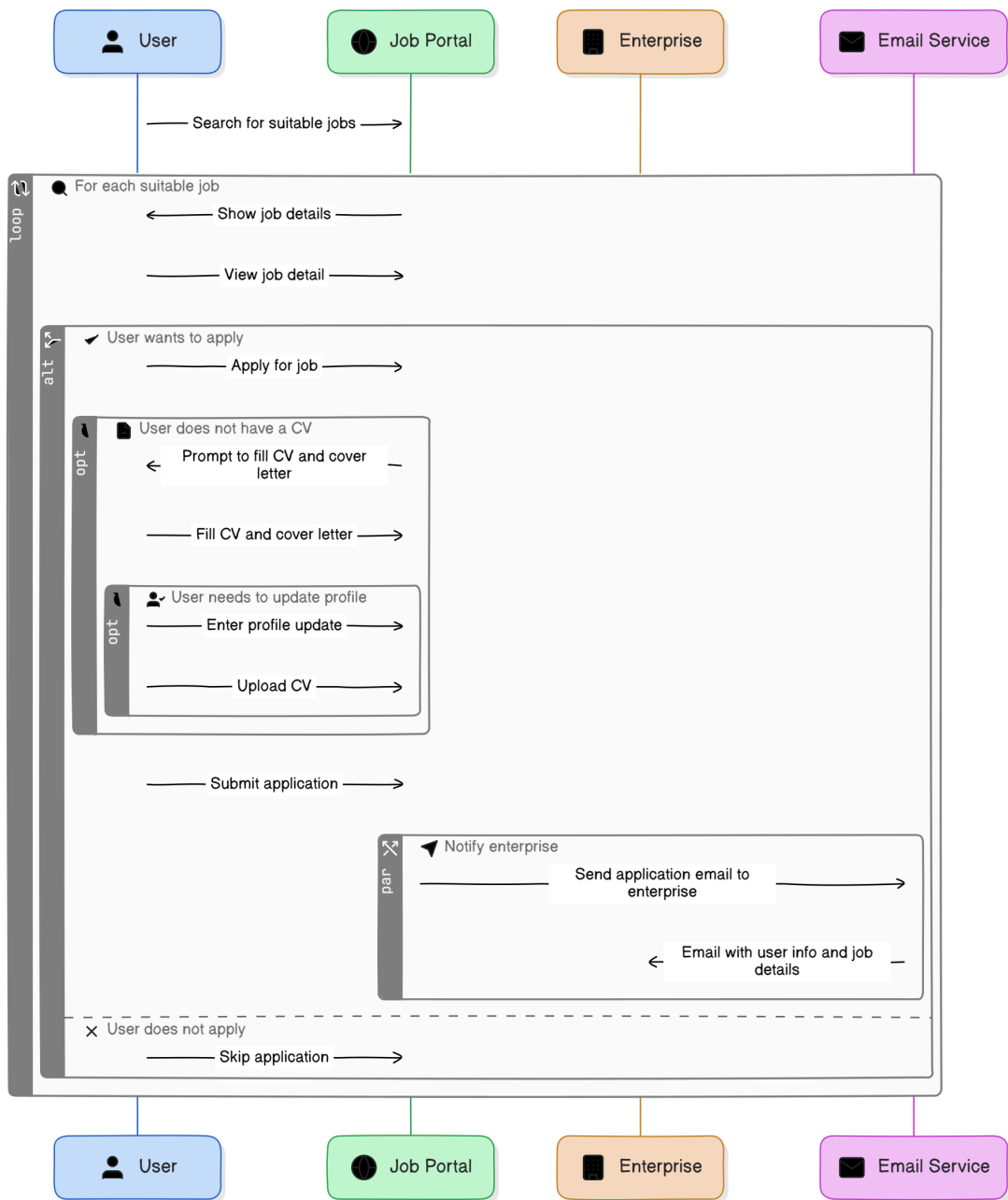


Figure 22: Sequence diagram for the Job Application Feature

0.5.5 Manage applicants

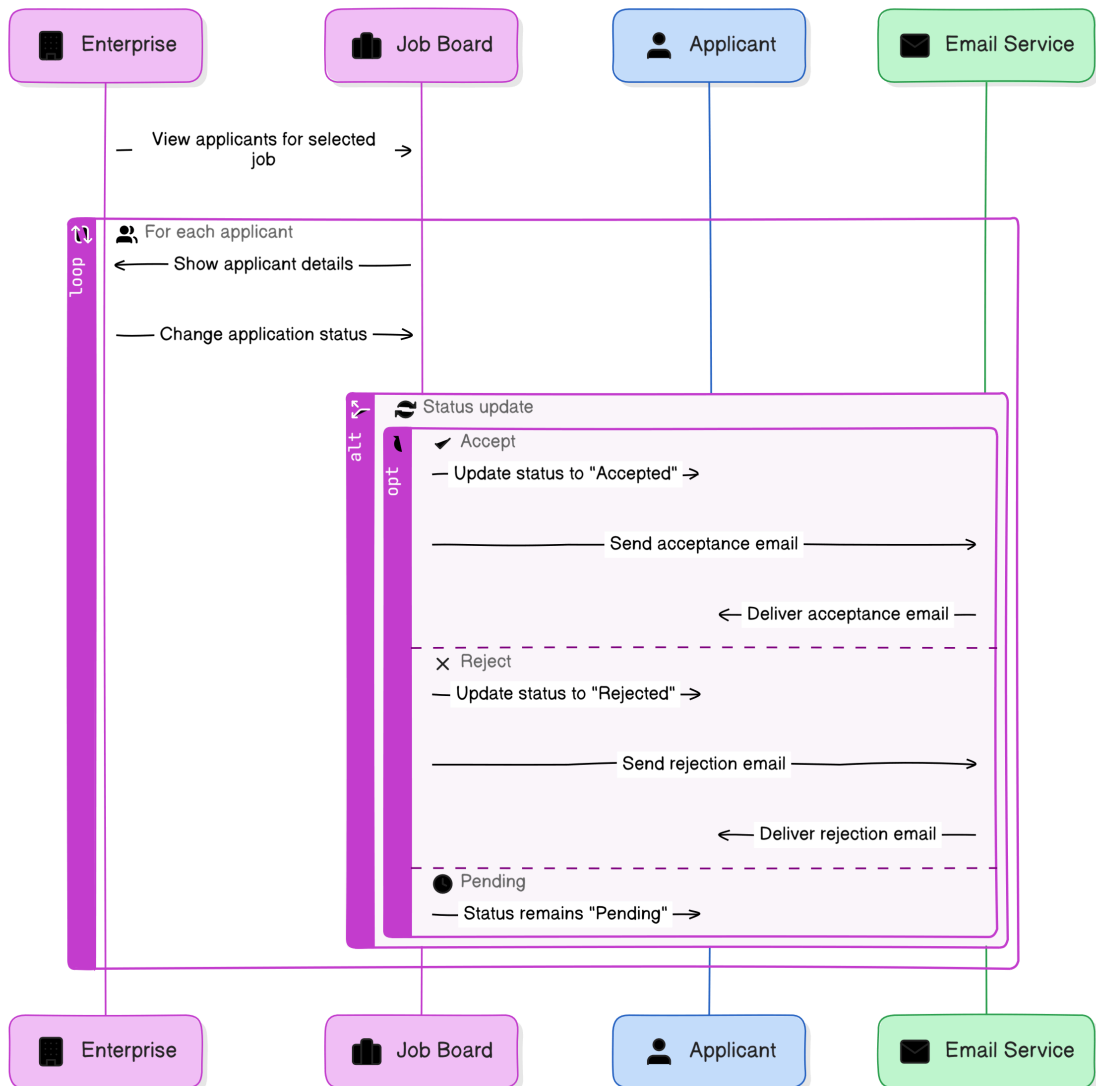


Figure 23: Sequence diagram for Managing Applicants Feature

0.5.6 Enterprise registration

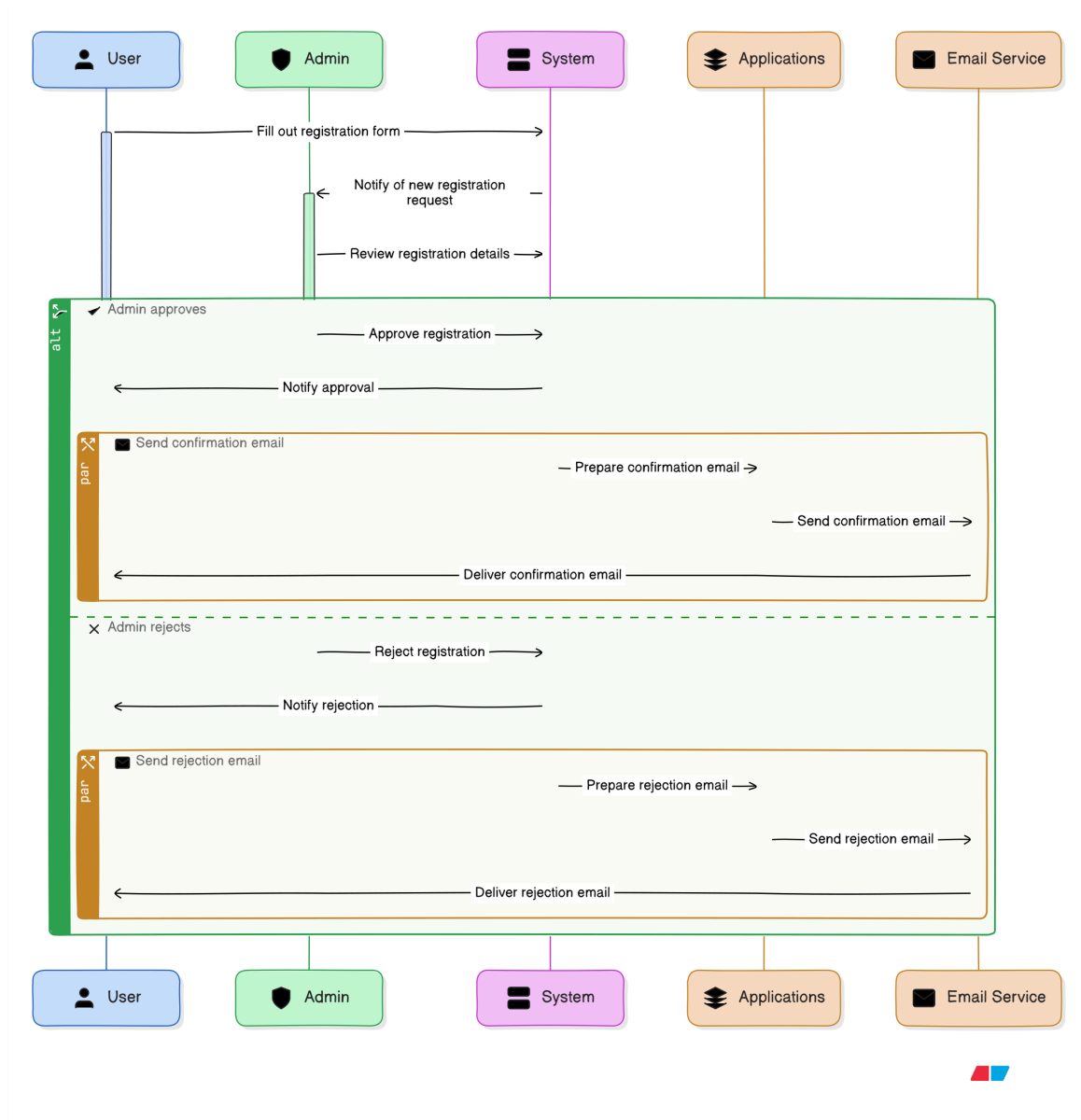


Figure 24: Sequence diagram for the Enterprise Registration Feature

0.6 Entity Diagram

0.6.1 Table Overview

Table 3: Details of Table Overview

No.	Table name	Short Description
1	Accounts	Stores account information for all users, including login credentials and roles
2	Profiles	Contains personal profile details of users such as name, birthday, gender, etc...
3	Addresses	General address table used across the system
4	Enterprises	Holds information about enterprise users
5	Jobs	Contains job postings created by enterprises
6	Tags	Master list of tags used across the system
7	Boosted Job	Contains data about job posts that have been promoted or boosted
8	Categories	Stores all general system-wide categories (if not only for jobs)
9	CVs	Stores uploaded CV documents and metadata linked to user profiles
10	Notifications	Stores notifications for users, including content and status (read/unread)
11	Transactions	Holds payment or system transaction records, including job boosts, upgrades, etc...
12	Website	Holds information about website companies

0.6.2 Entity Relationship Diagram

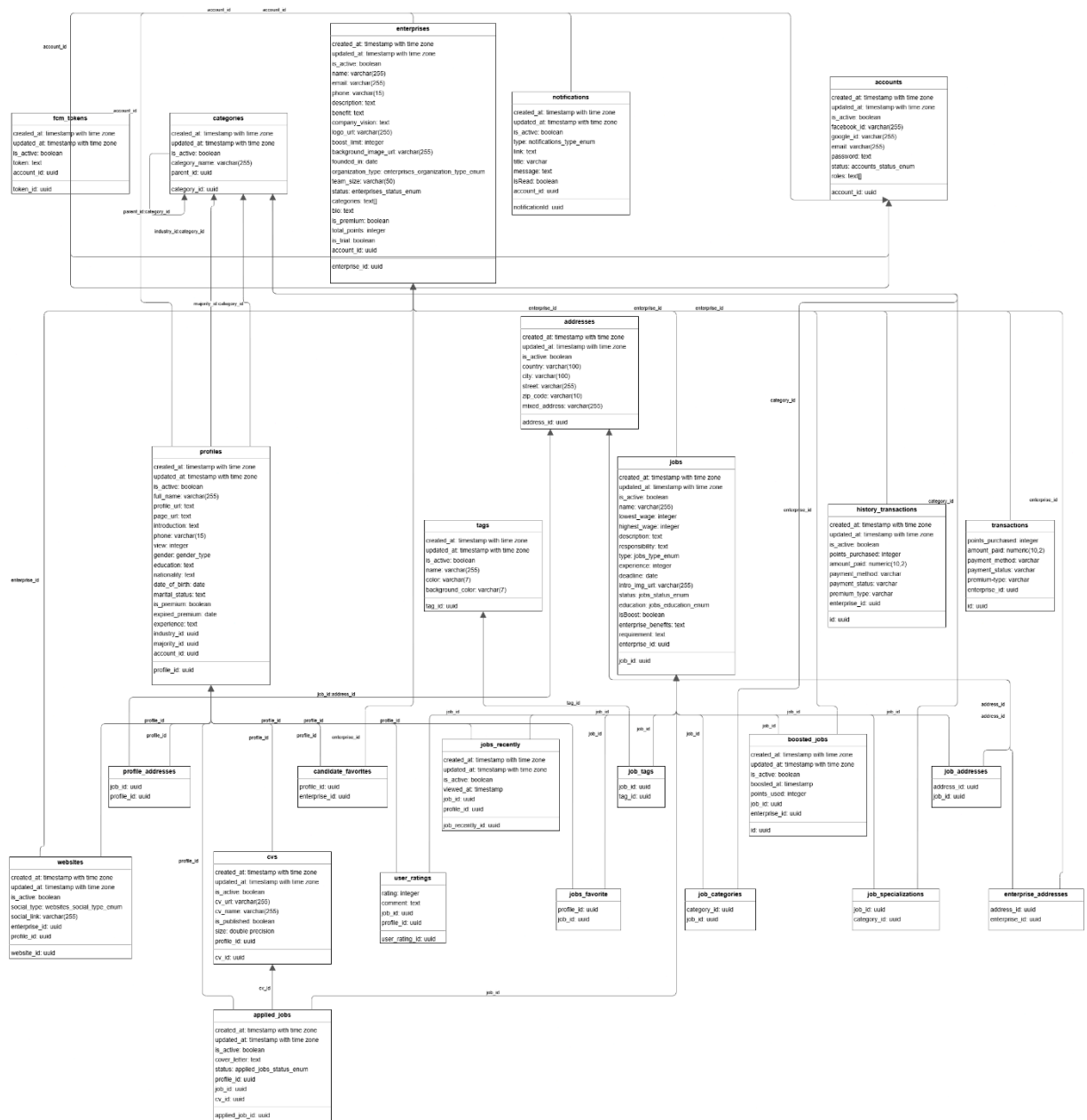


Figure 25: Entity diagram

0.6.3 Entity details

❖ Accounts

Table 4: Accounts entity detail

Attributes	Datatype	Null	Key	Default
accountId	UUID	NOT NULL	PK	
facebook_id	VARCHAR(50)			
google_id	VARCHAR(255)			
email	VARCHAR(50)			
password	VARCHAR(50)			
status	ENUM			PENDING
roles	TEXT			
profile	UUID		FK	
enterprise	UUID		FK	
notification	UUID		FK	
fcmToken	UUID		FK	

❖ Profile

Table 5: Profile entity detail

Attributes	Datatype	Null	Key	Default
profileId	UUID	NOT NULL	PK	
fullName	VARCHAR(255)	NOT NULL		
profileUrl	TEXT	NULL		
pageUrl	TEXT	NOT NULL		
introduction	TEXT	NULL		
phone	VARCHAR(15)	NULL		
view	INT	NOT NULL		
gender	ENUM	NULL		
education	TEXT	NULL		
nationality	TEXT	NULL		
dateOfBirth	DATE	NULL		

Application of a job suggestion system and integration of a chatbot based on RAG

Attributes	Datatype	Null	Key	Default
maritalStatus	TEXT	NULL		
isPremium	BOOLEAN	NOT NULL		
expiredPremium	DATE	NOT NULL		
experience	TEXT	NULL		
industry_id	UUID		FK	
majority_id	UUID		FK	
account_id	UUID		FK	
websites	UUID		FK	
jobs_favorite	UUID		FK	
cvs	UUID		FK	
appliedJob	UUID		FK	
addresses	UUID		FK	
candidate_favorite	UUID		FK	

❖ **Addresses**

Table 6: Address entity detail

Attributes	Datatype	Null	Key	Default
addressId	UUID	NOT NULL	PK	
country	VARCHAR(100)	NOT NULL		
city	VARCHAR(100)	NOT NULL		
street	VARCHAR(255)	NOT NULL		
zip_code	VARCHAR(10)	NOT NULL		
mixed_address	VARCHAR(255)	NOT NULL		
job_addresses	UUID		FK	
enterprise_addresses	UUID		FK	
profile_addresses	UUID		FK	

❖ **Enterprise**

Table 7: Enterprise entity detail

Attributes	Datatype	Null	Key	Default
enterprises_id	UUID	NOT NULL	PK	
name	VARCHAR(255)			
email	VARCHAR(15)			
phone	DATE	NOT NULL		
description	TEXT			
benefit	TEXT			
company_vision	TEXT	NULL		
logo_url	VARCHAR(255)			
boost_limit	INT			0
background_image_url	VARCHAR(255)	NULL		
founded_in	DATE	NULL		
organization_type	ENUM	NULL		
team_size	VARCHAR(50)	NULL		
status	ENUM	NOT NULL		PENDING
categories	TEXT	NOT NULL		
bio	TEXT	NULL		
is_premium	BOOLEAN	NOT NULL		FALSE
total_points	INT	NOT NULL		0
is_trial	BOOLEAN	NULL		
account_id	UUID		FK	
website	UUID		FK	
jobs	UUID		FK	
addresses	UUID		FK	
profiles	UUID		FK	
boostedJobs	UUID		FK	
transactions	UUID		FK	

❖ **Jobs**

Table 8: Jobs entity detail

Attributes	Datatype	Null	Key	Default
jobId	UUID	NOT NULL	PK	
name	VARCHAR(255)	NOT NULL		
lowest_wage	INT	NULL		
highest_wage	INT	NULL		
description	TEXT	NULL		
responsibility	TEXT	NULL		
type	ENUM	NOT NULL		CONTRACT
experience	INT			0
deadline	Datetime	NULL		
into_img_url	VARCHAR(255)			
status	ENUM			OPEN
education	ENUM	NULL		
isBoost	BOOLEAN			
enterprise_benefits	TEXT	NULL		
requirement	TEXT	NULL		
enterprise_id	UUID		FK	
job_tags	UUID		FK	
profiles	UUID		FK	
appliedJob	UUID		FK	
categories	UUID		FK	
addresses	UUID		FK	
job_specializations	UUID		FK	
boostedJob	UUID		FK	

❖ **Tags**

Table 9: Tag entity details

Attributes	Datatype	Null	Key	Default
tag_id	UUID		PK	
name	VARCHAR(255)			
color	VARCHAR(7)	NULL		
background_color	VARCHAR(7)	NULL		
jobs	UUID		FK	

❖ **Boosted Job**

Table 10: Boosted Job entity detail

Attributes	Datatype	Null	Key	Default
id	UUID		PK	
boosted_at	DATE			
points_used	INT	NOT NULL		
job	UUID		FK	
enterprise	UUID	NOT NULL	FK	

❖ **Categories**

Table 11: Categories entity detail

Attributes	Datatype	Null	Key	Default
category_id	UUID		PK	
category_name	VARCHAR(255)			
parent_id	UUID		FK	
children	UUID		FK	
jobs	UUID		FK	

❖ **CVs**

Table 12: CVs entity detail

Attributes	Datatype	Null	Key	Default
cv_id	UUID		PK	
cv_url	VARCHAR(255)			
cv_name	VARCHAR(255)			
is_published	BOOLEAN			
size	FLOAT			0
profile_id	UUID		FK	
appliedJob	UUID		FK	

❖ **Notifications**

Table 13: Notifications entity detail

Attributes	Datatype	Null	Key	Default
notificationId	UUID		PK	
type	ENUM			
link	TEXT	NULL		
title	VARCHAR(255)	NOT NULL		
message	TEXT	NOT NULL		
isRead	BOOLEAN	NOT NULL		
account_id	UUID		FK	

❖ **Transaction**

Table 14: Transaction entity detail

Attributes	Datatype	Null	Key	Default
id	UUID		PK	
points_purchased	INT			
amount_paid	DECIMAL			
payment_method	VARCHAR(255)			
payment_status	ENUM			PENDING
premium_type	ENUM	NULL		
enterprise_id	UUID		FK	

❖ Website

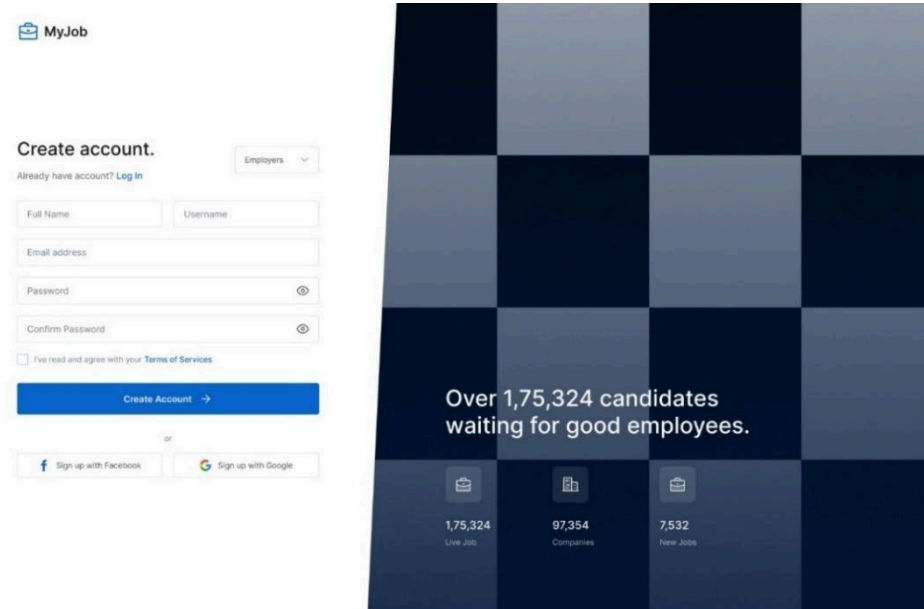
Table 15: Website entity detail

Attributes	Datatype	Null	Key	Default
website_id	UUID		PK	
social_type	ENUM	NULL		
social_link	VARCHAR(255)	NULL		
enterprise_id	UUID		FK	
profile_id	UUID		FK	

Chapter 3 IMPLEMENTATION AND EVALUATION

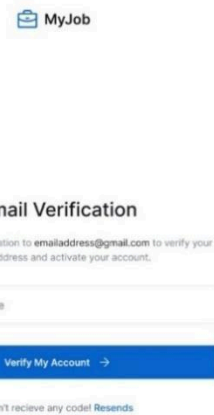
0.7 Implementation

0.7.1 UI for Authentication



The image shows the registration UI for 'MyJob'. It features a 'Create account.' section with a dropdown for 'Employers', a 'Log In' link for existing users, and input fields for 'Full Name', 'Username', 'Email address', 'Password', and 'Confirm Password'. There is a checkbox for 'I've read and agree with your Terms of Services' and a blue 'Create Account' button. Below this, there are social media login options for Facebook and Google. To the right, a large blue banner displays the text 'Over 1,75,324 candidates waiting for good employees.' and three statistics: '1,75,324 Live Jobs', '97,354 Companies', and '7,532 New Jobs'.

Figure 26: UI for registration



The image shows the email verification UI for 'MyJob'. It features a title 'Email Verification' and a message: 'We've sent an verification to emailaddress@gmail.com to verify your email address and activate your account.' Below this is a 'Verification Code' input field and a blue 'Verify My Account' button. At the bottom, there is a link for 'Didn't receive any code? Resends'.

Figure 27: UI for email verification

Application of a job suggestion system and integration of a chatbot based on RAG

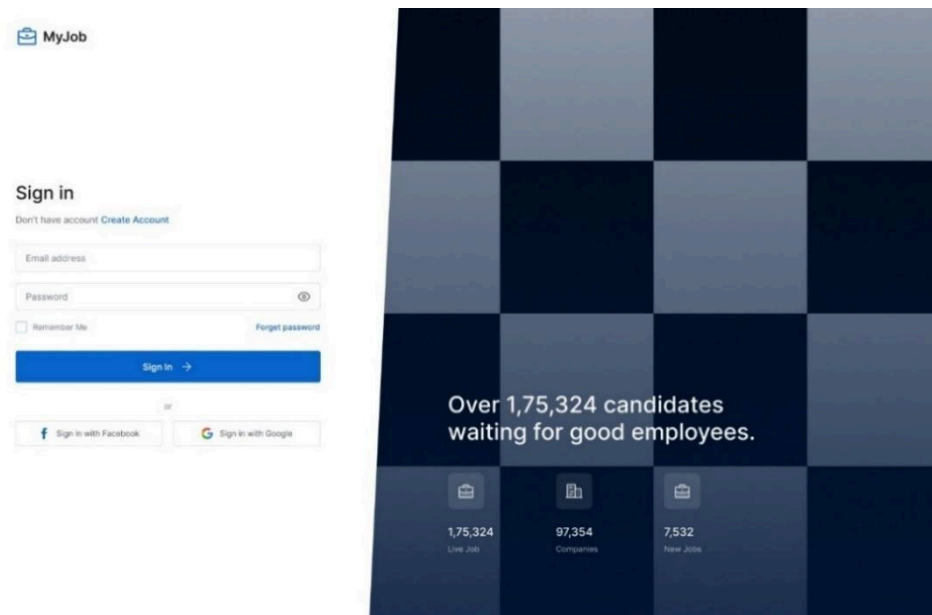


Figure 28: UI for login feature

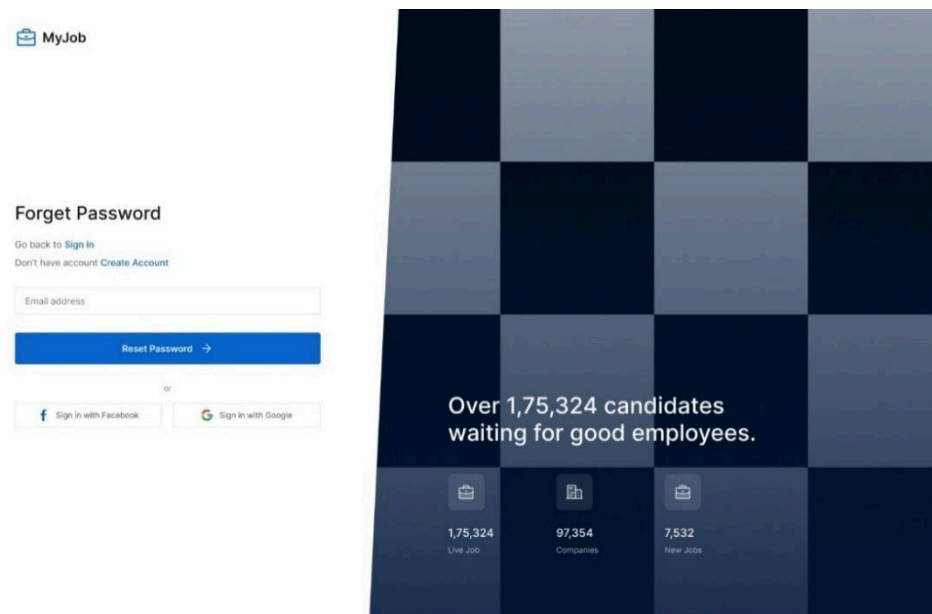
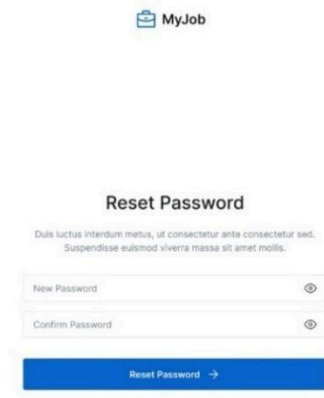


Figure 29: UI for forgot password feature

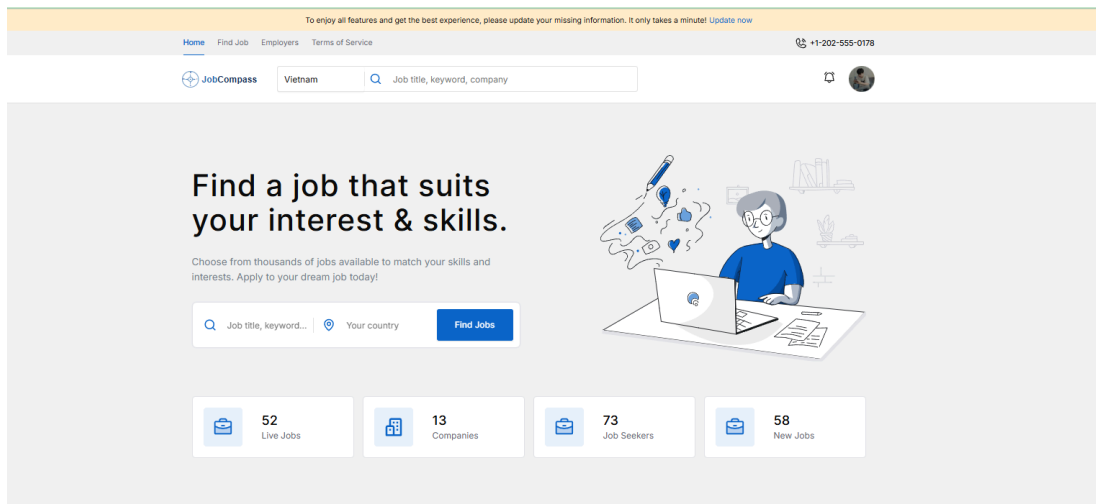
Application of a job suggestion system and integration of a chatbot based on RAG



The image shows a 'Reset Password' form. At the top, there is a 'MyJob' logo. Below it, the title 'Reset Password' is centered. A placeholder text reads: 'Duis luctus interdum metus, ut consectetur ante consectetur sed. Suspendisse euismod viverra massa sit amet mollis.' The form contains two input fields: 'New Password' and 'Confirm Password', each with a toggle icon for password visibility. At the bottom, there is a blue button labeled 'Reset Password' with a right-pointing arrow.

Figure 30: UI for reset password feature

0.7.2 UI for user role



The image shows the home page of a job portal. At the top, there is a navigation bar with links: 'Home', 'Find Job', 'Employers', and 'Terms of Service'. A notification banner says: 'To enjoy all features and get the best experience, please update your missing information. It only takes a minute! Update now'. A phone icon and the number '+1-202-555-0178' are also present. Below the navigation bar, there is a search bar with the text 'JobCompass' and a dropdown menu showing 'Vietnam'. The main content area features a large heading: 'Find a job that suits your interest & skills.' Below this, a subheading reads: 'Choose from thousands of jobs available to match your skills and interests. Apply to your dream job today!'. A search bar with a magnifying glass icon and a dropdown menu for 'Your country' is provided, along with a blue 'Find Jobs' button. To the right of the search bar, there is an illustration of a person sitting at a desk with a laptop, surrounded by various icons representing different job categories. Below the search bar, there are four statistics cards: '52 Live Jobs', '13 Companies', '73 Job Seekers', and '58 New Jobs'.

Most Popular Vacancies

Figure 31: UI home page

Application of a job suggestion system and integration of a chatbot based on RAG

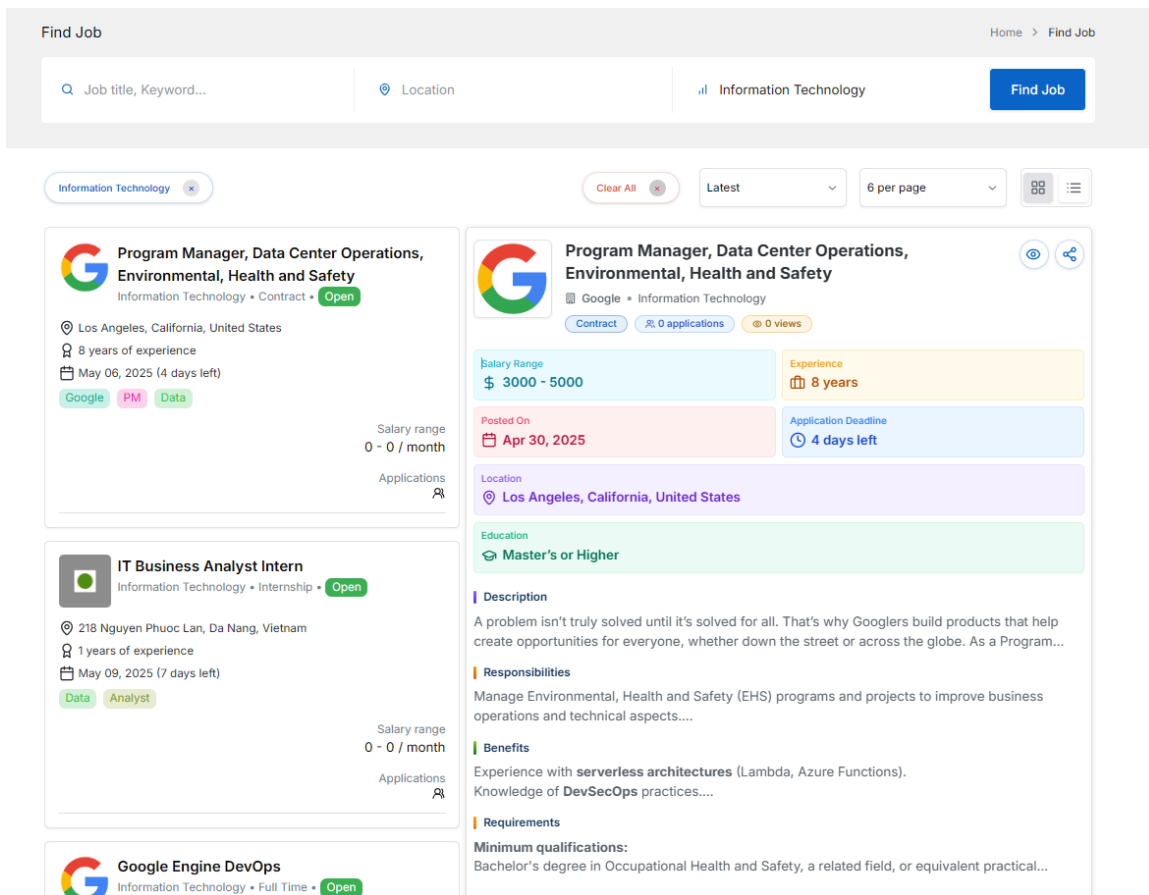


Figure 32: UI for find job feature

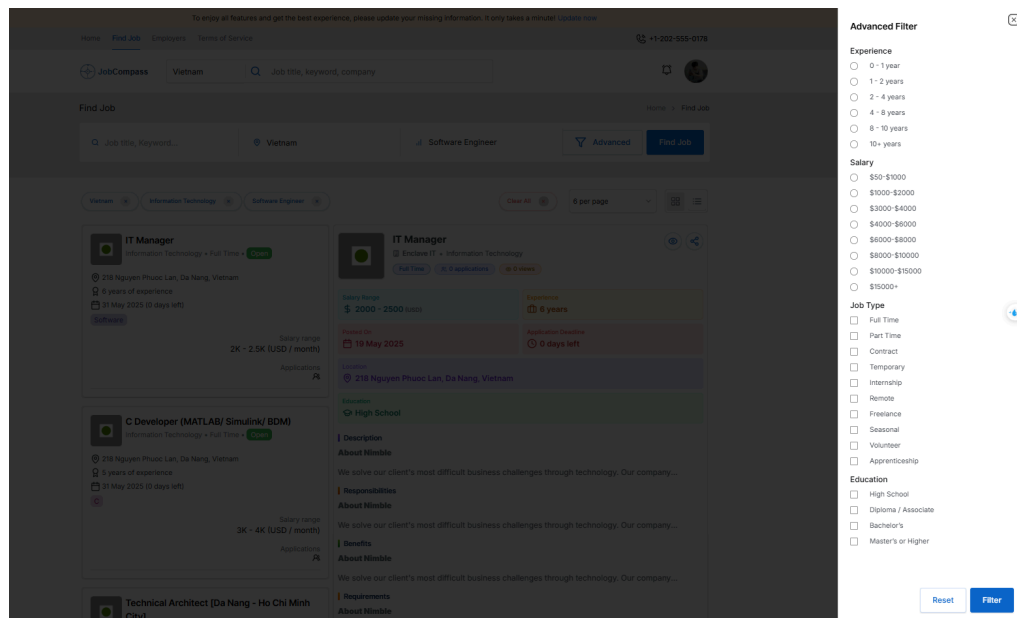


Figure 33: UI for filter job feature

Application of a job suggestion system and integration of a chatbot based on RAG

Find Job

Home > Find Job

Job title, Keyword...

Location

Information Technology

Find Job

Information Technology

Clear All

Latest

6 per page



Program Manager, Data Center Operations, Environmental, Health and Safety

Contract

Google

California, United States

\$ - \$

May 06, 2025

Apply Now



IT Business Analyst Intern

Internship

Enclave IT

Da Nang, Vietnam

\$ - \$

May 09, 2025

Apply Now



Google Engine DevOps

Full Time

Google

California, United States

\$ - \$

May 10, 2025

Apply Now



Senior Software Engineer, AI/ML GenAI, Google Cloud AI

Contract

Google

California, United States

\$ - \$

May 11, 2025

Apply Now

Figure 34: UI for list view job

Performed Student: Dang Quang Nhat Linh

Instructor: Ph.D. Vo Duc Hoang

48

Application of a job suggestion system and integration of a chatbot based on RAG



Program Manager, Data Center Operations, Environmental, Health and Safety

+84123456788 enterprise.account.1@gmail.com

PM

Data

Google



Apply Now >

Job Description

A problem isn't truly solved until it's solved for all. That's why Googlers build products that help create opportunities for everyone, whether down the street or across the globe. As a Program Manager at Google, you'll lead complex, multi-disciplinary projects from start to finish — working with stakeholders to plan requirements, manage project schedules, identify risks, and communicate clearly with cross-functional partners across the company. Your projects will often span offices, time zones, and hemispheres. It's your job to coordinate the players and keep them up to date on progress and deadlines.

Our goal is to build a Google that looks like the world around us — and we want Googlers to stay and grow when they join us. As part of our efforts to build a Google for everyone, we build diversity, equity, and inclusion into our work and we aim to cultivate a sense of belonging throughout the company.

The Data Center team designs and operates some of the most sophisticated electrical and HVAC systems in the world. We are a diverse, upbeat, creative, team-oriented group of engineers committed to building and operating powerful data centers.

Responsibilities

Manage Environmental, Health and Safety (EHS) programs and projects to improve business operations and technical aspects.

Identify, organize, and plan program activities and stakeholders in order to deliver business, operational, and technical improvements.

Ensure compliance with Authority Having Jurisdiction (AHJ) regulations, maintain permits, and monitor program Key Performance Indicators (KPI) performance.

Execute and manage site EHS compliance calendar tasks/requirements. Evaluate incident data and track corrective/preventive actions to closure.

Organize and execute meetings, reports, and site plans, identify and assess operational EHS risks.

Benefits

Experience with **serverless architectures** (Lambda, Azure Functions).

Knowledge of **DevSecOps** practices.

Certifications (AWS Certified DevOps Engineer, CKA, etc.).

Requirements

Minimum qualifications:

- Bachelor's degree in Occupational Health and Safety, a related field, or equivalent practical experience.
- 5 years of experience in program or project management.
- 5 years of experience in Environmental, Health and Safety (EHS) management, including management of EHS compliance for data center or other industrial facilities.
- Ability to communicate in English and Malay fluently in order to interact with contractors and stakeholders.

Preferred qualifications:

- Master's degree in Occupational Health and Safety, a related field, or equivalent practical experience.
- Certification from Board of Certified Safety Professionals, National Examination Board in Occupational Safety and Health (NEBOSH) or equivalent agency.
- Experience communicating with Federal, State and Local Authorities regarding EHS issues.
- Ability to advise, coach, influence, and collaborate with employees to achieve positive EHS outcomes.

Job Overview

 JOB POSTED 2025-04-30	 JOB EXPIRE IN 2025-05-06	 EDUCATION Master's or Higher
 SALARY \$3000 - 5000	 LOCATION United States - California	 JOB TYPE Contract



Google

Engineering Tomorrow: Innovate, Sustain, Succeed

Founded in:	2007-01-31
Organization type:	PUBLIC
Company size:	1-10
Phone:	+84123456788
Email:	enterprise.account.1@gmail.com
Industry:	



Share profile:  Facebook  X (Twitter)  LinkedIn

Figure 35: UI for job detail

Application of a job suggestion system and integration of a chatbot based on RAG

features and get the best experience, please update your missing information. It only takes a minute! [Update now](#)

×

Apply Job: Program Manager, Data Center Operations, Environmental, Health and Safety

Choose Resume

Select... ▾

B I U S | 🔗 | ☰ ☷ Aa

Cancel

Apply Now >

Figure 36: UI for apply job feature

To enjoy all features and get the best experience, please update your missing information. It only takes a minute! [Update now](#)

Home Find Job **Employers** Terms of Service

+1-202-555-0178

📍 Vietnam 🔍 Job title, keyword, company

Filters

Organization Type

☐ Private

☐ Public

☐ Flat

☐ Outsource

Location

☐ cn China

☐ tw India

☐ jp Japan

☐ kr South Korea

☐ ph Philippines

Show More

Enterprise Listings

Showing 6 results

Search: Search by name Show: 6 per page ▾



Skype Company

📍 123 Tech Lane - California - United States 📅 15 May 2003 📞 +15551234567

Skype Company is a leading software development firm founded in 2003, specializing in communication technologies.

View Detail →



Apple Inc.

📍 Los Altos - California - United States 📅 01 Apr 1976 📞 +123456781

<p>Founded by Steve Jobs, Steve Wozniak, and Ronald Wayne, Apple has grown from a garage startup to a global tech giant, revolutionizing personal computing, mobile devices, and digital services.</p>

View Detail →



Microsoft Corporation

📍 Round Rock - Texas - United States 📅 04 Apr 1975 📞 +123456782

<p>Founded by Bill Gates and Paul Allen, Microsoft is a leader in software and cloud computing, shaping modern technology with a focus on innovation and accessibility.</p>

View Detail →

Figure 37: UI enterprise list

Performed Student: Dang Quang Nhat Linh

Instructor: Ph.D. Vo Duc Hoang

50

Application of a job suggestion system and integration of a chatbot based on RAG

To enjoy all features and get the best experience, please update your missing information. It only takes a minute! [Update now](#)

[Home](#) [Find Job](#) [Employers](#) [Terms of Service](#)

JobCompass

Vietnam

Job title, keyword, company

+1-202-555-0178

Google

California - United States

Send Mail

General information

Company type

Information Technology

Country

United States

Company industry

PUBLIC

Company vision

Engineering Tomorrow:
Innovate, Sustain, Succeed

Company size

1-10

Phone

+84123456788

Company overview

Welcome to Enclave

Enclave, a company of and by software engineering professionals. We have been providing outstanding quality for software engineering and software testing services since 2007. Basing on demanding features collecting from many big names in IT and ITO industries, we – by ourselves – have created innovative working environment and effective solutions that are now available to all-sized companies.

With a pool of experienced IT engineers in developing, designing, and SQA engineering, we reliably offer clients the value-added solutions, professional skills, accountability, and industrial domain knowledge to reduce operating costs, eliminate risks, deliver solutions on time and on budget as well as ensure the right decision has been made as our best solutions and services.

Our expertise in developing applications and supporting professional testing services provided us the knowledge to develop solutions that meet worldwide clients' expectations nowadays. These solutions do not only help increase productivity, reduce the operating costs, and increase customer satisfaction, but also create sustainable cooperation and development. We are not providing Business Process Outsourcing (BPO) and Initial Public Offering (IPO) either; this helps us centralize our strengths and resources in software outsourcing to satisfy any client needs.

Moreover, we recently have been recognized as one of the top software companies of 100 Outstanding Enterprises of Danang city in 2014 by the Danang People's Committee for our outstanding performance, active contribution to local community, and considerable contribution to municipal budget. We are also one of the best partners of Danang University in providing professional courses in software engineering and annual job opportunities for senior students.

Job openings

Digital Marketing Specialist

Contract • Open

Los Angeles, California, United States

2 years of experience

11 Jul 2025 (41 days left)

Salary range

3K - 5K (USD / month)

Applications

Marketing Specialist

Full Time • Open

Los Angeles, California, United States

2 years of experience

10 Jul 2025 (40 days left)

Salary range

5K - 8K (USD / month)

Figure 38: UI enterprise detail

To enjoy all features and get the best experience, please update your missing information. It only takes a minute! [Update now](#)

[Home](#) [Find Job](#) [Employers](#) [Terms of Service](#)

JobCompass

Vietnam

Job title, keyword, company

+1-202-555-0178

CANDIDATE DASHBOARD

Overview

Applied Jobs

Favorite Jobs

Settings

Logout

Hello, Nhật Linh Đặng Quang

Here is your daily activities and applications

1

Applied Jobs

1

Favorite Jobs

22

Job Alerts

Your profile

Show Profile Details

Edit Profile

Recently Applied

View all

JOB	DATE APPLIED	STATUS	ACTION
Technical Architect [Da Nang - Ho Chi Minh City]	APPROVED	28 May 2025	Active View Details

Figure 39: UI user profile

Performed Student: Dang Quang Nhat Linh

Instructor: Ph.D. Vo Duc Hoang

51

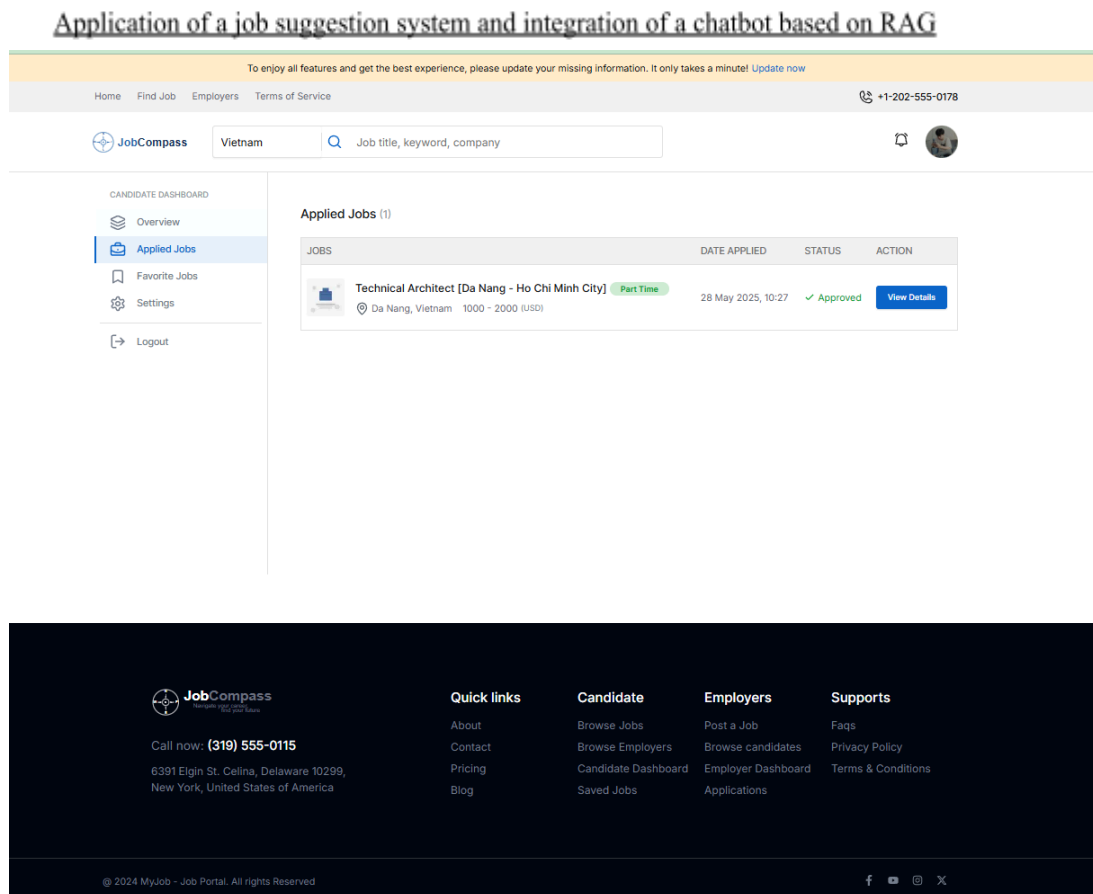


Figure 40: UI for view applied job

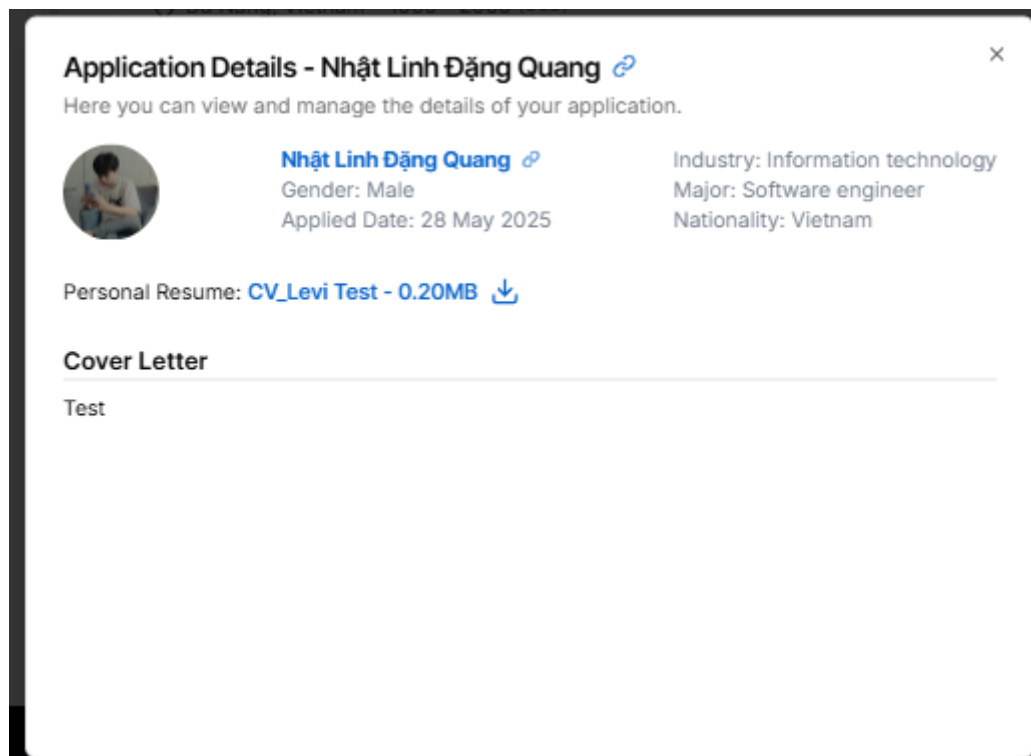


Figure 41: UI for view detail applied job

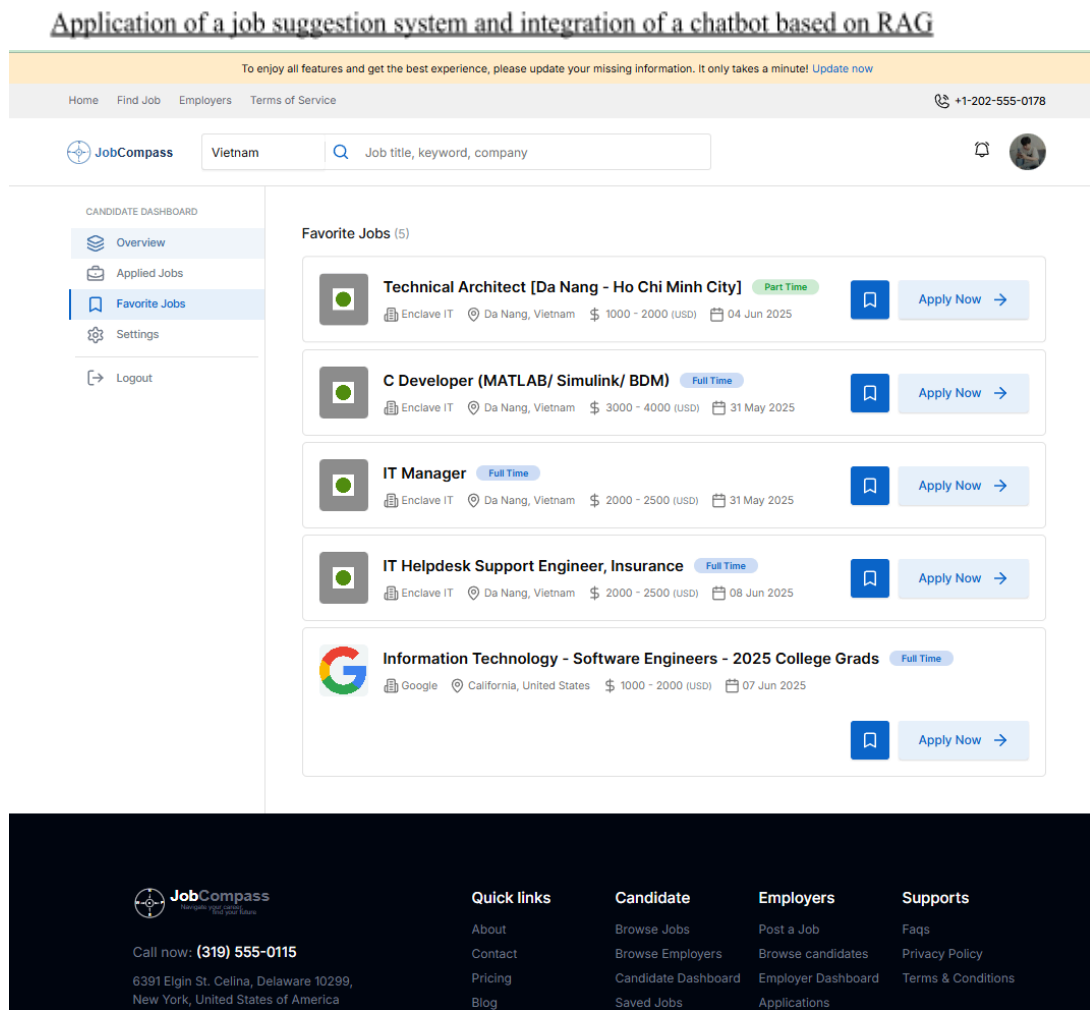


Figure 42: UI list favorite job view

Application of a job suggestion system and integration of a chatbot based on RAG

To enjoy all features and get the best experience, please update your missing information. It only takes a minute! [Update now](#)

Home Find Job Employers Terms of Service +1-202-555-0178

JobCompass Vietnam Job title, keyword, company

CANDIDATE DASHBOARD

- Overview
- Applied Jobs
- Favorite Jobs
- Settings**
- Logout

Settings

Personal Profile CV/Resume Social Links Account Settings

Basic Information

[Edit](#)

Profile Picture



0.06 MB Remove Replace

Background Picture



Full name

Phone

Date Of Birth

Marital Status

Education

- Bachelor's Degree In Information Technology**
University of Technology, 2020 – 2024
Relevant Coursework: Software Engineering, Computer Networks, Database Systems, Cloud Computing
- Google IT Support Professional Certificate**
Coursera – Google, 2023
Covered topics like networking, system administration, security, and troubleshooting.
- AWS Certified Cloud Practitioner**

Experience

- Backend Developer Intern – TechNova Solutions**
(2023 – 2024)
 - Developed and maintained RESTful APIs using NestJS and PostgreSQL.
 - Implemented Docker-based microservices for scalable deployments.
 - Collaborated in an Agile team to deliver new features on time.

Figure 43: UI for set up personal user

To enjoy all features and get the best experience, please update your missing information. It only takes a minute! [Update now](#)

Home Find Job Employers Terms of Service +1-202-555-0178

JobCompass Vietnam Job title, keyword, company

CANDIDATE DASHBOARD

- Overview
- Applied Jobs
- Favorite Jobs
- Settings**
- Logout

Settings

Personal **Profile** CV/Resume Social Links Account Settings

Candidate Information

[Edit](#)

Nationality

Gender

Industry

Majority

Introduction (Bio)

I'm a dedicated **Information Technology** professional with a deep passion for solving complex problems and delivering innovative digital solutions. With hands-on experience across a broad spectrum of technologies, I specialize in crafting scalable backend systems, designing cloud-native architectures, and optimizing performance in real-world applications.

Core Competencies

- Backend Development (NestJS, Node.js)
- Cloud Infrastructure (AWS, GCP, Docker, CI/CD)
- Database Management (PostgreSQL, Redis)
- RESTful APIs & Microservices Architecture



Call now: (319) 555-0115

6391 Elgin St. Celina, Delaware 10299,
New York, United States of America

Quick links

About
Contact
Pricing
Blog

Candidate

Browse Jobs
Browse Employers
Candidate Dashboard
Saved Jobs

Employers

Post a Job
Browse candidates
Employer Dashboard
Applications

Supports

Faqs
Privacy Policy
Terms & Conditions

Figure 44: UI for setting profile

Application of a job suggestion system and integration of a chatbot based on RAG

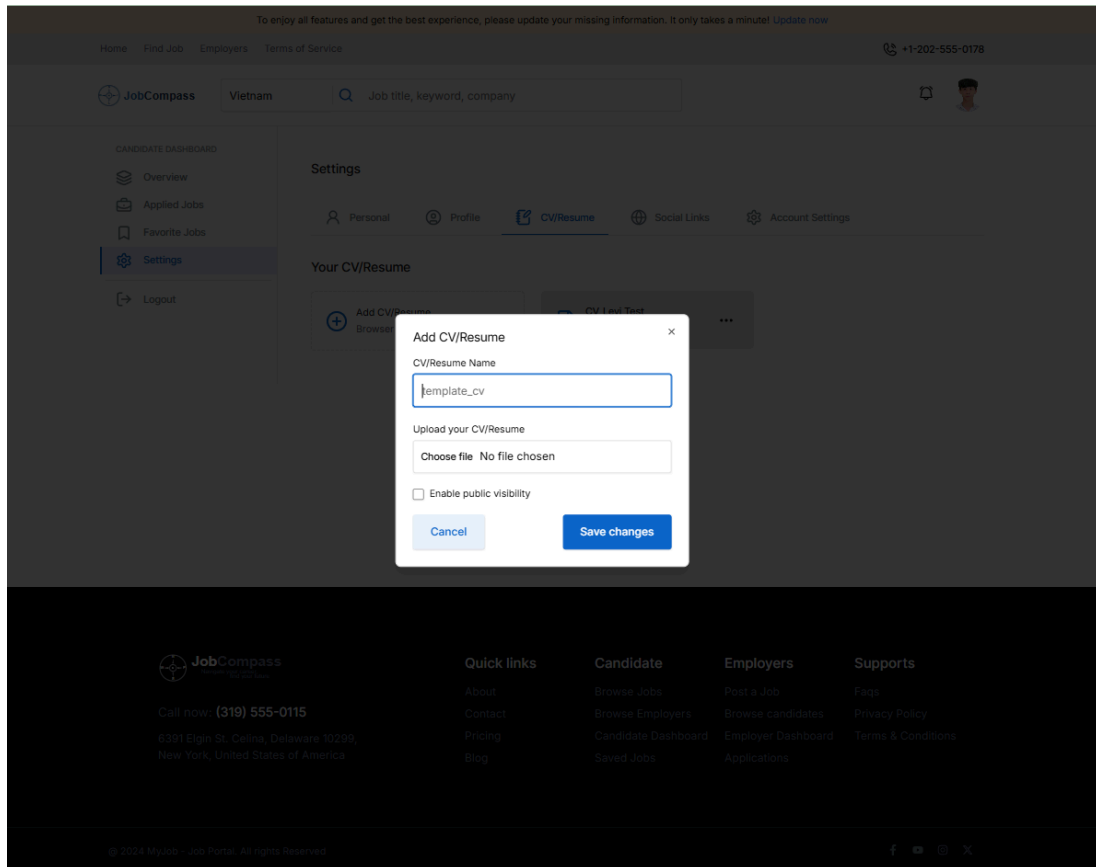


Figure 45: UI for posting CV

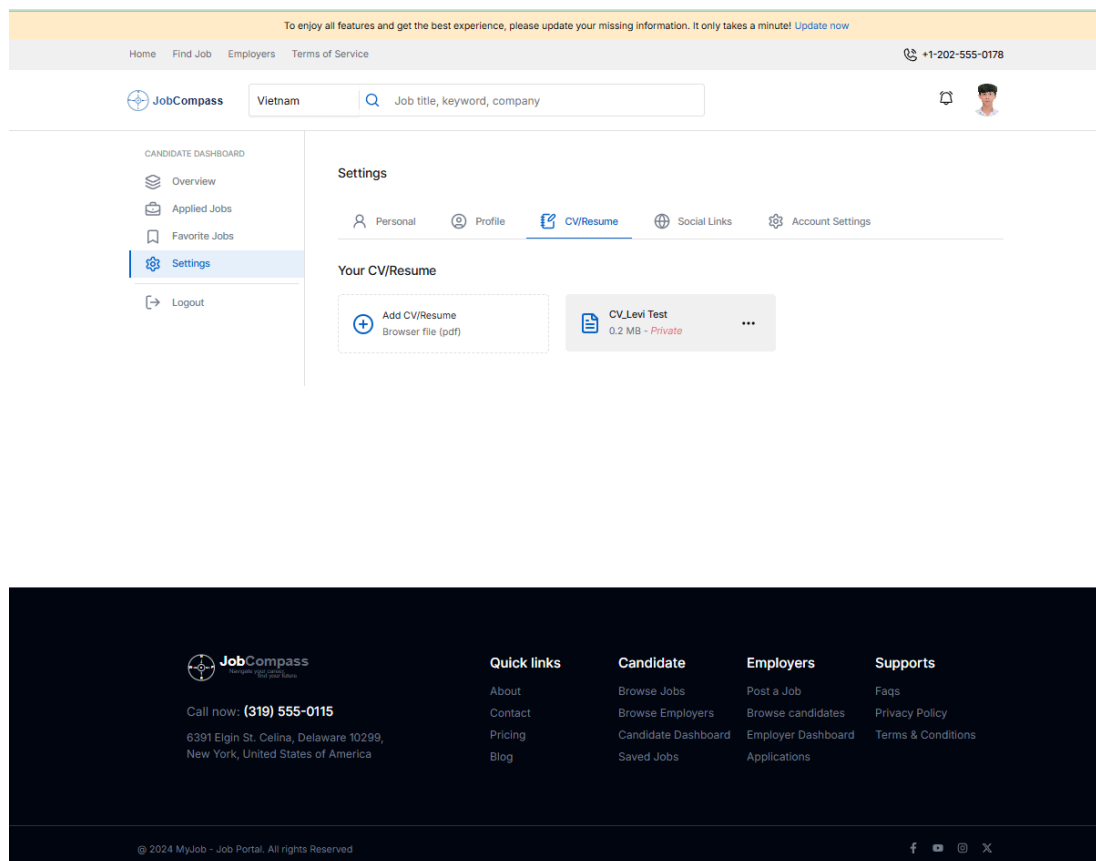


Figure 46: UI for managing CV

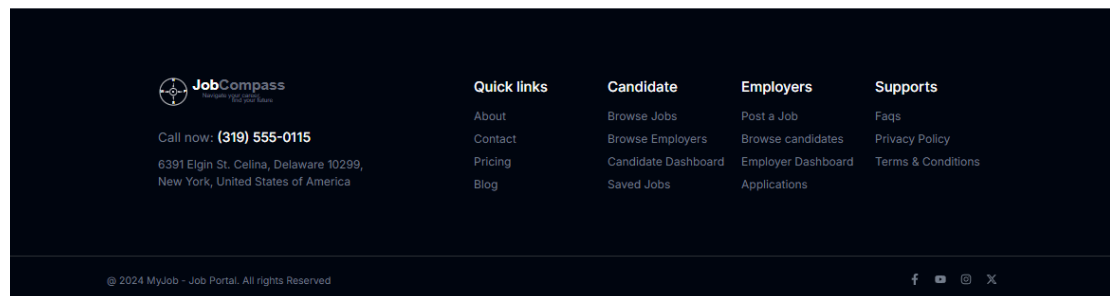
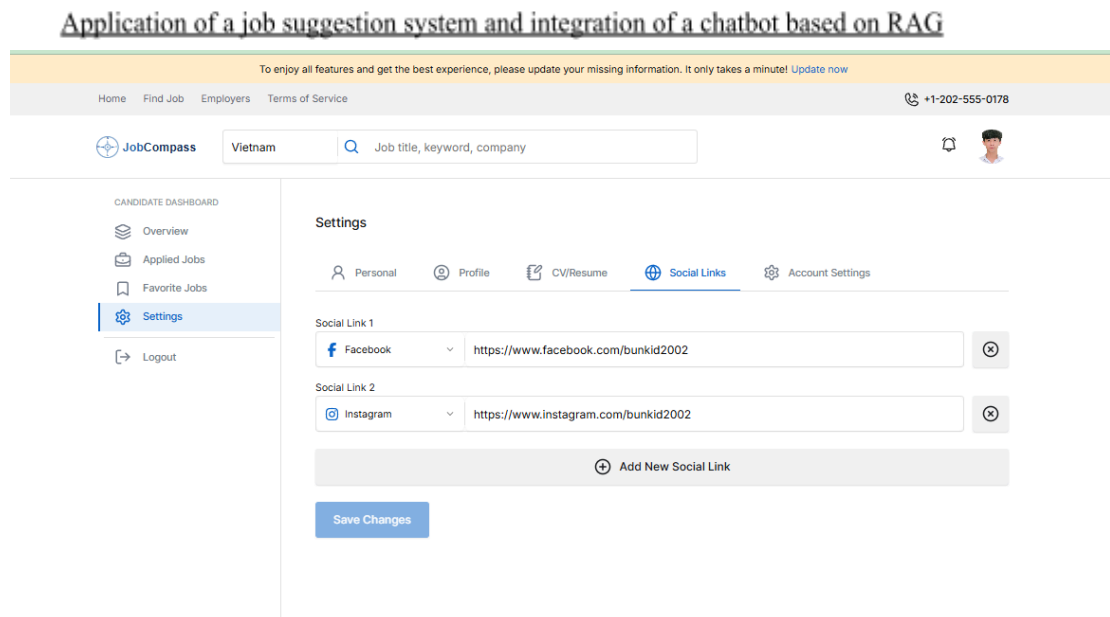


Figure 47: UI for managing social network

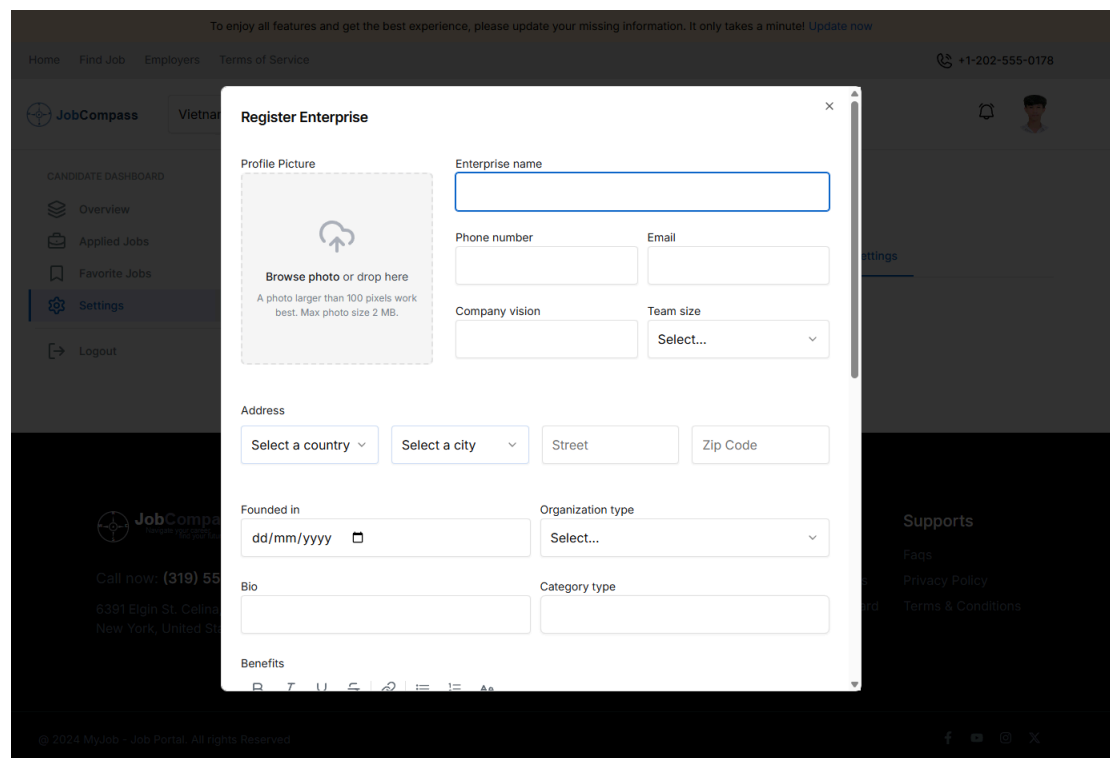


Figure 48: UI for enterprise registration

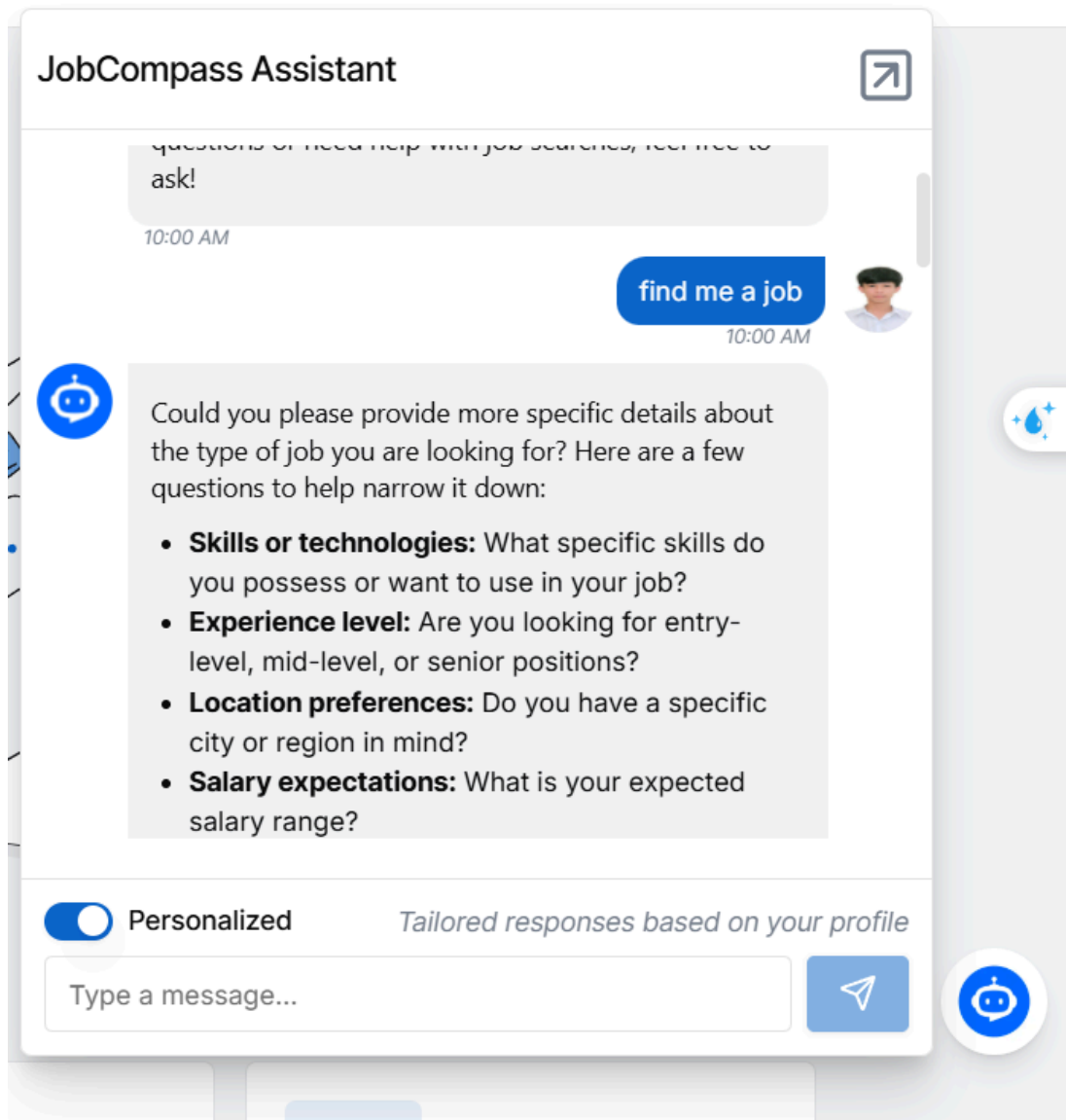


Figure 49: UI chat bot

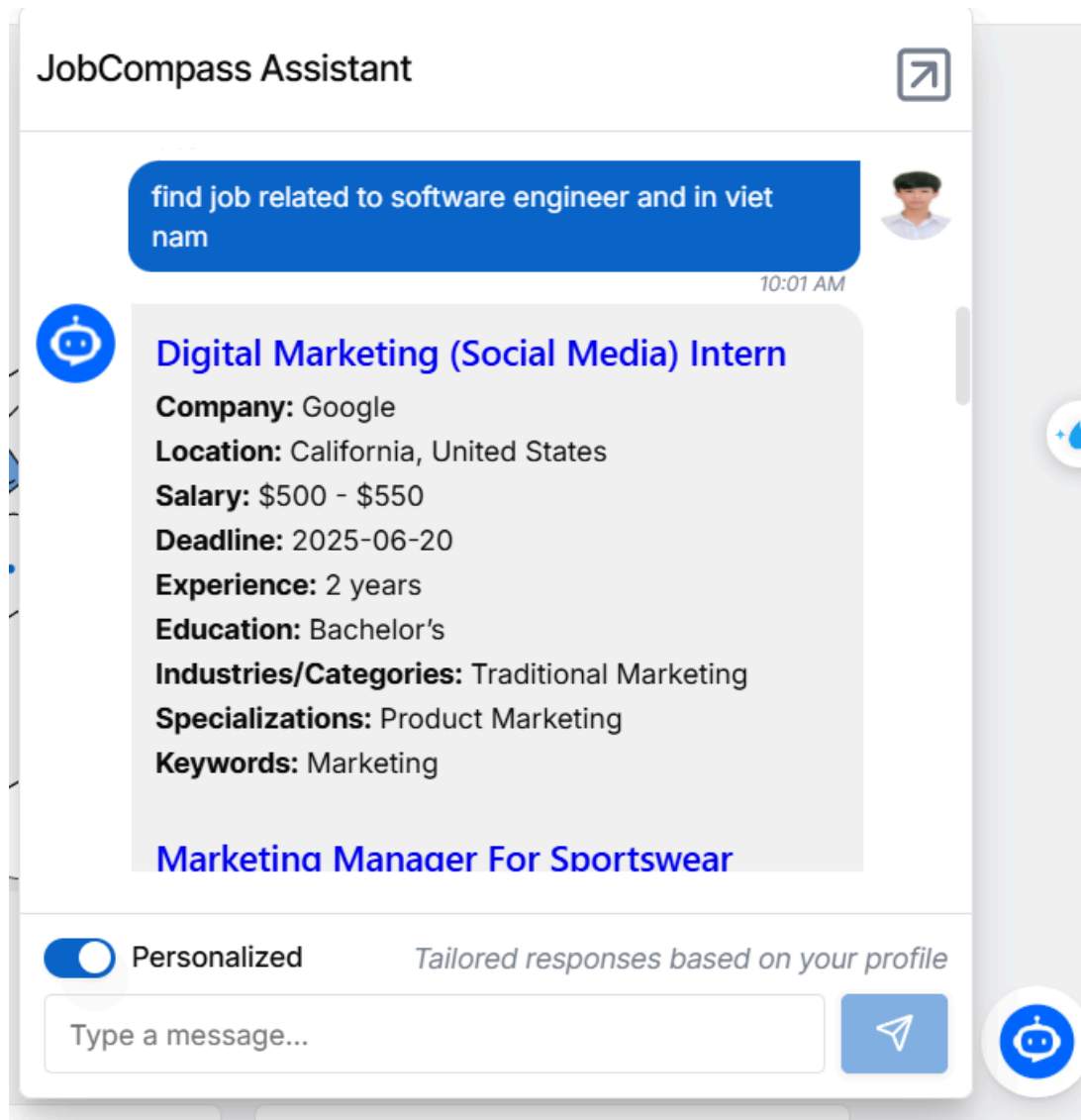


Figure 50: Find job follow user experience

0.7.3 UI for enterprise role

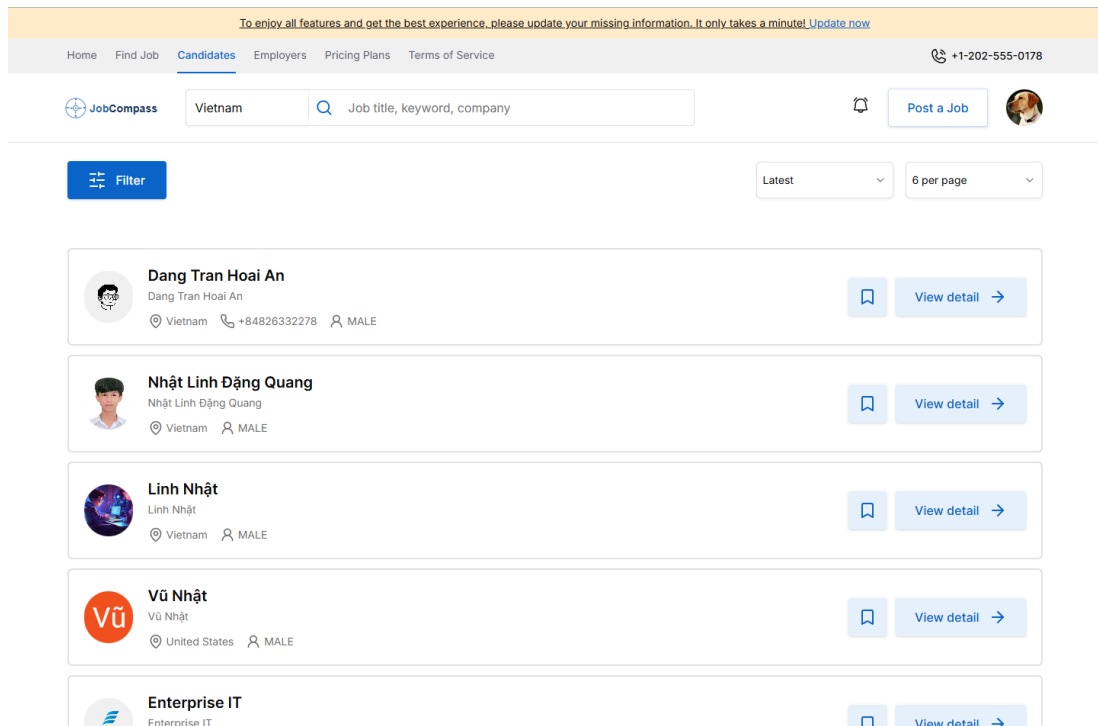


Figure 51: UI candidate list

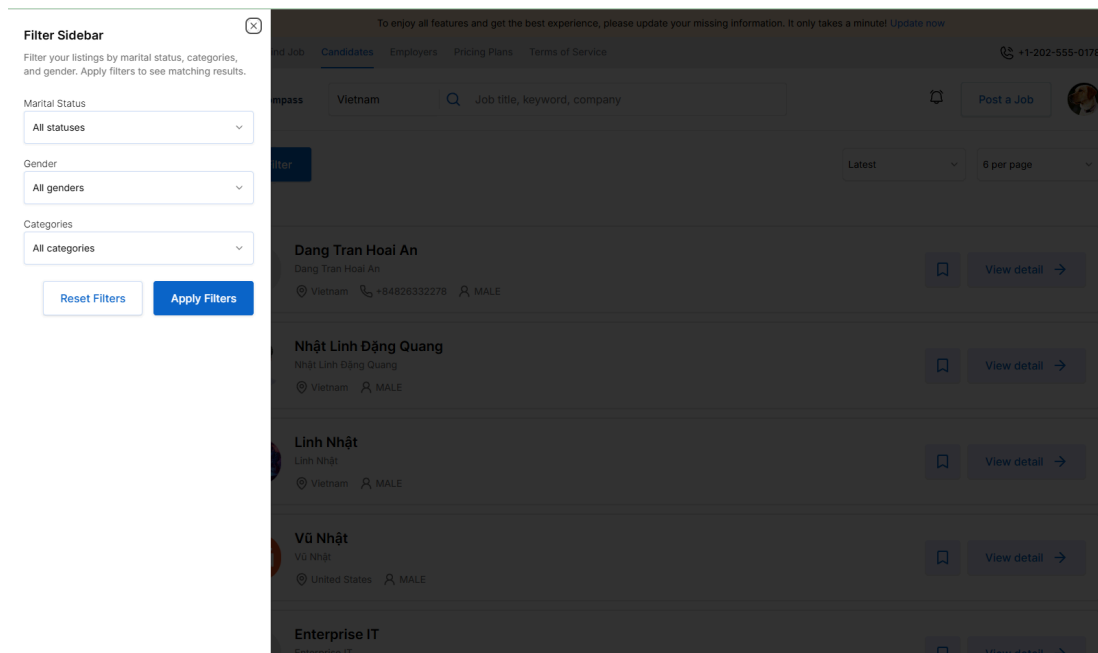


Figure 52: UI for filtering user

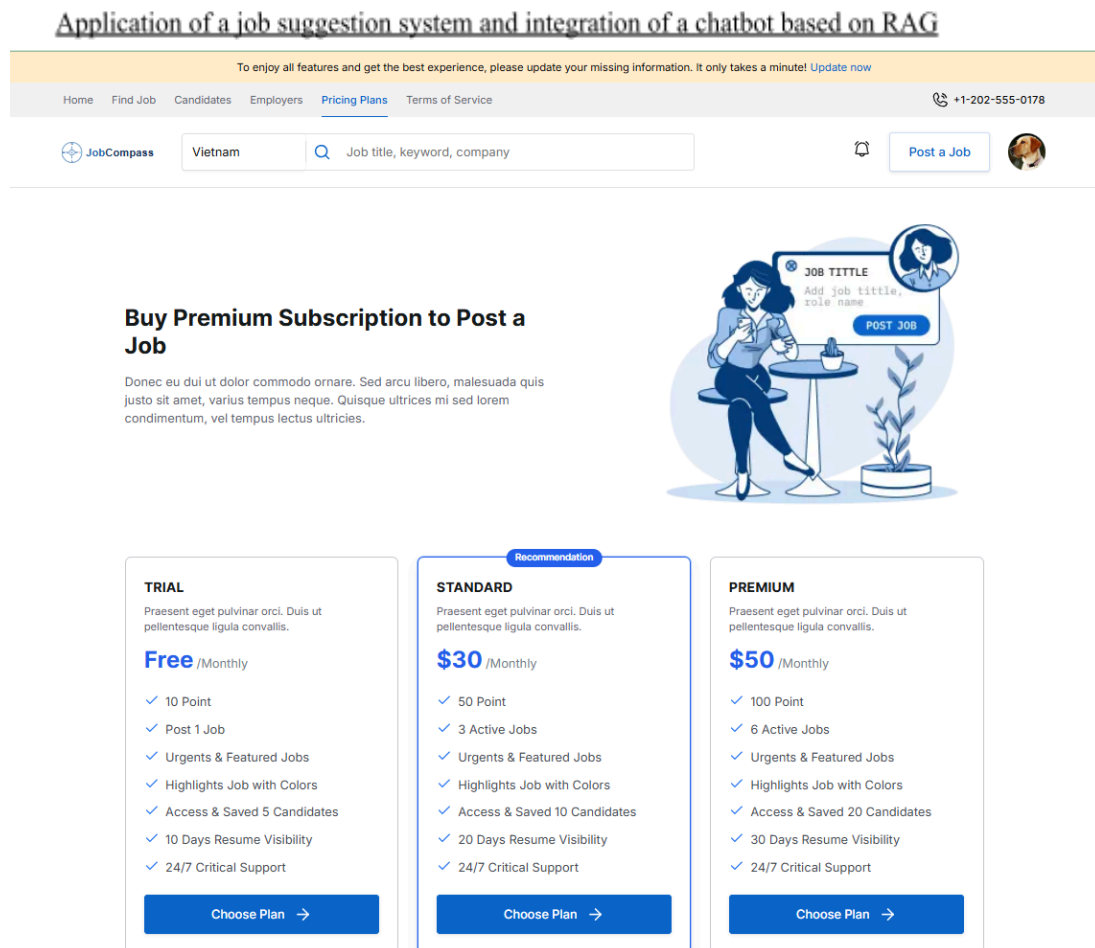


Figure 53: UI for pricing plan

LP **PayPal**

☒ **Pay in full**
\$30.00 today

☐ **Pay in 4**
4 interest-free payments of \$7.50. No late fees.
[Terms](#)

☐ **Pay Monthly**
Available for eligible purchases of \$49.00-\$10,000.00.

Pay with ^

☒ **CREDIT UNION 1** \$30.00
Checking **3995

☐ **Visa**
Credit **5076

+ Add card

Order \$30.00 v

Pay | \$30.00

[About payment methods](#)

[Return to JobCompass](#)

Figure 54: UI for complete payment

Application of a job suggestion system and integration of a chatbot based on RAG

To enjoy all features and get the best experience, please update your missing information. It only takes a minute! [Update now](#)

Home Find Job Candidates Employers Pricing Plans Terms of Service +1-202-555-0178

JobCompass Vietnam Job title, keyword, company Post a Job

EMPLOYER DASHBOARD

- Overview
- Post a Job
- My Jobs
- Plans & Billing
- Settings
- Logout

Hello, Enclave IT
Here is your daily activities and applications

Enterprise Points 110

8 Total jobs 9 Total applicants 3 Promoted Jobs

Your profile Show Enterprise Details Edit Profile

Recently Posted Jobs View all

JOB	CREATE AT	STATUS	ACTION
Information Technology - Software Engineers - 2025 College Grads	28 May 2025	Open	View
IT Helpdesk Support Engineer, Insurance	26 May 2025	Open	View
IT Manager	19 May 2025	Expired	View
IT Manager	19 May 2025	Expired	View
IT Manager	19 May 2025	Open	View

Figure 55: UI for enterprise overview

To enjoy all features and get the best experience, please update your missing information. It only takes a minute! [Update now](#)

Home Find Job Candidates Employers Pricing Plans Terms of Service +1-202-555-0178

JobCompass Vietnam Job title, keyword, company Post a Job

EMPLOYER DASHBOARD

- Overview
- Post a Job
- My Jobs
- Plans & Billing
- Settings
- Logout

Post a job

Job Title Tag
e.g. "Software Engineer" Select tags... Add Tag

Address
Select an address...

Salary
Min Salary Max Salary
USD e.g. "1000" USD e.g. "2000"

Advanced Information
Education Experience (Year) Job Type
Select... e.g. "2" Select...

Expiration Date Job Category Job Specialization
dd/mm/yyyy Select... Select categories...

Description & Responsibility
Description
B I U S Link List Bulleted List Aa

Figure 56: UI for posting job

Application of a job suggestion system and integration of a chatbot based on RAG

To enjoy all features and get the best experience, please update your missing information. It only takes a minute! [Update now](#)

[Home](#) [Find Job](#) [Candidates](#) [Employers](#) [Pricing Plans](#) [Terms of Service](#)

[📞 +1-202-555-0178](#)

[JobCompass](#)

Vietnam

[🔔](#) [Post a Job](#) 

EMPLOYER DASHBOARD

[Overview](#)

[Post a Job](#)

[My Jobs](#)

[Plans & Billing](#)

[Settings](#)

[Logout](#)

My Jobs (8)

Newest

Filter

1 of 2 pages



Technical Architect [Da Nang - Ho Chi Minh City]

Information Technology

Part Time

Open

📍 218 Nguyen Phuoc Lan, Da Nang, Vietnam

👤 4 years of experience

📅 04 Jun 2025 (4 days left)

Architect

Salary range

1K - 2K (USD / month)

Applications

6

Not boosted

View Applications



Information Technology - Software Engineers - 2025 College Grads

Information Technology

Full Time

Open

📍 218 Nguyen Phuoc Lan, Da Nang, Vietnam

👤 2 years of experience

📅 22 Jun 2025 (22 days left)

IT

Salary range

1K - 1.5K (USD / month)

Applications

0



Technical Architect [Da Nang - Ho Chi Minh City]

Enclave IT

Information Technology

Part Time

6 applications

0 views

Salary Range

\$ 1000 - 2000 (USD)

Experience

4 years

Posted On

02 May 2025

Application Deadline

4 days left

Location

218 Nguyen Phuoc Lan, Da Nang, Vietnam

Education

High School

Description

PRIMARY JOB RESPONSIBILITIES

Development Oversight: Lead the software development life cycle,...

Responsibilities

INNOVATIVE, CREATIVE - Think outside of the box

KEEP CHALLENGING - Never stay in your comfort zone...

Benefits

More on OTSV - what is it like to work in OTSV?

At OTSV, work is a form of self-fulfillment, we encourage staff to leave the...

Requirements

POSITION QUALIFICATIONS AND REQUIREMENT

Full-stack Development Expertise: Extensive experience in both...

TAGS

Figure 57: UI for managing job

Performed Student: Dang Quang Nhat Linh

Instructor: Ph.D. Vo Duc Hoang

63

Application of a job suggestion system and integration of a chatbot based on RAG

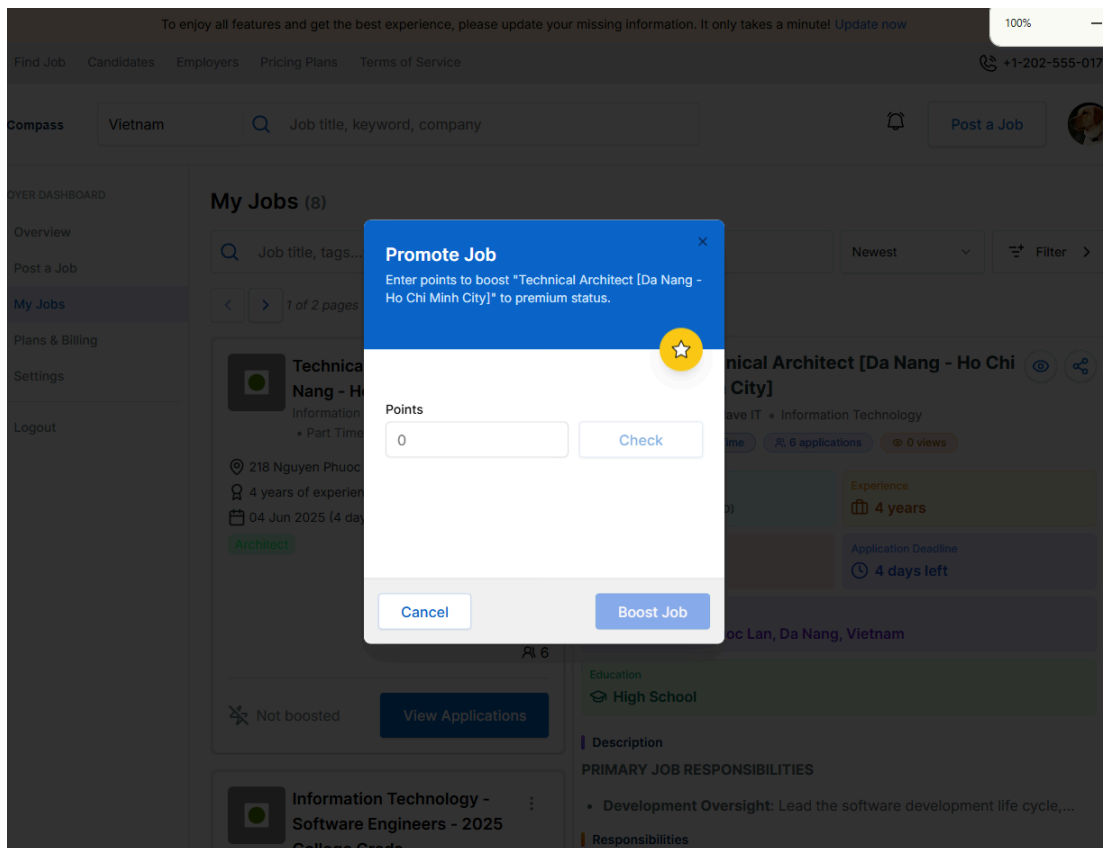


Figure 58: UI for boosting job

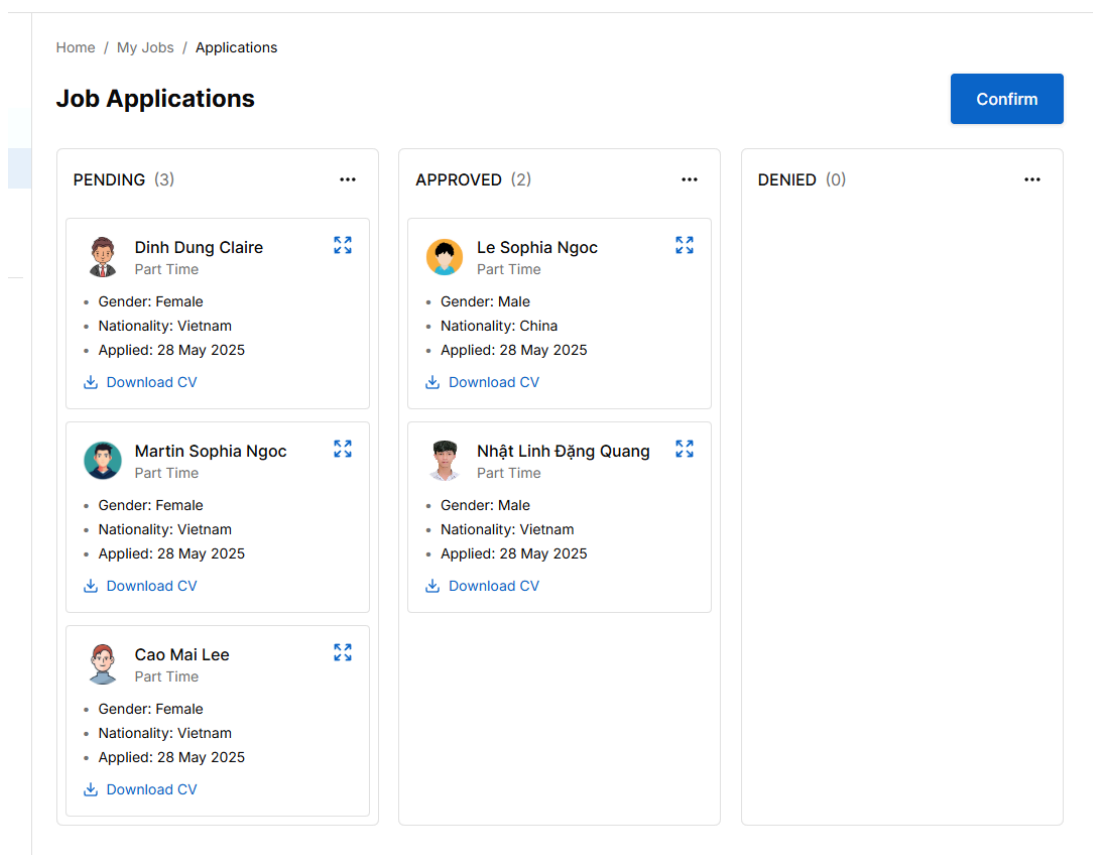


Figure 59: UI for managing applications

Application of a job suggestion system and integration of a chatbot based on RAG

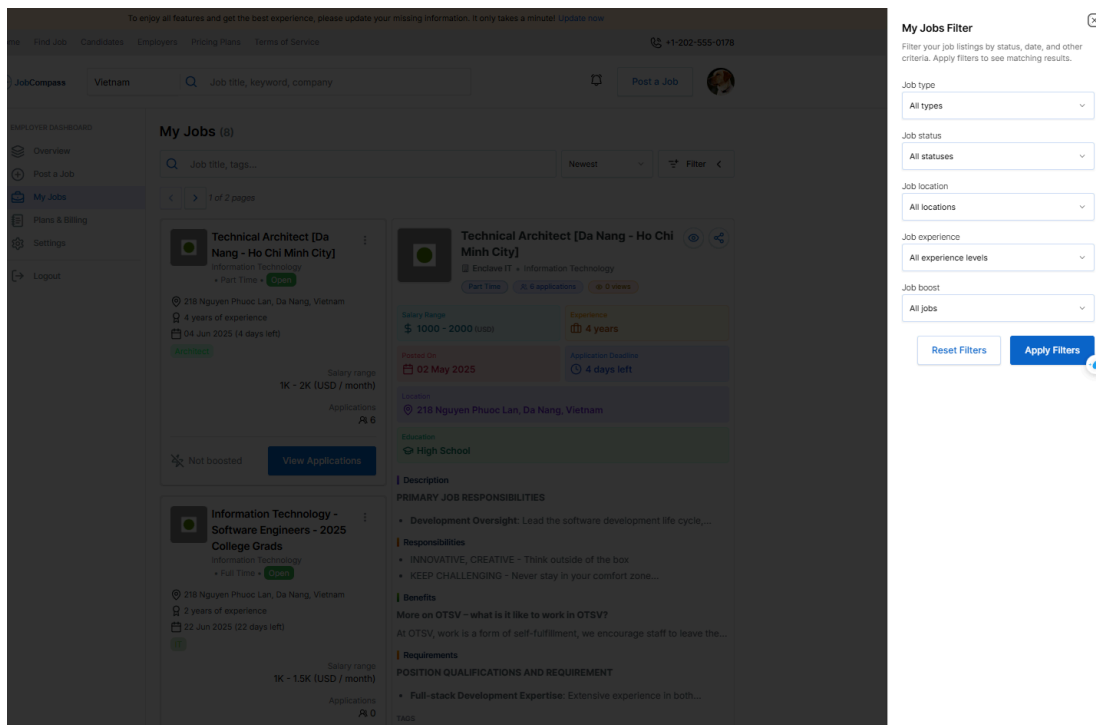


Figure 60: UI for filter job

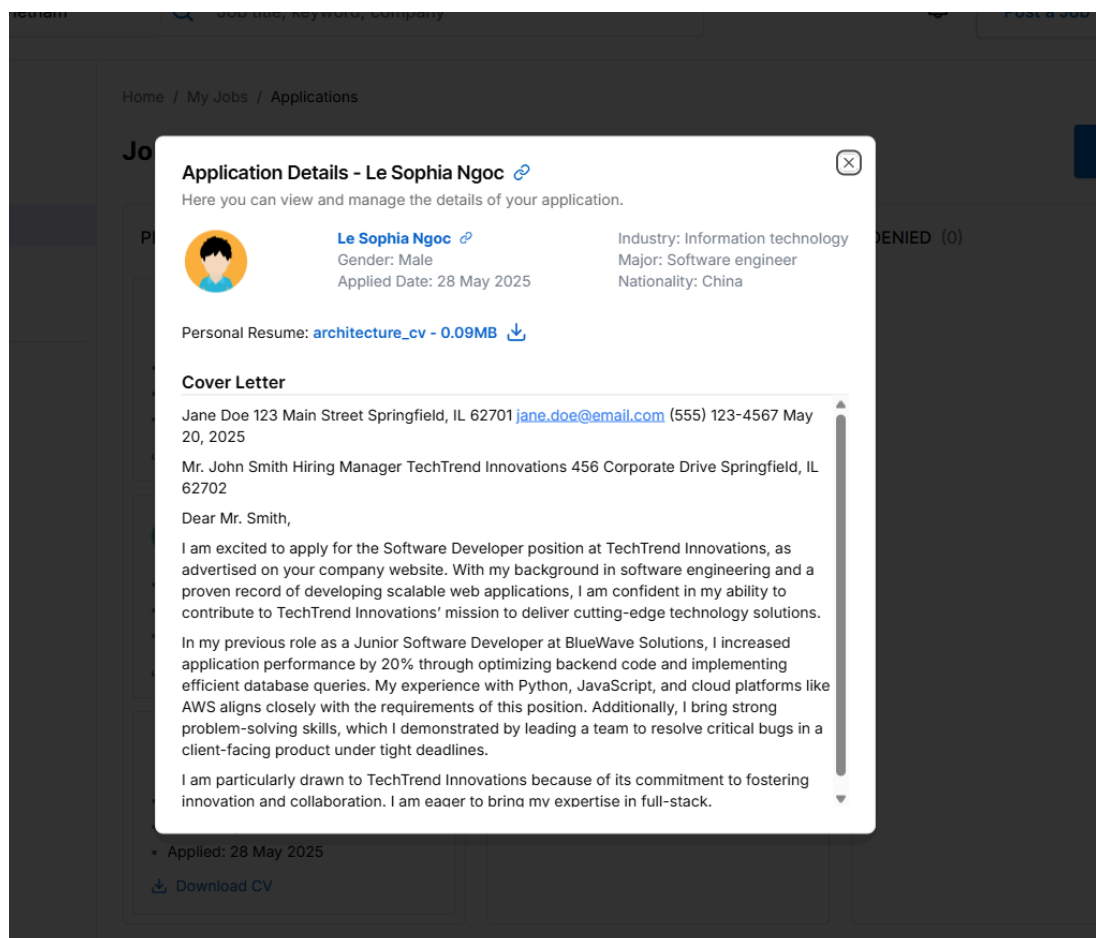


Figure 61: UI for application detail

Application of a job suggestion system and integration of a chatbot based on RAG

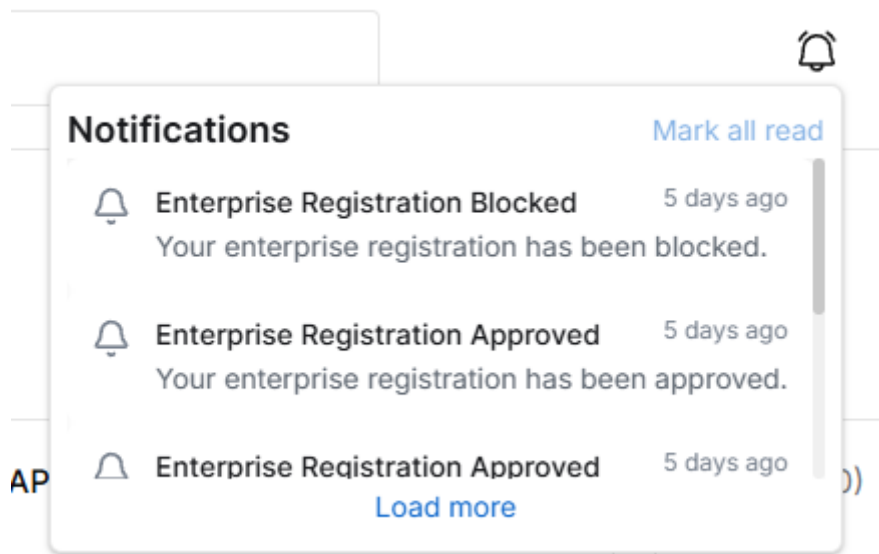


Figure 62: UI for notifications

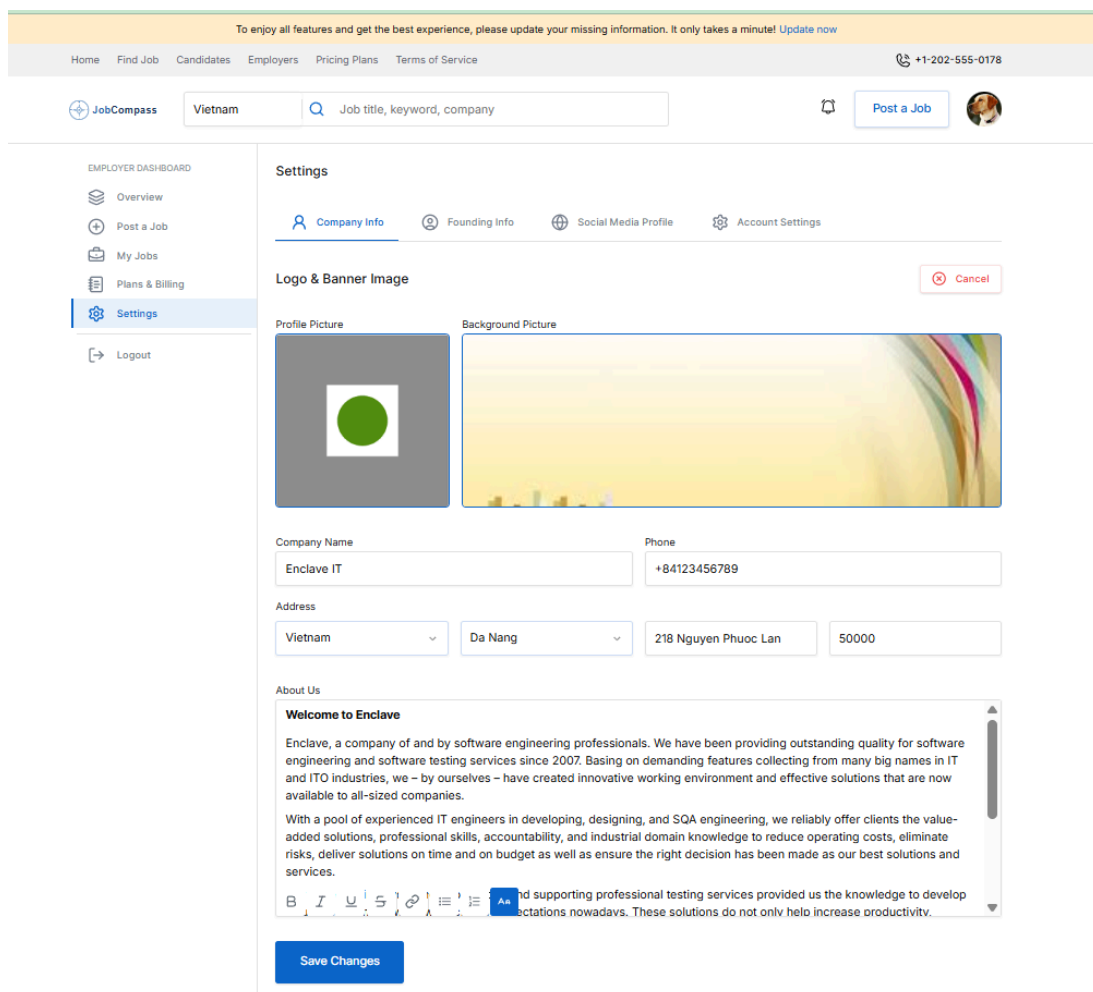


Figure 63: UI for setting enterprise

Application of a job suggestion system and integration of a chatbot based on RAG

To enjoy all features and get the best experience, please update your missing information. It only takes a minute! [Update now](#)

Home Find Job Candidates Employers Pricing Plans Terms of Service
+1-202-555-0178

Vietnam

[Post a Job](#)

EMPLOYER DASHBOARD

- Overview
- Post a Job
- My Jobs
- Plans & Billing
- Settings
- Logout

Settings

Company Info
 Founding Info
 Social Media Profile
 Account Settings

Founding Information [Edit](#)

Organization Type

Flat

Industry Type

Information T...

Team Size

1-10

Year of Establishment

31/01/2007

Company Website

anmixmax03@gmail.com

Introduction (Bio)

Engineering Tomorrow: Innovate, Sustain, Succeed

Company Vision

Engineering Tomorrow: Innovate, Sustain, Succeed

Description

Welcome to Enclave

Enclave, a company of and by software engineering professionals. We have been providing outstanding quality for software engineering and software testing services since 2007. Basing on demanding features collecting from many big names in IT and ITO industries, we – by ourselves – have created innovative working environment and effective solutions that are now available to all-sized companies.

With a pool of experienced IT engineers in developing, designing, and SQA engineering, we reliably offer clients the value-added solutions, professional skills, accountability, and industrial domain knowledge to reduce operating costs, eliminate risks, deliver solutions on time and on budget as well as ensure the right decision has been made as our best solutions and services.

And supporting professional testing services provided us the knowledge to develop solutions nowadays. These solutions do not only help increase productivity.

[Save Changes](#)

Figure 64: UI for setting up a founding enterprise

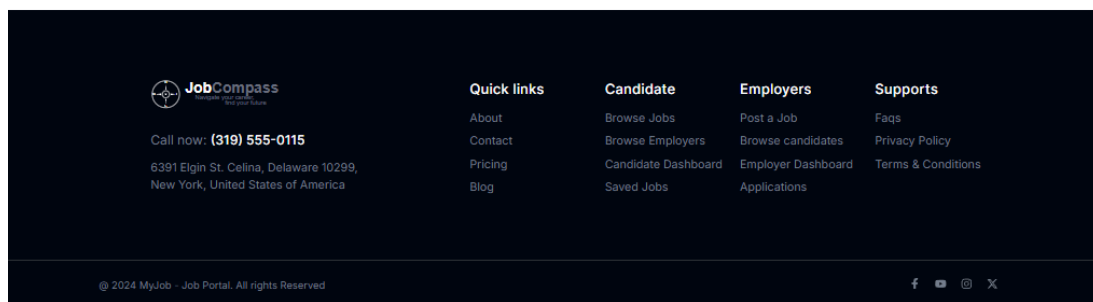
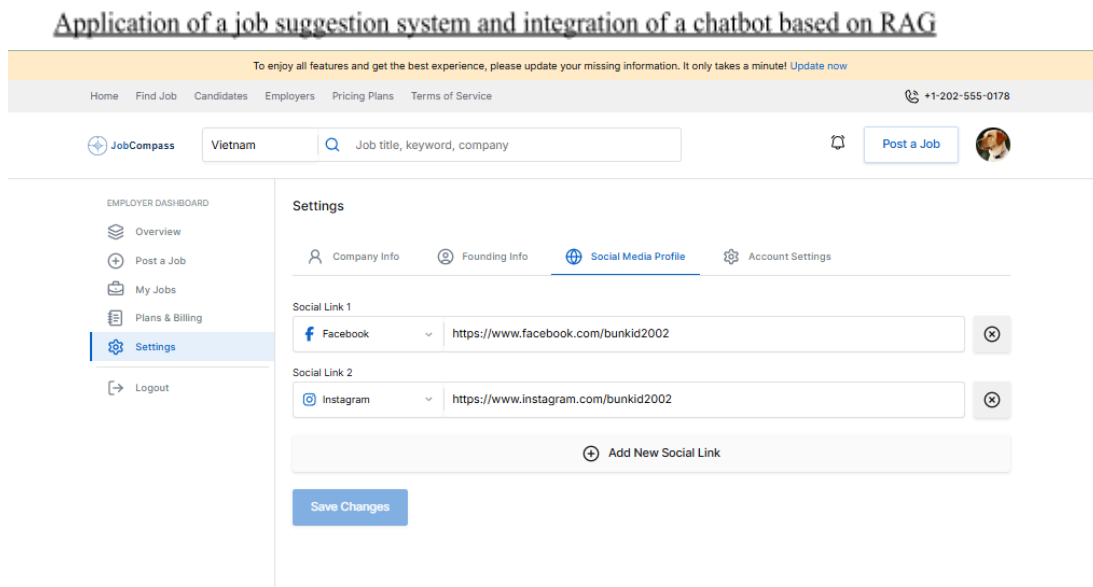


Figure 65: UI for social networking

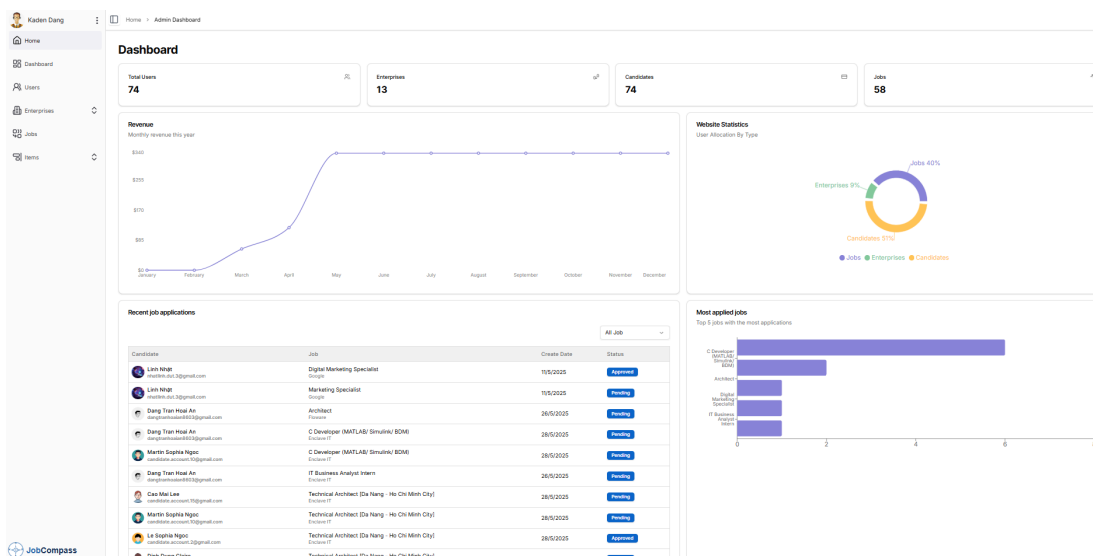
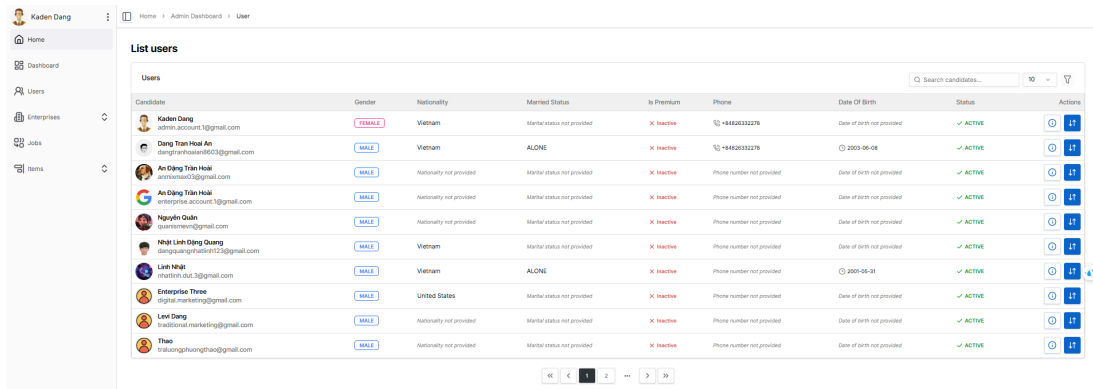


Figure 66: UI for admin dashboard

Application of a job suggestion system and integration of a chatbot based on RAG

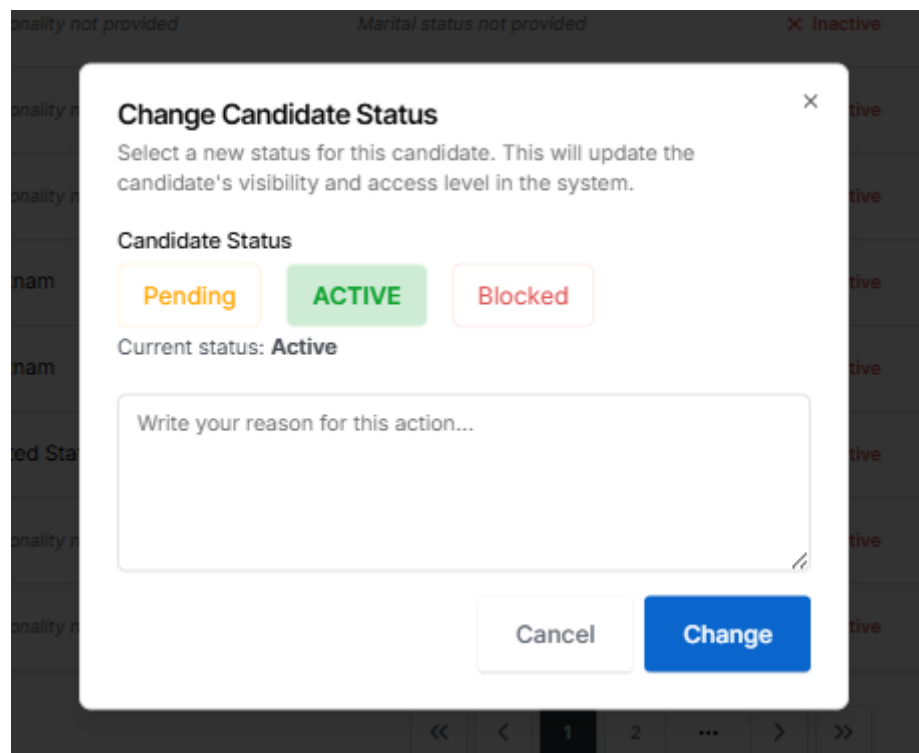


List users

Search candidates... 10

Candidate	Gender	Nationality	Married Status	% Premium	Phone	Date of Birth	Status	Actions
Kaden Dang kaden.dang@gmail.com	MALE	Vietnam	Marital status not provided	Inactive	+8420332276	Date of birth not provided	ACTIVE	
Dang Tran Hoai An dangtranhoai867@gmail.com	MALE	Vietnam	ALONE	Inactive	+8420332276	2003-06-08	ACTIVE	
An Dang Tran Hoai an.dangtranhoai@gmail.com	MALE	Nationality not provided	Marital status not provided	Inactive	Phone number not provided	Date of birth not provided	ACTIVE	
Nguyễn Quân quangnhan@gmail.com	MALE	Nationality not provided	Marital status not provided	Inactive	Phone number not provided	Date of birth not provided	ACTIVE	
Nhiệt Linh Đặng Quang dangquangnhan192@gmail.com	MALE	Vietnam	Marital status not provided	Inactive	Phone number not provided	Date of birth not provided	ACTIVE	
Linh Nhựt linhnhat192@gmail.com	MALE	Vietnam	ALONE	Inactive	Phone number not provided	2001-05-31	ACTIVE	
Enterprise Three digital.marketing@gmail.com	MALE	United States	Marital status not provided	Inactive	Phone number not provided	Date of birth not provided	ACTIVE	
Levi Dang traditional.marketing@gmail.com	MALE	Nationality not provided	Marital status not provided	Inactive	Phone number not provided	Date of birth not provided	ACTIVE	
Thao traoquanghuy@gmail.com	MALE	Nationality not provided	Marital status not provided	Inactive	Phone number not provided	Date of birth not provided	ACTIVE	

Figure 67: UI for listing users



Change Candidate Status

Select a new status for this candidate. This will update the candidate's visibility and access level in the system.

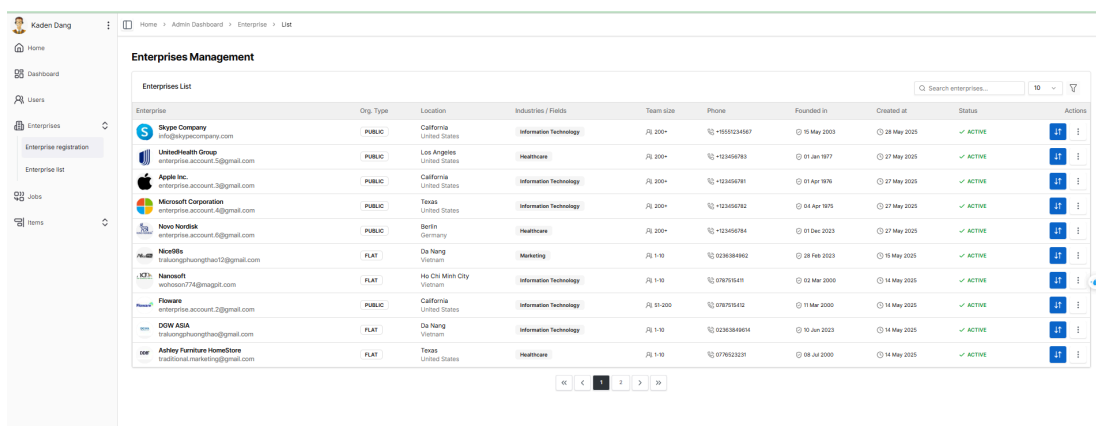
Candidate Status

☐ Pending
 ☒ **ACTIVE**
☐ Blocked

Current status: **Active**

Write your reason for this action...

Figure 68: UI for change status



Enterprises Management

Search enterprises... 10

Enterprise	Org. Type	Location	Industries / Fields	Team size	Phone	Founded in	Created at	Status	Actions
Skype Company info@skypecompany.com	PUBLIC	California United States	Information Technology	100+	+15551234567	15 May 2003	28 May 2025	ACTIVE	
UnitedHealth Group enterprise.account.5@gmail.com	PUBLIC	Los Angeles United States	Healthcare	100+	+13456789	01 Jan 1977	01 May 2025	ACTIVE	
Apple Inc. enterprise.account.3@gmail.com	PUBLIC	California United States	Information Technology	100+	+13456789	01 Apr 1976	07 May 2025	ACTIVE	
Microsoft Corporation enterprise.account.4@gmail.com	PUBLIC	Texas United States	Information Technology	100+	+13456789	04 Apr 1975	07 May 2025	ACTIVE	
Novo Nordisk enterprise.account.6@gmail.com	PUBLIC	Berlin Germany	Healthcare	100+	+13456789	01 Dec 2023	07 May 2025	ACTIVE	
Niceville traoquanghuy192@gmail.com	PLAT	Da Nang Vietnam	Marketing	1-10	0238384862	28 Feb 2023	10 May 2025	ACTIVE	
Nanosoft wohson774@gmail.com	PLAT	Ho Chi Minh City Vietnam	Information Technology	1-10	07670349	02 Mar 2020	14 May 2025	ACTIVE	
Floware enterprise.account.2@gmail.com	PUBLIC	California United States	Information Technology	91-200	07670349	11 Mar 2020	14 May 2025	ACTIVE	
DOW ASA traoquanghuy@gmail.com	PLAT	Da Nang Vietnam	Information Technology	1-10	0238384862	10 Jan 2023	14 May 2025	ACTIVE	
Ashley Furniture HomeStore traditional.marketing@gmail.com	PLAT	Texas United States	Healthcare	1-10	076622323	08 Jul 2000	14 May 2025	ACTIVE	

Figure 69: UI for managing enterprise

Application of a job suggestion system and integration of a chatbot based on RAG

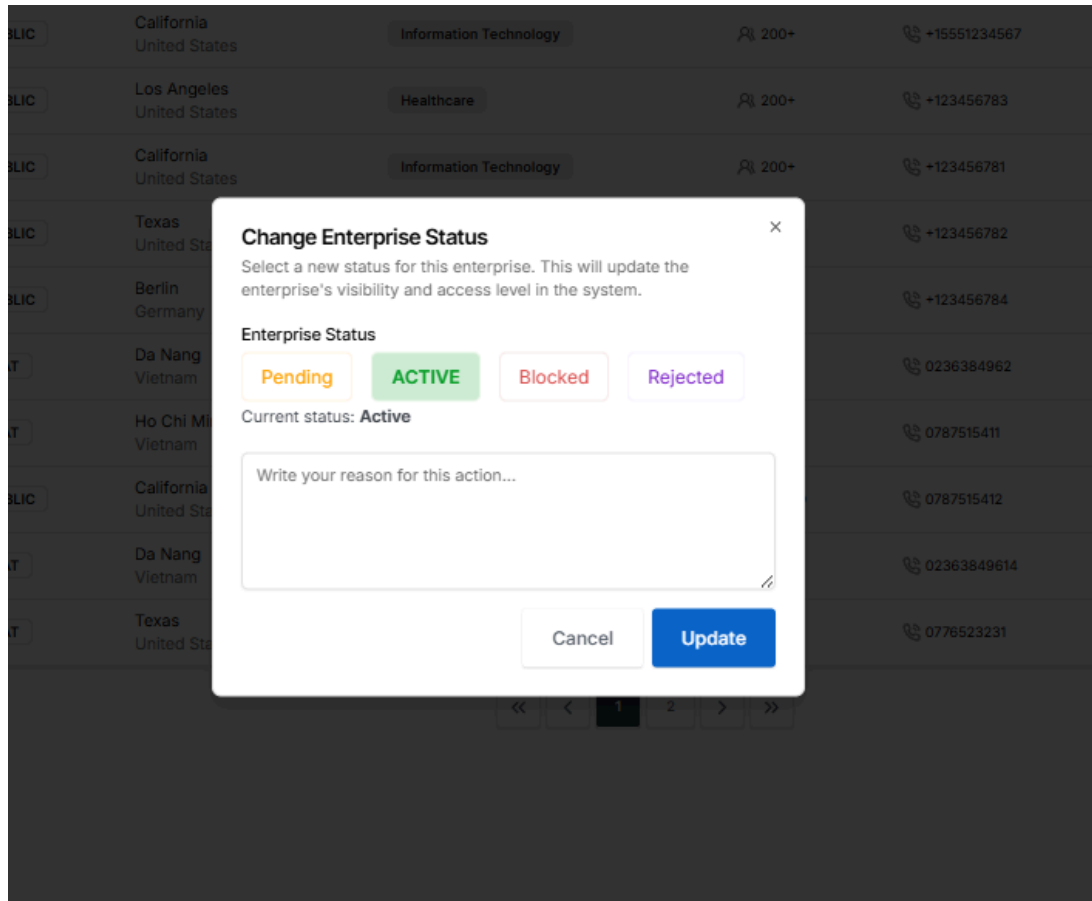


Figure 70: UI for change status enterprise

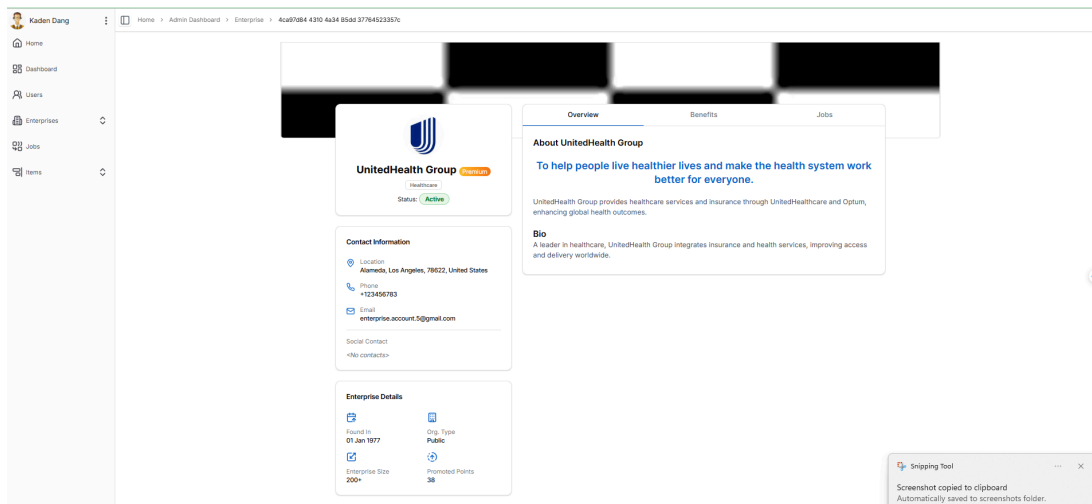


Figure 71: UI for enterprise detail

Application of a job suggestion system and integration of a chatbot based on RAG

Jobs Management

Job List

Job	Status	Type	Industry	Specializations	Education	EXP.	Salary	Tags	Address	Deadline	Created at	Actions
IT Manager Encore IT	Open	Full Time	Information Technology	Software Engineer, Software Architecture	High School	6 years	\$2K - \$2.5K	Software	Da Nang, Vietnam	31 May 2025	19 May 2025	[Edit] [Delete]
C Developer (MATLAB/ Simulink) SOM	Open	Full Time	Information Technology	Software Engineer, Software Architecture	Bachelor's	5 years	\$3K - \$4K	C	Da Nang, Vietnam	31 May 2025	02 May 2025	[Edit] [Delete]
Technical Architect (Da Nang - Ho Chi Minh City)	Open	Part Time	Information Technology	Software Engineer, Software Architecture	High School	4 years	\$1K - \$2K	Architecture	Da Nang, Vietnam	04 Jun 2025	02 May 2025	[Edit] [Delete]
Digital Product Marketing Manager Digital Marketing company	Open	Temporary	Marketing	Advertising	High School	1 years	\$1.4K - \$2K	Marketing, Advertising	California, United States	05 Jun 2025	13 May 2025	[Edit] [Delete]
IT Business Analyst Intern Encore IT	Open	Internship	Information Technology	Data Analyst, Software Engineer	Bachelor's	1 years	\$1K - \$1.5K	Data, Analytics	Da Nang, Vietnam	06 Jun 2025	30 Apr 2025	[Edit] [Delete]
Information Technology - Software Engineers - 2025 College Grads Google	Open	Full Time	Information Technology	Software Engineer	Bachelor's	2 years	\$1K - \$2K	IT	California, United States	07 Jun 2025	28 May 2025	[Edit] [Delete]
IT Manager Encore IT	Expired	Full Time	Information Technology	Software Engineer, Software Architecture	High School	6 years	\$2K - \$2.5K	Software	Da Nang, Vietnam	07 Jun 2025	19 May 2025	[Edit] [Delete]
IT Manager Encore IT	Expired	Full Time	Information Technology	Software Engineer, Software Architecture	High School	6 years	\$2K - \$2.5K	Software	Da Nang, Vietnam	07 Jun 2025	19 May 2025	[Edit] [Delete]
IT Helpdesk Support Engineer, Insurance Encore IT	Open	Full Time	Information Technology	Software Engineer	Master's or Higher	2 years	\$2K - \$2.5K	IT	Da Nang, Vietnam	08 Jun 2025	26 May 2025	[Edit] [Delete]
Marketing Assistant Astory Furniture HomeDecor	Open	Contract	Marketing	Digital Marketing, Public Relations	High School	2 years	\$3K - \$4K	Marketing, Assistant	Texas, United States	14 Jun 2025	14 May 2025	[Edit] [Delete]

Figure 72: UI for list job

Tags Management

Name	Color	Preview	Created At	Updated At	Actions
☐ AI		AI	30 Apr 2025	29 May 2025	[Edit] [Delete]
☐ Analyst		Analyst	30 Apr 2025	30 Apr 2025	[Edit] [Delete]
☐ Analytics		Analytics	15 May 2025	15 May 2025	[Edit] [Delete]
☐ Architect		Architect	02 May 2025	02 May 2025	[Edit] [Delete]
☐ Assistant		Assistant	14 May 2025	14 May 2025	[Edit] [Delete]
☐ Automation Testing		Automation Testing	30 Apr 2025	30 Apr 2025	[Edit] [Delete]
☐ AWS		AWS	30 Apr 2025	30 Apr 2025	[Edit] [Delete]
☐ Azure		Azure	30 Apr 2025	30 Apr 2025	[Edit] [Delete]
☐ C		C	02 May 2025	02 May 2025	[Edit] [Delete]
☐ Data		Data	30 Apr 2025	30 Apr 2025	[Edit] [Delete]
☐ DevOps		DevOps	30 Apr 2025	30 Apr 2025	[Edit] [Delete]
☐ DM		DM	13 May 2025	13 May 2025	[Edit] [Delete]
☐ Frontend		Frontend	15 May 2025	15 May 2025	[Edit] [Delete]

Figure 73: UI for managing tag

Categories Management

Name	Subcategories	Created At	Updated At	Actions
☐ Information Technology	7 subcategories	30 Apr 2025	24 May 2025	[Edit] [Delete] [Add]
☐ Software Engineer		30 Apr 2025	30 Apr 2025	[Edit] [Delete]
☐ Software Architecture		30 Apr 2025	30 Apr 2025	[Edit] [Delete]
☐ Quality Assurance		30 Apr 2025	30 Apr 2025	[Edit] [Delete]
☐ DevOps Engineer		30 Apr 2025	30 Apr 2025	[Edit] [Delete]
☐ Cybersecurity Engineer		30 Apr 2025	30 Apr 2025	[Edit] [Delete]
☐ Machine Learning Engineer		30 Apr 2025	30 Apr 2025	[Edit] [Delete]
☐ Data Analyst		30 Apr 2025	30 Apr 2025	[Edit] [Delete]
☐ Marketing	7 subcategories	30 Apr 2025	24 May 2025	[Edit] [Delete] [Add]
☐ Healthcare	7 subcategories	30 Apr 2025	24 May 2025	[Edit] [Delete] [Add]

Figure 74: UI for managing category

0.7.4 Notifications outside

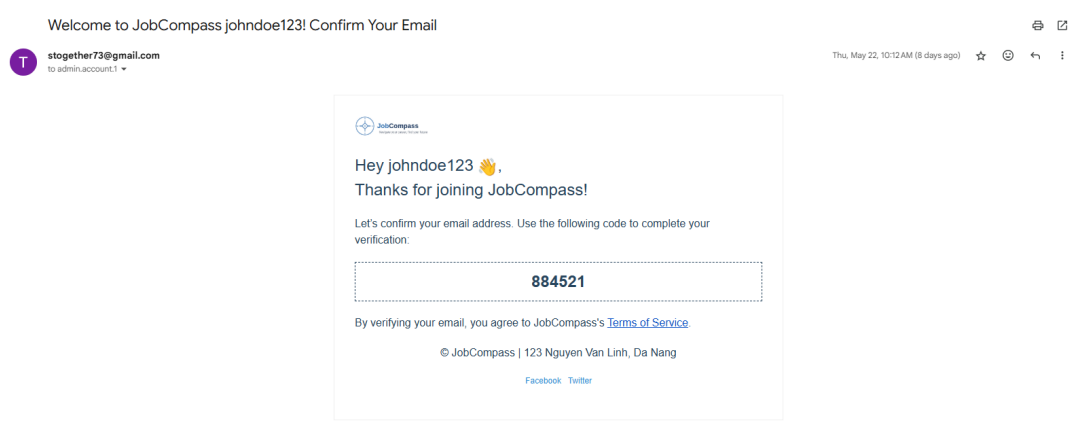


Figure 75: UI for code verification email

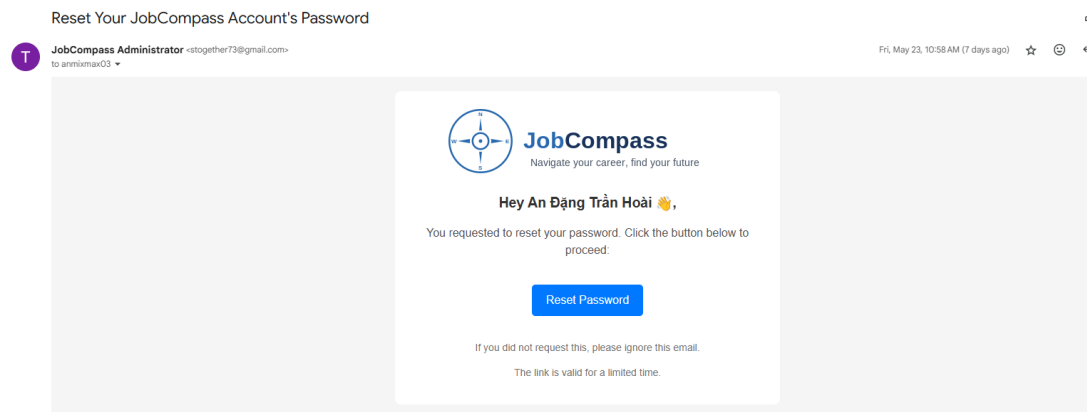


Figure 76: UI for reset password

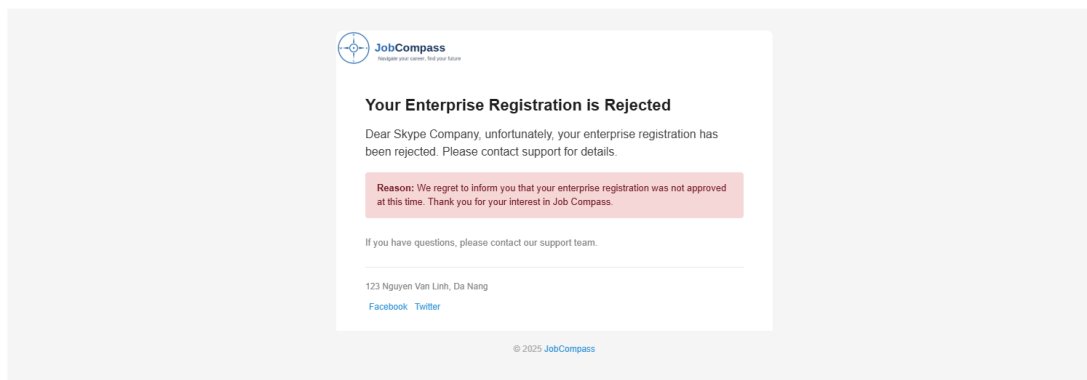


Figure 77: UI for admin reject form enterprise registration

Application of a job suggestion system and integration of a chatbot based on RAG

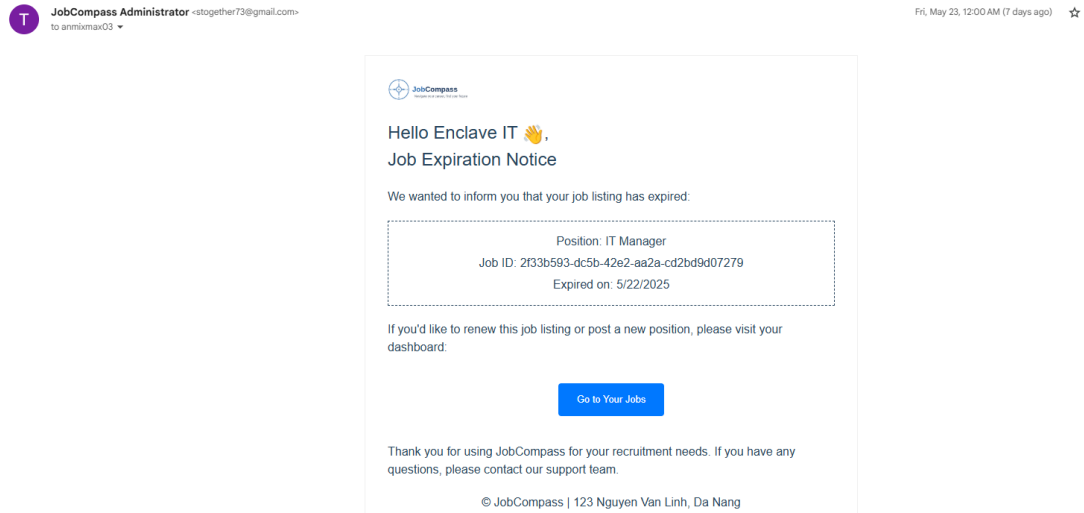


Figure 78: UI for job expired

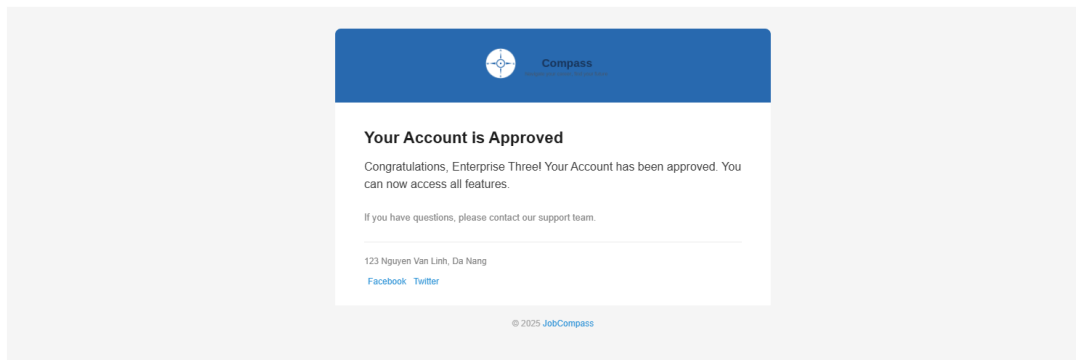


Figure 79: UI when admin accepted enterprise form

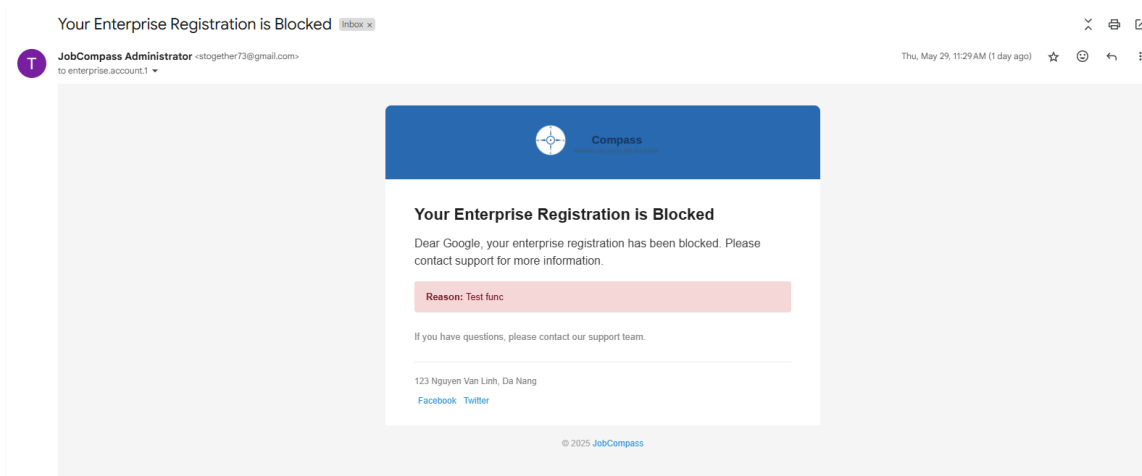


Figure 80: UI when admin block enterprise

0.8 Evaluation

0.8.1 The effectiveness of TF-IDF algorithms

The TF-IDF-based model was developed to extract meaningful keywords from job-related text data and assist in job classification and relevance scoring. The model was assessed based on the quality of keyword extraction, alignment with actual job categories, and its utility in downstream tasks such as industry prediction and semantic search.

Preprocessing Strategy

```
df['processed_text'] = df.apply(lambda row: ''.join([
    preprocess_text(row['Job Title']) * 6,
    preprocess_text(row['Job Description']),
    preprocess_text(row['Job Requirements']),
    preprocess_text(row['Industry']) * 8,
    preprocess_text(row['Career Level']) * 2,
    preprocess_text(row['Job Type'])
]), axis=1)
```

Weighting rationale:

- **Industry:** Multiplied by 8 — to emphasize the job domain strongly.
- **Job Title:** Weighted 6 times — as a concise summary of the position.
- **Career Level:** Given moderate emphasis with a weight of 2 — indicating seniority or experience level.
- **Job Description, Requirements, and Job Type:** Included with default weight — to retain general context.

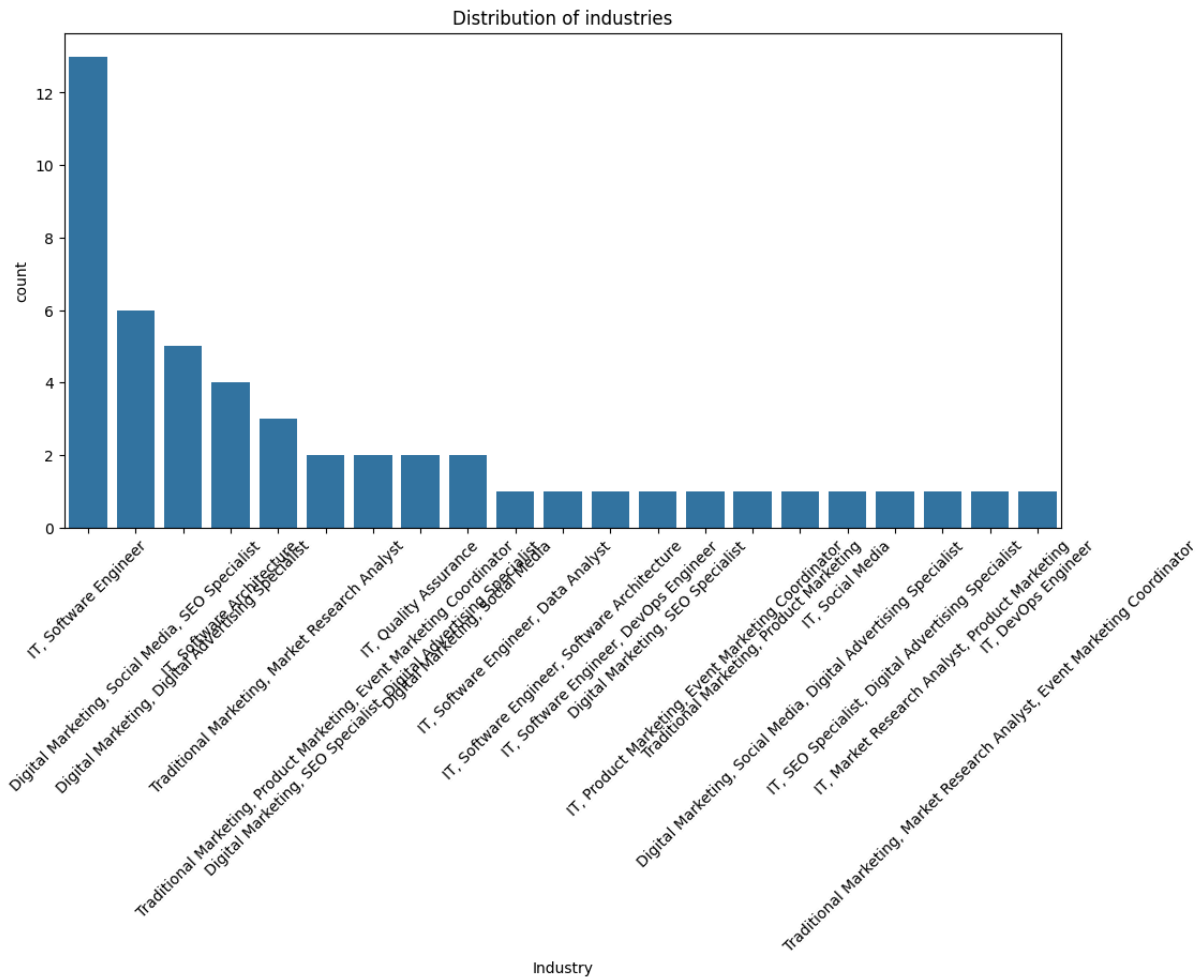


Figure 82: Industry type distribution

A bar chart showing the frequency of job postings across various industries. The consistency between frequently occurring industries (e.g., *"IT, Software Engineer"*, *"Digital Marketing"*, *"SEO Specialist"*) and the top TF-IDF keywords indicates strong alignment between extracted terms and actual domain representation.

0.8.2 The embedding models

In the model evaluation phase of our recommendation system, we selected and tested several text embedding models based on key criteria such as retrieval performance, inference cost, and deployability. Notably, the **Snowflake/snowflake-arctic-embed-m** model was chosen as the primary candidate due to its outstanding performance and balanced trade-off between accuracy and computational efficiency.

Model Description

Snowflake Arctic Embed is a family of embedding models developed by Snowflake, optimized for text retrieval tasks. The **arctic-embed-m** model, with 110 million parameters and a 768-dimensional embedding size, achieves a

NDCG@10 score of 54.90 on the MTEB (Massive Text Embedding Benchmark) leaderboard. This outperforms many well-known models such as bge-base-en-v1.5 (53.25) and gte-base (52.31).

The model is trained using a multi-stage pipeline:

- **Pretraining** on 400 million query-document pairs using in-batch negative sampling.
- **Fine-tuning** on around 1 million triplets (query, positive, negative) with hard-negative mining to improve accuracy.
- The training data includes both open-source corpora and proprietary web search data, enhancing the model's ability to handle natural user queries effectively.

Table 16: Performance Evaluation

Model	MTEB NDCG@10	Parameters (M)	Embedding Dim
Arctic-embed-m	54.90	110	768
bge-base-en-v1.5	53.25	~110	768
omic-embed-text-v1.5	53.25	~100	768
GIST-Embedding-v0	52.31	~90	768
gte-base	52.31	~110	768

The results show that Arctic-embed-m clearly outperforms comparable models in its class, proving its effectiveness in encoding both queries and documents for retrieval tasks.

0.8.3 The LLMs

In the large language model (LLM) selection process, the key criteria were: reasoning ability, response latency, context window, deployment flexibility (cloud or local), and cost-effectiveness for production use. This section evaluates leading LLMs suitable for inference and reasoning tasks in our system, with a special focus on **GPT-4o mini**, OpenAI's newest lightweight model released in 2025.

Table 17: Other LLMs Evaluated

Model	Provider	Context	Cost (\$/1k tokens)	Notes
GPT-4o	OpenAI	128k	~\$0.005–0.015	State-of-the-art, best overall

Application of a job suggestion system and integration of a chatbot based on RAG

Model	Provider	Context	Cost (\$/1k tokens)	Notes
GPT-3.5-turbo	OpenAI	16k	~\$0.001–0.002	Best value for moderate tasks
Mistral 7B/8x7B	Mistral AI	32k	Free (self-hosted)	Fast, cost-free for local use
LLaMA 3 (8B / 70B)	Meta	8k–32k	Free (self-hosted)	Competitive with GPT-3.5 when fine-tuned

Conclusion

GPT-4o mini was selected as the default LLM for the current phase of our system due to its:

- Strong balance of reasoning ability and speed.
- Compatibility with OpenAI's API and embedding models.
- Significantly lower cost compared to GPT-4, while still offering high-quality answers.

For open-source or offline deployments, **LLaMA 3 (8B or 70B)** or **Phi-3 Mini** may be considered in future iterations, especially for privacy-sensitive or resource-constrained environments.

0.8.4 Advantages

In terms of UI and UX, my application provides a simple UI with a good UX, so users find it easy full-featured experience.

Regarding functionalities, Time to implement and complete the process easily and quickly (in line with the criteria of shortening time and traditional recruitment process) the system provides all necessary features for users, which is declared in initial requirements:

- Job seekers must complete registration operations to have a user account in the system.
- Job seekers must login into the system with a valid account to use the features for job seekers.
- The system suggests jobs for candidates based on user behavior.
- Candidates can search for job information by name, location, salary, job.

Application of a job suggestion system and integration of a chatbot based on RAG

- Every time, candidate apply for a job, this CV will be automatically submitted to that job posting. They may be able to change the CV uploaded to the system if needed.
- Job seekers can see the recent job view, favorite job, applied job.
- They can also see the status of that job and will be notified if it is approved or rejected by the Enterprise or when the job has ended the recruitment period.

0.8.5 Disadvantages

Besides the advantages mentioned above, our website still has some disadvantages:

First, The UI on the web application does not well Since, in some page I only focus on full screen responsive. Furthermore, the number of jobs data is quite small.

Secondly, the data to serve the recommender system is quite a bit small so that the model of recommender may perform well according to the required problem which was introduced before. Nevertheless, when the data get bigger in the future, the model will be facing some problems. In addition, the application has not been strongly popularized for cooperation yet and with the development of technology in 4.0 area the connection and unity between the systems is inevitable.

Finally, the system is only in charge of maximum 100 connection at the same time. When the number connection exceeds at that time, the system may occur overloaded server.

Chapter 4 CONCLUSION AND FUTURE WORK

0.9 Result

Through the completion of this project, I have gained valuable knowledge and practical experience from both research and experimentation.

Theoretical Knowledge

- Have basic knowledge about Machine learning.
- Have knowledge of system analysis and design programming, libraries, framework.
- Have experience in using Framework and language, version control system
- I have knowledge about recommendation algorithms.
- In practical term, I have created an application operating both mobile and web platform which helps to resolve the current employment issue.

Practical Achievements

- Successfully developed an application that operates on both mobile and web platforms, aiming to address current employment challenges.

0.10 Future work

As outlined in the evaluation section, there are still some limitations in the current system. Future improvements for this project may include:

- Strengthening collaboration with businesses to implement more professional and authoritative features, including a comprehensive customer support system.
- Integrating AI-powered CV assessment to assist enterprises in evaluating candidates more efficiently.
- Adding an online testing and interview system to enhance the connection between candidates and companies, making the recruitment process faster and more streamlined.

REFERENCES

1. <https://www.geeksforgeeks.org/understanding-tf-idf-term-frequency-inverse-document-frequency/>
2. <https://viblo.asia/p/rag-ai-agents-va-agentic-rag-chung-la-gi-va-hoat-dong-nhu-the-nao-vlZL9KjW4QK>
3. <https://www.datacamp.com/tutorial/agentic-rag-tutorial>
4. <https://nextjs.org/>
5. <https://nextjs.com/>
6. <https://www.langchain.com/>
7. <https://flask.palletsprojects.com/en/stable/>
8. <https://viblo.asia/p/tim-hieu-ve-word-embedding-Ljy5VD7kZra>